# Advanced Quantum Mechanics 02/01/26

1 electron should see its value of wavefunction value to be zero at the position of another fermion, so that the electron does not go there.

- We solve this by writing an antisymmetrised wavefunction.

* (How?)

- What are the dimensions of Hilbert space for 1 single free particle problem (1-D)?

**Ans:** Infinite.

- If infinite, countably or uncountably?

**Ans:** Uncountably infinite.

* **Extra:** For identical fermions, total wavefn must be antisymmetric under exchange of any two particles : $\boxed{\Psi(r_1, r_2) = -\Psi(r_2, r_1)}$

If $2e^-$ want to be in same quantum state, $\boxed{r_1 = r_2 = r} \Rightarrow \Psi(r, r) = -\Psi(r, r)$

$$\Rightarrow \Psi(r, r) = 0$$

State of a system is encoded in a vector which belongs to the vector space named Hilbert's space.

$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$

$|\psi\rangle = \alpha|\uparrow\rangle_z + \beta|\downarrow\rangle_z$

- For normalization:

$|\alpha|^2 + |\beta|^2 = 1$

$|\uparrow\rangle_z$ and $|\downarrow\rangle_z$ are linearly independent (no overlap).

Now we know,

$\hat{S}_z|\uparrow_z\rangle = +1|\uparrow\rangle_z$

So, Hilbert space here is 2-dimensional as $|\uparrow\rangle_z$ has eigenvalue +1 and $|\downarrow\rangle_z$ has eigenvalue −1 under $\hat{S}_z$ operator.

- Why is $|x\rangle$ a bad choice for Hilbert space of a free particle?

Ans: $|x\rangle$ is not an eigenstate of free-parti

Hamiltonian.

$|x\rangle$ is a bad choice of Hilbert space for free particle because it is not an eigenstate of $\hat{H}$ of free particle.

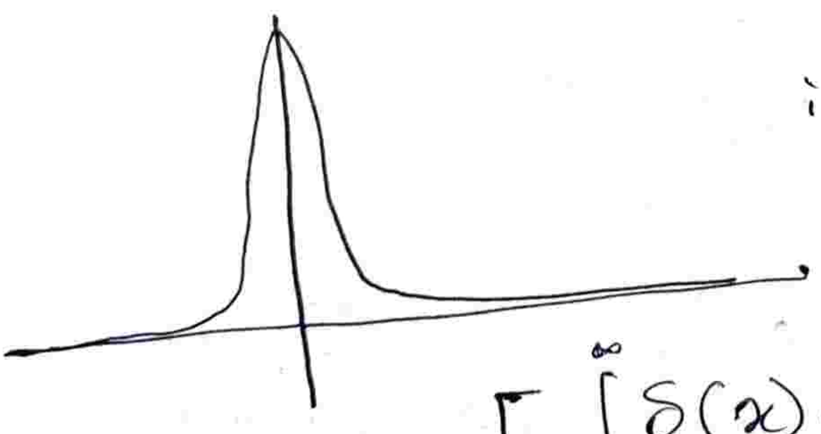
A better choice is $|k\rangle$ [Wavevector]

- This is an eigenstate of free particle Hamiltonian.

- As all Hermitian operators have two eigenvalues and two eigenstates which are orthogonal to each other.

Now, $\langle\uparrow|\uparrow\rangle_z = 1$

$\langle\downarrow|\uparrow\rangle_z = 0$



Suppose a system is initially prepared like a delta.

$\langle x|x'\rangle \propto \delta(x-x')$

$$\left[\int_{-\infty}^{\infty} \delta(x)\,dx = 1\right]$$

This orthonormality tells us, there is a continuum of eigenstates. This explains uncountably infinite dimension of Hilbert space.

Now, let us think of another 1-D problem, simple harmonic oscillators.

- $V(\hat{x}) = \frac{1}{2} k \hat{x}^2$ [Potential of this form]

- $E_n = \hbar\omega\left(n + \frac{1}{2}\right)$

$$\hat{H} \equiv \begin{bmatrix} E_1 & \cdot & \cdot & \cdot \\ \cdot & E_2 & \cdot & \cdot \\ \cdot & \cdot & \ddots & \cdot \\ \cdot & \cdot & \cdot & E_n \end{bmatrix}$$

$\therefore$ Hilbert's space is countably infinite in this case, as $n$ labels the eigenvalues discretely.

(Still, verify).

- To show: $\mathbb{I}$ is basis-independent

$$\mathbb{I} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} {}_z\langle\uparrow|\hat{\mathbb{I}}|\uparrow\rangle_z & {}_z\langle\uparrow|\mathbb{I}|\downarrow\rangle_z \\ {}_z\langle\downarrow|\mathbb{I}|\uparrow\rangle_z & {}_z\langle\downarrow|\mathbb{I}|\downarrow\rangle_z \end{bmatrix}$$

• Will Identity matrix look the same even in $|\uparrow\rangle_x$ basis?

Ans: Yes. (Verify). $\mathbb{I}$ is basis independent.

Now, $E_n = \hbar\omega\left(n + \frac{1}{2}\right)$

Now, $\sum_n |n\rangle\langle n| = \mathbb{I}_{n \times n}$ [Countably ∞ basis]

$\updownarrow$

$\int dx\, |x\rangle\langle x| = \mathbb{I}$ [Uncountably ∞ basis]

Now, $\delta(x)$ is not square integrable. Hence, we put the particle in a box, wavefunction becomes normalizable and eigenstates discretize into countably infinite eigenstate. We take the box size to infinite and get a continuum, but still we have a count.

# Space of Hamiltonians

Hamiltonian has some parameters, varying which gives you a different Hamiltonian (parameters such as $\hat{S}_z$, $m$ etc).

Let's say,

$$\boxed{\hat{H} = \# \vec{B} \cdot \vec{S}}, \text{ where } \boxed{\vec{S} = \frac{1}{2}\hbar\vec{\sigma}}$$

$[\vec{\sigma} = (\sigma_x, \sigma_y, \sigma_z)]$

So, $H = B_x \sigma_x + B_y \sigma_y + B_z \sigma_z + \alpha \mathbb{I}$

Any $2\times2 \rightarrow$ Hermitian matrix It can be expressed as a linear combination of $\sigma_x, \sigma_y, \sigma_z, \alpha\mathbb{I}$

Now, $\sigma_z \rightarrow \{|\uparrow\rangle_z, |\downarrow\rangle_z\}$

$$\sigma_z = \begin{bmatrix} +1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$\sigma_x = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$\sigma_z = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$$

[Verify]

The space of 2-D Hermitian matrices forms a vector space over the field of real numbers.

* If we put $i$ in front of $\sigma_x, \sigma_y, \sigma_z$
$\Rightarrow$ Becomes generator of SU(2) algebra.

▫ Let us define Trace product:

$$\mathrm{Tr}[\sigma_i \sigma_j] = 2 \quad [\sigma_i^2 = \mathbb{I}, \mathrm{Tr}(\mathbb{I}) = 2]$$

↳ means product of $\sigma_i$, $\sigma_j$ and taking their trace.

Now, $\mathrm{Tr}(\sigma_i) = 0$, $i \in \{x, y, z\}$

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- Perturbation Theory:

if we know how to solve eigenvalues and eigenstates of a Hamiltonian, and if we introduce a potential in it, can we still solve the new "perturbed Hamiltonian".

- How are Matrices big or small?

Ans: How large the largest eigenvalue of one matrix is relative to the largest eigenvalue of the second matrix.

⇒ Notion of "smallness" of perturbation Hamiltonian.

- No sense of "smaller" or "bigger" unless there is a ratio involved between two quantities of same dimension.

- We look for physically meaningful ~~power~~ perturbation for which the power series is asymptotically convergent ⇒ we have to truncate the series at correct place.

SHO: convergent to a limit but not beyond it.

$$\left\{\begin{array}{l} H_0 \\ \mathcal{E}_0^n \\ |n\rangle_0 \end{array}\right\} \begin{array}{l} \rightarrow \text{exactly solvable Hamiltonian} \\ \rightarrow \text{Series of eigenvalues} \\ \rightarrow \text{series of eigenvectors} \end{array}$$

$$\begin{array}{c} + \\ H_1 \end{array} \equiv \left\{\begin{array}{c} H_0 + H_1 \\ \mathcal{E}_0^n \\ |n\rangle_0 \end{array}\right\}$$

$[H_0, H_1] = 0$

Let's say, hamiltonian of a two level system: $H = \vec{B} \cdot \vec{\sigma} = B_z \sigma_z$

where, $\sigma_z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$

Let us add, two perturbations:

$$\boxed{H_1 \begin{array}{l} \nearrow \lambda \sigma_z \\ \searrow \lambda \sigma_x \end{array}}$$

$\boxed{\text{Case I}: H_1 = \lambda \sigma_z}$

$$H_0 + H_1 = \begin{bmatrix} B_z & 0 \\ 0 & -B_z \end{bmatrix} + \begin{bmatrix} \lambda & 0 \\ 0 & -\lambda \end{bmatrix}$$

$$= \begin{bmatrix} B_z + \lambda & 0 \\ 0 & -(B_z - \lambda) \end{bmatrix}$$

$\sigma_z$

$\boxed{\text{It does not change eigenfunctions, but shifts eigenvalues}} \rightarrow \boxed{|\uparrow\rangle_z, |\downarrow\rangle_z}$

○ As the eigenfunction does not change, the physics does not change.

□ ○ Has the dynamics changed?

Ans: $e^{-\frac{i}{\hbar}Ht}|\Psi\rangle$

$$\Rightarrow e^{-\frac{i}{\hbar}[B_z\sigma_z + \lambda\sigma_z]t}|\Psi\rangle$$

$$= \left(e^{-\frac{i}{\hbar}tB_z\sigma_z}\right)\left(e^{-\frac{i}{\hbar}\lambda\sigma_z t}\right)|\Psi\rangle$$

*[Will there be a −ve sign in front of $-\frac{i}{\hbar}$? Sir wrote +ve from 2nd step onwards!]

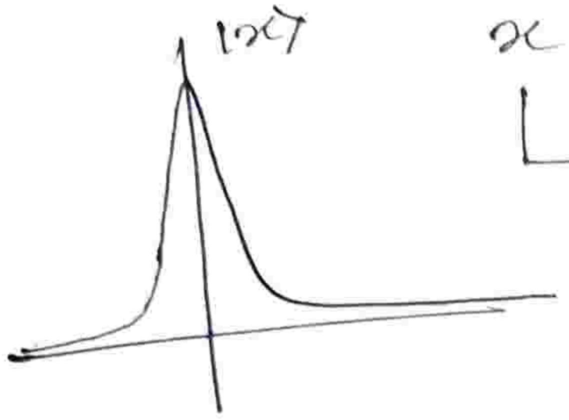
$$e^A \cdot e^B \neq e^{A+B}, \text{ if } [A,B]=0$$

□ If we prepare an eigenstate of $\hat{x}$, what is interesting is not the expectation value of p, rather the standard deviation value of p.

$$H = \frac{\hat{P}_x}{2m} \quad (1-D)$$

$$t=0 \rightarrow |x=0\rangle$$

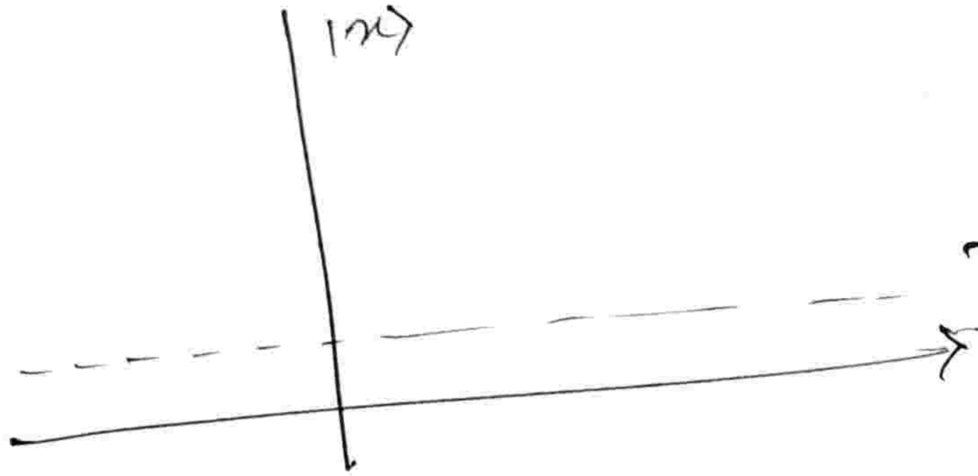
If number of trials in an experiment is low : datapoints peaked near ~~x=0~~.



$x = \cancel{\langle x \rangle} \langle x=0|\hat{x}|x=0\rangle$

↳ particle position eigenfunction peaks here.

- But if no. of trials very high:



Data evenly scattered.

- What is the standard deviation/ fluctuation of the data?

<u>Ans:</u> Massive ☺

The "average" or "expectation value", is useless if we do not know the standard deviation.

Quantum systems have fluctuations even in 0 temperature.

Now: $H_1 = \lambda \sigma_x$ switches on:

$${}_z\langle \uparrow | \sigma_x | \downarrow \rangle_z \neq 0$$

But $H_1 = \lambda \sigma_z$ did not switch on:

$${}_z\langle \uparrow | \sigma_z | \downarrow \rangle_z = 0$$

[Given, system was prepared in the z-basis]

The state is going to gradually "leak out".

$$\mathcal{E}_0^n \rightarrow \mathcal{E}^n$$

$$|n\rangle_0 \rightarrow |n\rangle$$

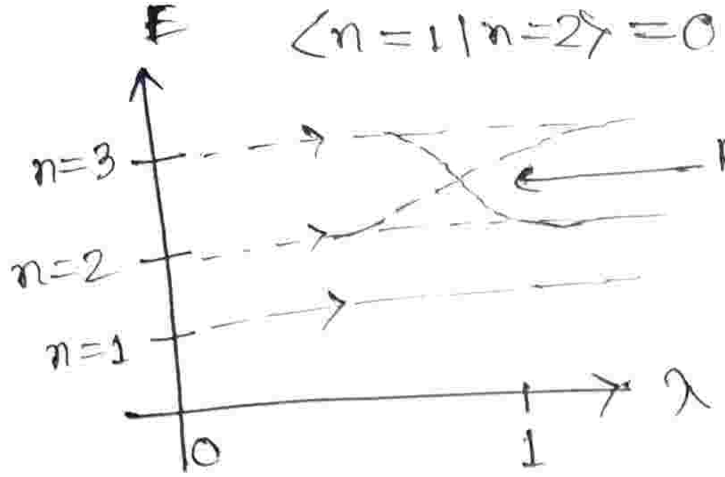
$\longrightarrow$

A sense of one-one, onto mapping

(Adiabatic continuity required → slowly turning on the perturbation).

↳ connected by the label "n".

$E \quad \langle n=1 | n=2 \rangle = 0$



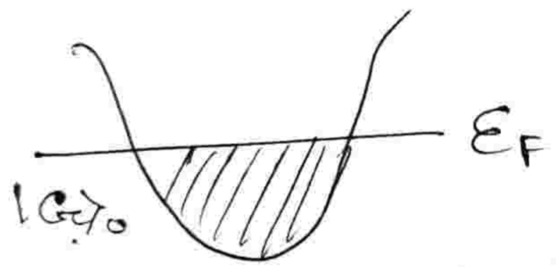

No level crossing in perturbation theory !

$H = H_0 + \lambda H_1$

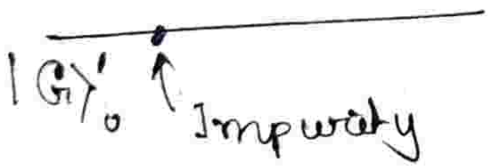
- $\lambda = 0$ goes to $H_0$ which I can exactly solve
- $\lambda = 1$ goes to $H_1$ which I want to solve . 😁

- Anderson orthogonality catastrophe

(Explains meaning of adiabatic continuity)




Let there be a pristine metal block with fully filled Fermi-Level.




$\langle G | G_0 \rangle' = 0$
$_0$

For perturbation, one state "tilts" a bit, but does not become completely perpendicular ! Even with slight perturbation, complete perpendicularity ! $\Rightarrow$ Catastrophic !!

$$(H_0 + H_1)|n\rangle = \mathcal{E}^n |n\rangle$$

- $|n\rangle = |n\rangle_0 + \lambda|n\rangle_1 + \lambda^2|n\rangle_2 + \dots$

[We are going to put $\lambda = 1$ in the end anyways]

- $\mathcal{E}_n = \mathcal{E}^n_0 + \lambda \mathcal{E}^1_n + \lambda^2 \mathcal{E}^2_n + \dots$

$\mathcal{E}^n_0$ → unperturbed

- $\hat{H}_0 |n\rangle_0 = \mathcal{E}^n_0 |n_0\rangle$

$$_0\langle n| \quad \underbrace{H_0|n\rangle_1}_{0} + \underbrace{H_1|n\rangle_0}_{_0\langle n|H_1|n\rangle_0} = \mathcal{E}^n_0 |n\rangle_1 + \mathcal{E}^n_1 |n\rangle_0$$

- For all terms, they are 1st order

$$\mathcal{E}^n_1 = {}_0\langle n|H_1|n\rangle_0$$

Now, $\underset{=5}{|n\rangle_1} = \sum_n c_n |n\rangle_0$ ← Complete set of state

$= c_1 |n=1\rangle_0 + c_2|n=2\rangle_0 + \dots + \dots c_5|n=5\rangle_0$

- Will $|n=5\rangle_1$ and $|n=5\rangle_0$ have overlap? → No. But why?

$$ {}_0\langle m|n\rangle_1 = \frac{{}_0\langle m|H_1|n\rangle_0}{\mathcal{E}_0^n - \mathcal{E}_0^m} \quad \Bigg| \sum_n C_n |n\rangle_0 $$

$|m\rangle_0 \Rightarrow$ eigenstates of unperturbed Hamiltonians.

- Matrix representation :

$$ H_0 + H_1 = \begin{bmatrix} \mathcal{E}_1 & 0 & 0 & 0 \\ 0 & \mathcal{E}_2 & 0 & 0 \\ 0 & 0 & \mathcal{E}_3 & 0 \\ 0 & 0 & 0 & \mathcal{E}_4 \end{bmatrix} $$

$\uparrow$

$\{|n\rangle_0\}$

$$ + \begin{bmatrix} \square & {}_0\langle 1|H_1|2\rangle_0 & & \\ & \square & & \\ & & \square & \\ & & & \square \end{bmatrix} $$

- So, If we divide ${}_0\langle 1|H_1|2\rangle_0$ [The off-diagonal element] is divided by the difference between two diagonal elements of $H_0$, and the ratio comes out to be much lesser than one, the perturbation is small enough.

$$|n\rangle = |n_0\rangle + \sum_{m\neq n} \left\{ \frac{{}_0\langle m|H_1|n\rangle_0}{\mathcal{E}_0^m - \mathcal{E}_0^n} |m_0\rangle \right\}$$

$$\mathcal{E}_2^n = - \sum_{m\neq n} \frac{|\langle m|H_1|n\rangle|^2}{\mathcal{E}_0^m - \mathcal{E}_0^n}$$

$$|\langle m|H_1|n\rangle|^2 = \langle m|H_1|n\rangle\langle n|H_1|m\rangle$$

↳ Virtual transition

→ It goes to a state, gains a perturbation, comes back to its original eigenstate.

- This process of "going" becomes harder and harder as energy difference between levels increase.

Now, $\mathcal{E}_n^0 + \mathcal{E}_n^1 + \dots \rightarrow \mathcal{E}_n$

- Now, does $\mathcal{E}_n^1$ of power series truncated at $\mathcal{E}_n^1$, will these be overestimated or underestimated?

⇒ It always overestimates the actual number [For the ground state only].

$$\mathcal{E}_0^{n=0} + \mathcal{E}_1^{n=0}$$

$$= {}_0\langle n=0 | H_0 | n \rangle_0 + {}_0\langle n=0 | \lambda H_1 | n=0 \rangle_0$$

$$= {}_0\langle n=0 | H_0 + \lambda H_1 | n=0 \rangle_0$$

$$= {}_0\langle n=0 | H | n=0 \rangle_1$$

- **Statement:** From variational theory, when we optimize the parameters to reach the ground state, we always approach from the ~~st~~ top (no states exist below the ground state). Hence, the number is always an overestimate.

$$\mathcal{E}_2^n = {}_0\langle n | H_1 | n_1 \rangle \quad [2^{nd} \text{ order correction}]$$

$$= \underbrace{\sum_{n \neq m} \frac{|{}_0\langle n | H_1 | n \rangle_0|^2}{\mathcal{E}_0^n - \mathcal{E}_0^m}}$$

$-ve \rightarrow$ It pulls down the overestimation.

# Advanced Quantum Mechanics:

08/01/2026

$H^0 + H'$ $\quad |n\rangle = |n\rangle_0 + |n\rangle_1 + |n\rangle_2 + \dots$

$|n\rangle^0, \mathcal{E}_n^0 \quad {}_0\langle m|n\rangle_1$

- What is component of $|n_1\rangle$ onto initial basis of eigenstates of $H^0$?

Ans: It is the leakage of the wavefunction onto other wavefunctions.

$$\therefore \underbrace{{}_0\langle m|n_1\rangle}_{\boxed{\text{Leakage}}} = \frac{{}_0\langle m|H'|n\rangle_0}{\cancel{E^{(n)}}\ E_n^{(0)} - E_m^{(0)}}$$

- So, how close is the new state $|n\rangle_0$ [Due to perturbed Hamiltonian] to the original eigenstate ${}_0|m\rangle_0$ (Due to unperturbed Hamiltonian)?

⇒ Basically, the matrix element ${}_0\langle m|H'|n_0\rangle$ gives the probability of $H'|n_0\rangle$ being in ${}_0\langle m|$. [Is it?]

We know ${}_0\langle m | n \rangle_0 = 0$

what we do? we time evolve $|n\rangle_0$ a bit : $e^{-iH'\delta t} |n\rangle_0$

So, ${}_0\langle m | \left\{ |n\rangle_0 - i\delta t H' |n\rangle_0 \right\}$

$$= {}_0\langle m | n \rangle_0 - i\delta t \, \underbrace{{}_0\langle m | H' | m \rangle_0}_{0}$$

- G.S. energy of $H(\lambda)$:

$$\therefore E_0 = \underbrace{\mathcal{E}_0}_{\text{exact ground state energy of } H_0.} + \langle n=0 | H' | n=0 \rangle + \dots$$

$$\Rightarrow E_0 < \mathcal{E}_0 + \langle n=0 | H' | n=0 \rangle + \mathcal{E}^{(2)}$$

- Depending on symmetries, leading 1st order correction might vanish (Read about it).

- $\mathcal{E}^{(2)}$ for ground state is always negative - definite.

$$\mathcal{E}^{(2)}_{n=0} = + \sum_{n \neq m} \frac{|{}_0\langle n | H' | m \rangle_0|^2}{\underbrace{E_0^{(0)} - E_m^{(0)}}_{-\text{ve}}}$$

$$\mathcal{E}_n^{(2)} = - \sum_{\substack{n \neq m \\ (m > n)}} \frac{|{}_0\langle n|H'|m\rangle_0|^2}{\mathcal{E}_m^{(0)} - \mathcal{E}_n^{(0)}} + \sum_{\substack{n \neq m \\ (n > m)}} \frac{|{}_0\langle n|H'|m\rangle_0|^2}{\mathcal{E}_n^{(0)} - \mathcal{E}_m^{(0)}}$$

1st correction pushes below.

2nd correction pushes above.

$\Rightarrow$ Level repulsion.

Certain perturbations can not give rise to transitions (different symmetry sectors, selection rule).

$\hookrightarrow$ In 1st order especially.

- $H_0 = \frac{\hat{p}^2}{2m} + \frac{1}{2} m \omega^2 \hat{x}^2 \rightarrow$ Symmetric

$H_1 = e E \hat{x}$

$\hookrightarrow$ Is it Time-translationally invariant? $\rightarrow$ Yes! Energy conserved

# Read about Noether's Theorem.

- For space-translation symmetry $\rightarrow$

→ momentum is a conserved quantity.

For this Hamiltonian, space translation NOT invariant ⇒ broken symmetry

⇒ momentum is NOT a good quantum number.

- Discrete Symmetry: Time reversal.

→ Yes! Invariant.

- Give an example where time-reversal symmetry is broken.

Ans: Time reversal symmetry of electron spin system is broken by the introduction of some magnetic field.

- Now, Parity symmetry:

Does perturbation respect Parity Symmetry? No!

→ Consequence → 1st order correction will vanish! $H_1 = -eE\hat{x}$

$\mathcal{E}_n^{(1)} = {}_0\langle n|H_1|n\rangle_0 = -eE\,{}_0\langle n|\hat{x}|n\rangle_0$

$\hat{x} \propto (\hat{a}+\hat{a}^\dagger)\Big| = -eE\{{}_0\langle n|\hat{a}|n\rangle_0 + {}_0\langle n|\hat{a}^\dagger|n\rangle_0\}$

Now we insert resolution of $\mathbb{I}$ operators twice (in the continuum sense).

$$ {}_0\langle n|\hat{x}|n\rangle_0 = {}_0\langle n|\mathbb{I}\hat{x}\mathbb{I}|n\rangle_0 $$

$$ = \iint {}_0\langle n|x\rangle\langle x|\hat{x}|x'\rangle\langle x'|n\rangle_0 \, dx\,dx' $$

$$ = \int \Psi_n^*(x)\left[x\delta(x-x')\right]\Psi_n(x')\,dx\,dx' $$

$$ {}_0\langle n|\hat{x}|n\rangle_0 = \int_{-\infty}^{\infty} |\Psi_n(x)|^2 x\,dx = 0 $$

- With this we conclude TIPT.

# Advanced Quantum : 09/01/2026

## → Time dependent perturbation theory:

$$H(t) = \vec{B}(t) \cdot \vec{S}$$

Now, as $\vec{B}(t)$ is a vector

→ Two types of possible time dependences.

Case 2 ▫ Direction changes in time.. magnitude does not.

Case 1 ▫ Magnitude changes in time, direction does not.

▫ We are not going to turn on both at the same time!

Case (1) $|\vec{B}(t)| = f(t)$ , $\hat{n}(t) = \hat{n}$

Case (2) $|\vec{B}(t)| = c$ , $\hat{n}(t) = (\cos \omega t)\hat{x} + (\sin \omega t)\hat{y}$

If $\hat{n} \to \hat{z}$ , we have, $e^{-i|\vec{B}(t)|\sigma_z (N\delta t)}$

$$= e^{-i|B(t)|\sigma_z \delta t} \cdot e^{-i|B(t)|\sigma_z \delta t}$$

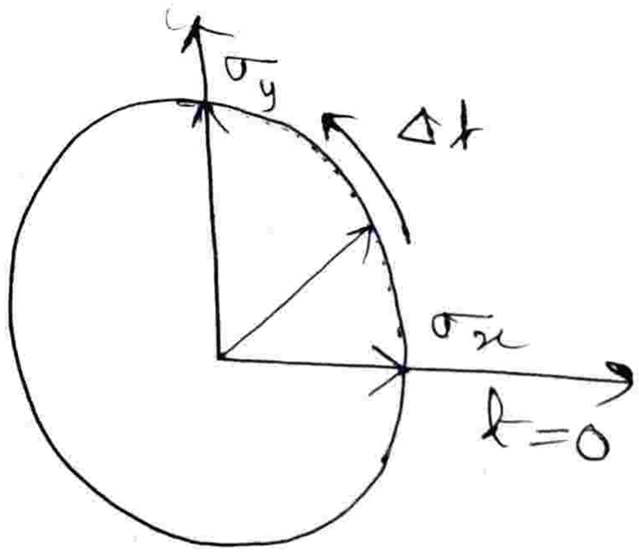
Now suppose, direction changes in

[Can treat them like numbers if they commute]

time. Hence, Hamiltonians at different times do not commute! Hence, this decomposition can not be performed.

$$\left[\text{If } [A,B] = 0 \Rightarrow e^{A+B} \neq e^{A}e^{B} \neq e^{B}e^{A}\right]$$

Now,




Can we decompose it into:

$$\simeq e^{i|\vec{B}|\sigma_x \frac{\Delta t}{2}} \cdot e^{i|\vec{B}|\sigma_y \frac{\Delta t}{2}} \rightarrow \boxed{\text{[Time ordered]}}$$

$\therefore$ For $|\vec{B}(t)| = f(t)$, we can not do the $e^{-i\int H(t)\,dt}$ integration thing. But for case 2, can not use this integration $\leftarrow$ $\boxed{\text{Time ordering comes into play}}$

So, $\hat{T} \cdot e^{-i\int H(t)\,dt}$

We have a time dependent Hamiltonian

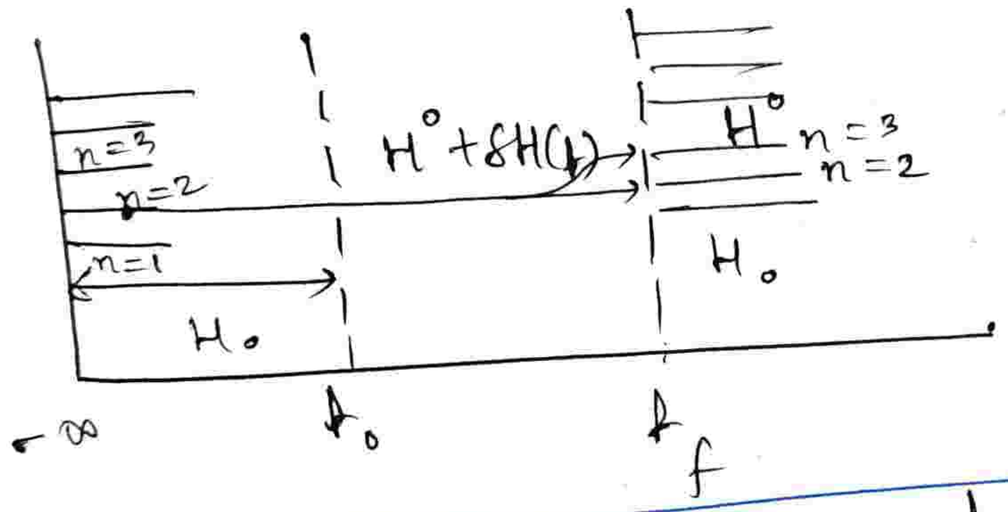
$\hat{H}(t) = H_0 + \delta H(t)$

When we do not have time dependence,

$$|\psi_n(x,t)\rangle = e^{-iE_n t/\hbar}\,|\psi_n(x)\rangle$$

$$|\psi(t)\rangle = \sum C_n e^{-iE_n t/\hbar}\,|\psi_n(x)\rangle$$

$$i\hbar\frac{\partial}{\partial t}|\Psi(t)\rangle = (H^0 + \delta H(t))|\Psi(t)\rangle \quad \text{(i)}$$




We see the probability of, if initially we prepare the system in an initial eigenstate, the state will remain in the same eigenstate at a time $t_f$? And if it is not in same state transition is due to perturbation.

- Why is $|\Psi_n(x,t)\rangle = e^{-iE_nt/\hbar}|\Psi_n(x)\rangle$ not relevant for:

$$i\hbar\frac{\partial}{\partial t}|\Psi(t)\rangle = (\hat{H}^0 + \delta H(t))|\Psi(t)\rangle ?$$

Ans: Read about it.

We see the "interaction picture" now.

$$\langle\Psi(t)|\hat{O}_s|\Psi(t)\rangle$$

$$|\Psi(t)\rangle = \underbrace{\hat{U}(t,t_0)}_{\text{unitary operator.}}|\Psi(t_0)\rangle$$

$$= \langle\Psi(t)|\hat{O}_s|\Psi(t)\rangle$$

$$= \langle\Psi(t_0)|\hat{U}^\dagger(t,t_0)\,\hat{O}_s U(t,t_0)|\Psi(t_0)\rangle$$

- From my state, we are going to take away the forward time evolution → we apply ~~only~~ a backward time evolution, which is only a function of H int, and not of $H_0$. We "undo" effects of $H_0$.

$$|\tilde{\Psi}(t)\rangle = \underbrace{e^{+iH^0 t/\hbar}}_{\text{Backward time evolution.}} |\Psi(t)\rangle \quad \text{--- (2)}$$

$$i\hbar \frac{d}{dt}|\tilde{\Psi}(t)\rangle = (i\hbar)\left(\frac{iH^0}{\hbar}\right) + e^{iH^0 t/\hbar} \frac{d}{dt}|\Psi(t)\rangle$$

$$\Rightarrow i\frac{d}{dt}|\tilde{\Psi}(t)\rangle = -H^0|\tilde{\Psi}(t)\rangle \cdot + e^{iH^0 t}\{(H^0+\delta H)|\Psi(t)\rangle\}$$

$$\Rightarrow i\frac{d}{dt}|\tilde{\Psi}(t)\rangle = \underbrace{e^{iH^0 t}\,\delta H\, e^{-iH^0 t}}_{\delta\tilde{H}}|\tilde{\Psi}(t)\rangle$$

$$\Rightarrow i\frac{d}{dt}|\tilde{\Psi}(t)\rangle = \tilde{\delta} H\,|\tilde{\Psi}(t)\rangle \quad \text{--- (5)}$$

eigenstate of $H^0$.

$$|\tilde{\Psi}(t)\rangle = \sum_n c_n(t)|n\rangle$$

$$\Rightarrow i\frac{d}{dt}\sum_n c_n(t)|n\rangle = \mathbb{1}\,\delta\tilde{H}\sum_j c_j(t)|j\rangle$$

$$\Rightarrow i\sum_n\left[\frac{d}{dt}c_n(t)\right]|n\rangle = \left\{\sum_k \mathbb{1}|k\rangle\langle k|\right\}\delta\tilde{H}\sum_j c_j(t)|j\rangle$$

$$= \sum_{k,j}\langle k|\delta\tilde{H}|j\rangle c_j(t)|k\rangle$$

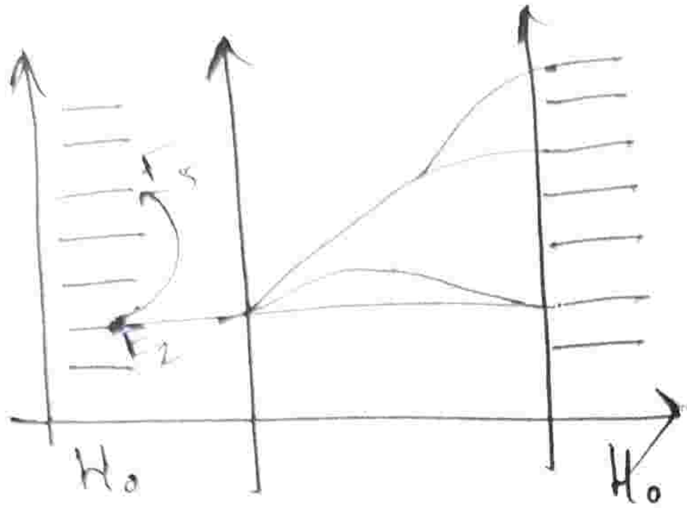
[P.T.O.]

$$\Rightarrow i \sum_n \left[\frac{d}{dt} C_n(t)\right]_{|n\rangle} = \sum_{k,j} \delta\tilde{H}_{k,j}\, C_j(t)\, |k\rangle$$

$$\Rightarrow i \sum_n \left[\frac{d}{dt} C_n(t)\right] \underbrace{\langle l|n\rangle}_{\to\, \delta_{l,n}} = \sum_{k,j} \delta\tilde{H}_{k,j}\, C_j(t) \langle l|n\rangle$$

$$\boxed{i \frac{d}{dt} C_l(t) = \sum_j \delta\tilde{H}_{lj}(t)\, C_j(t)}$$

Quantum Mechanics 14/01/26




- $i\frac{d}{dt}|\tilde{\Psi}(t)\rangle = \delta\tilde{H}|\tilde{\Psi}(t)\rangle$
- $|\tilde{\Psi}(t)\rangle = \sum_n c_n(t)|n\rangle$

- $i\frac{d}{dt}|\tilde{\Psi}(t)\rangle = \delta\tilde{H}\,|\tilde{\Psi}(t)\rangle$
- $|\tilde{\Psi}(t)\rangle = \sum_n c_n(t)|n\rangle$

Now, $\tilde{\Psi}(t) = \exp\left(+\frac{iH^{(0)}t}{\hbar}\right)|\Psi(t)\rangle$

$$\Rightarrow |\Psi(t)\rangle = \exp\left(-\frac{iH^{(0)}t}{\hbar}\right)|\tilde{\Psi}(t)\rangle$$

$$\Rightarrow |\Psi(t)\rangle = \exp\left(-\frac{iH^{(0)}t}{\hbar}\right)\sum_n c_n(t)|n\rangle$$

$$= \sum_n c_n(t)\,e^{-i\hat{H}^{(0)}t}|n\rangle$$

$$= \sum_n c_n(t)\,e^{-iE_n t}|n\rangle$$

Now, $i\frac{d}{dt}c_l(t) = \sum_j \delta H_{ij}(t)\,c_j(t)$

$$\delta\tilde{H}_{i,j} = \langle l|e^{i\hat{H}_0 t}\,\delta\tilde{H}(t)\,e^{-i\hat{H}_0 t}|j\rangle$$

$$\delta\hat{H}_{l,j}(t) = e^{\frac{i(E_l - E_j)t}{\hbar}} \langle l | \delta H(t) | j \rangle$$

$$\boxed{i\frac{dC_l(t)}{dt} = \sum_j e^{i\omega_{lj} t}\, \delta H_{lj}(t)\, C_j(t)}$$

Suppose we want to guide a particle at $E=2$ ~~and~~ to $E=5$. We bring in "light", to set up a harmonic trap. What Frequency should be the light? Yes, "$\omega$"!!! True.

This is resonant

In the RHS, no zeroth order of $|\Psi(t)\rangle$

$$H(t) = H_0 + \lambda\, \delta H(t) \quad \text{—(i)}$$

$$\boxed{i\frac{d}{dt}|\tilde{\Psi}(t)\rangle = \lambda\, \delta\tilde{H}\, |\Psi(t)\rangle}$$

As it is a smooth function of $\lambda$, we can expand it as:

~~$\Psi(t)$~~ ("Taylor series)

$$|\tilde{\Psi}(t)\rangle = |\tilde{\Psi}^{(0)}(t)\rangle + \lambda|\tilde{\Psi}^{(1)}(t)\rangle + \lambda^2|\tilde{\Psi}^{(2)}(t)\rangle + \ldots \quad \text{—(ii)}$$

Because if no continuity (smoothness), even a small perturbation blows up the mixing, i.e. Taylor series invalid.

$$i\hbar \frac{d}{dt} |\tilde{\Psi}(t)\rangle^0 = 0 \rightarrow$$ Is a constant, hence, does not evolve.

$$i\hbar \frac{d}{dt} |\tilde{\Psi}(t)\rangle^1 = \delta\tilde{H} |\tilde{\Psi}(t)\rangle^0$$

$$i\hbar \frac{d}{dt} |\tilde{\Psi}(t)\rangle^2 = \delta\tilde{H} |\tilde{\Psi}(t)\rangle^1$$

$$\vdots$$

$$i\hbar \frac{d}{dt} |\tilde{\Psi}(t)\rangle^{n+1} = \delta\tilde{H} |\tilde{\Psi}(t)\rangle^n$$

[We multiply (i) and (ii), compare and equate the powers of lambda],

$$|\tilde{\Psi}(t)\rangle^1 = \frac{1}{i}\int_0^t \delta\tilde{H}(t') |\Psi(0)\rangle^0 dt' + C$$

$$\boxed{= \frac{1}{i}\int_0^t \delta\tilde{H}(t') |\Psi(0)\rangle dt' + C}$$

$\rightarrow = |\Psi(0)\rangle$ [Boundary condition for this ODE, as $|\Psi(t)\rangle^0$ does not evolve in time]

- Use of boundary conditions:

$$|\tilde{\Psi}(0)\rangle^1 = 0$$

$$|\tilde{\Psi}(t)\rangle^2 = \frac{1}{i}\int_0^t \delta\tilde{H}(t')\,|\tilde{\Psi}(t')\rangle^1 dt' + c$$

$$= \frac{1}{i}\int_0^t \delta\tilde{H}(t')\left[\frac{1}{i}\int_0^{t'} \delta\tilde{H}(t'')\,|\Psi(0)\rangle\, dt''\right] + c(=0)$$

$t'$ and $t''$ are "different" time variables, and $\delta\tilde{H}(t')$ and $\delta H(t'')$ do not commute! $\left([\delta\tilde{H}(t'), \delta\tilde{H}(t'')] \neq 0\right)$

∴ Sense of time ordering is included in the 2nd order correction

$$H(t) = H^0 + \lambda\,\delta H(t)$$

$$i\frac{d}{dt}|\tilde{\Psi}(t)\rangle = \lambda\,\delta\tilde{H}\,|\tilde{\Psi}(t)\rangle$$

$$|\tilde{\Psi}(t)\rangle = |\tilde{\Psi}(t)\rangle^0 + \lambda|\tilde{\Psi}(t)\rangle^1 + \lambda^2|\tilde{\Psi}(t)\rangle^2$$

$$P_{m\leftarrow m} = |\langle m|\tilde{\Psi}(t)\rangle^1_n|^2$$ → If initially ~~proportional~~ prepared in $|n\rangle$, where in $t$?

$$|\Psi(t)\rangle = |m\rangle + |\tilde{\Psi}\rangle^1 + |\tilde{\Psi}\rangle^2 + \dots$$

$$\therefore P_{m\leftarrow m} = |\langle m|\tilde{\Psi}(t)\rangle^1_n|^2$$

$$= |\langle m|e^{-iH^{(0)}t}\{|n\rangle + |\tilde{\Psi}(t)\rangle^1 + \}|^2$$

$$\boxed{P_{n\leftarrow m} = \left|\frac{1}{i\hbar}\int_0^t \langle m|\delta\tilde{H}(t')|n\rangle\, dt'\right|^2}$$

Now, we undo the interaction picture.

Now we have to develop a perturbation theory for $C_n$'s in

$$|\tilde{\Psi}(t)\rangle = \sum_n C_n(t)|n\rangle$$

$$|\tilde{\Psi}(t)\rangle = |\tilde{\Psi}(t)\rangle^0 + \lambda|\tilde{\Psi}(t)\rangle^1 + \lambda^2|\tilde{\Psi}(t)\rangle^2$$

$$\Rightarrow |\tilde{\Psi}(t)\rangle = \sum_n [C_n^{(0)}(t) + \lambda C_n^1(t) + \lambda^2 C_n^2(t) + \dots]|n\rangle \quad \text{——(iii)}$$

In the Formulation:

$$P_{m\leftarrow n} = \frac{1}{i\hbar}\left|\int_0^t \langle m|\delta\tilde{H}(t')|n\rangle dt'\right|^2$$

asks, if system is prepared in eigenstate $|m\rangle$ at $t < t_0$, what is the probability the state will remain in $|m\rangle$ eigenstate in $t > t_f$ (survival probability) and what is the probability it will transit to some other state? (Transition probability)

For the $C_n$ formulation, we do not look at eigenstates, rather at arbitrary states (Read):

Substituting (iii) in Schrodinger eq$^n$: and compare order by order:

$$|\tilde{\Psi}(t)\rangle^0 = \sum_n c_n^0(t)|n\rangle$$

$$|\tilde{\Psi}(t)\rangle^1 = \sum_n c_n^1(t)|n\rangle$$

$$\vdots$$

$$|\tilde{\Psi}(t)\rangle^0 = |\tilde{\Psi}(0)\rangle^0 = |\Psi(0)\rangle$$

$$\Rightarrow c_n^0(t) = c_n^0(0) = c_n(0)$$

and $c_n^i(0) = 0$ when $i \neq 0$,

If we just focus on 1st order:

$$|\tilde{\Psi}(t)\rangle^1 = \sum_n c_n^1(t)|n\rangle$$

$$= \frac{1}{i}\int_0^t \delta\tilde{H}(t')\,|\Psi(0)\rangle dt'$$

$$\Rightarrow \boxed{|\tilde{\Psi}(t)\rangle^1 = \frac{1}{i}\int_0^t dt'\,\tilde{H}(t')\left[\sum_n c_n(0)|n\rangle\right]}$$

$$P'_{m\leftarrow n}(t) = |\langle m | \Psi(t)\rangle|^2$$

$$= |\langle m | \left( |\Psi(t)\rangle^0 + |\Psi(t)\rangle^1 \right)|^2$$

$$= |{}_m\langle m | \tilde{\Psi}(t)\rangle^0 + \underbrace{\frac{1}{i}\sum_n c_n(0) \int_0^t dt' \langle m|\delta\tilde{H}(t')|n\rangle}_{\text{oscillatory leakage}}|^2$$

overlap of ~~mth~~ $m^{th}$ eigenstate with state we prepared it in

- For time dependent perturbation, the leakage is oscillatory due to the frequency, $\omega_{mn} = \frac{(E_m - E_n)}{\hbar}$.

So, how does one reach steady state? As in, how to make sure a particle that goes from $n^{th}$ state $m^{th}$ state, and "stays" there (and not oscillates!)?

Answer is, we need a thermal bath, which acts as sink for dissipated energy

Can we have steady state without a bath?

Ans: Yes, global wavefunction will be oscillatory, but a tiny subsystem will look thermal, and will attain steady state in a finite time. Local observables, in long time limit, shows thermalization.

- Thermalization Theory
- Floquet Theory

Possible term paper topics.

- What is intrinsic time scale of a quantum system (i.e. of transition from one state to another?)

Ans: $\omega^{-1} \simeq \frac{(\Delta E)_{min}}{\hbar} \equiv T$

[Minimum time in which a quantum system reacts to a perturbation]

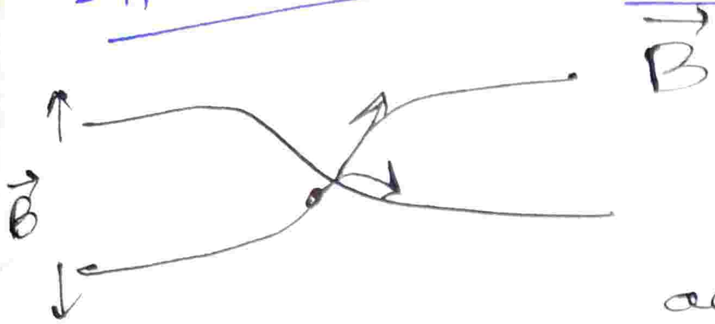
If $\underbrace{T''}_{\frac{(\Delta E)_{min}}{\hbar}} \gg \tau$, $|\Psi(t=0)\rangle \neq \{|m\rangle\}$
$= \sum_n c_n |n\rangle$

If perturbation is turned on faster (much shorter timescale) than the characteristic time scale of the problem, we get the exactly same eigenstate of the system after $t > t_f$ as we initially prepared it in $(t < t_0)$, no longer how strong the perturbation is.

Now, in $\Delta \hat{E} \Delta t$ and $\Delta \hat{x} \Delta \hat{p}$:

- $\Delta t$ is not an operator, it is a parameter.
- The $\Delta t$ gives us the time range, which gives us the limits for which no mixing happens at all, and for which the mixing makes the states completely perpendicular (horrible mixing).

## ▫ Adiabaticity: Adiabatic perturbation Theory




We must avoid level crossing in adiabatic perturbation theory.

- Here, no transition occurs between states A and B. But then why calculate it at all?

Ans: Because of non-trivial phase contribution. Concept of Berry phase comes in.

# Advanced Quantum Mechanism

15/01/2026 :

$$H = \begin{cases} H_0 & \text{for } t \le 0 \\ H_0 + V & \text{for } t \ge \tau \end{cases}$$

↳ $\tau$

$$\tau << \frac{\hbar}{(\Delta E)_{min}}$$

[Characteristic Time length
↳ perturbation is linearly turned on "during" this timegap]

Now, for $t \le 0$,

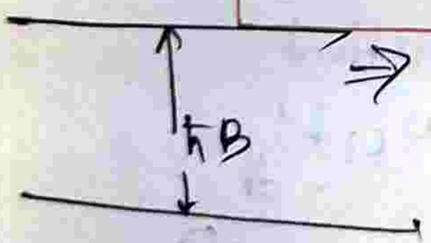
$$H_0 |n\rangle_0 = \varepsilon_0 |n\rangle_0$$

After time $\tau$, $e^{-i(H_0 + V)t/\hbar} |n_0\rangle$

spin $= \frac{1}{2}$, $\uparrow B_x$, $H = \# \vec{B} \cdot \vec{S}$

$$= \# \hbar \vec{B}_x \frac{\sigma_x}{2}$$

$$\boxed{e^{i \# B_z \sigma_z / \hbar} |+\rangle_z}$$




$\Rightarrow$ The energy splitting will be, $\Delta E = \frac{\hbar B}{2} - \left(-\frac{\hbar B}{2}\right)$

$$= \hbar B$$

[$\sigma_x$ has eigenvalues $\pm 1$]

If the time length is much smaller than $\tau$, even if perturbation of magnetic field is turned on, whatever state we prepared does not change. [Energy splitting does not occur].

$$H = \begin{cases} H_0 , & t \leq 0 \\ H_0 + V, & t > 0 \end{cases}$$

[Perturbation switched on almost instantaneously].

$H = H_0 + V\Theta(t)$ [$\Theta(t)$ is step function]

[The worst possible scenario, fastest possible way]

$$c_m'(t_0) = \frac{1}{i\hbar}\int_0^{t_0} \underbrace{\langle f|\hat{V}|i\rangle}_{\text{Constant}} e^{i[\omega_{fi}/t]t'} dt'$$

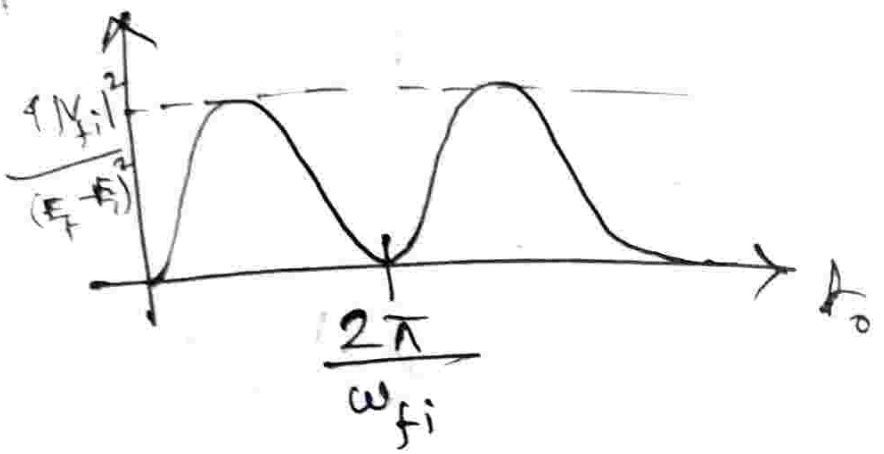
$$= \frac{1}{i\hbar} V_{fi} \int_0^{t_0} e^{i\omega_{fi}t'} dt'$$

$$= \frac{1}{i} \frac{V_{fi}}{\hbar\omega_{fi}} \left(e^{i\omega_{fi}t_0} - 1\right)$$

$\Theta$

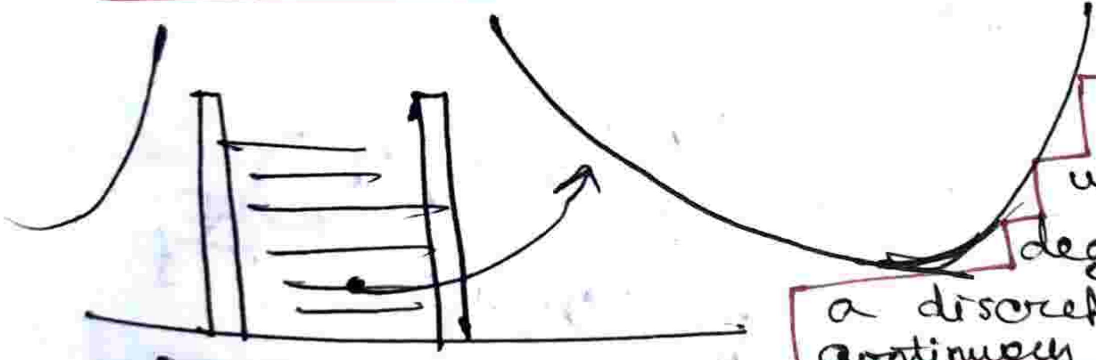
$$= (e^{i\theta} - 1) \equiv \left(e^{(i\theta/2)} - e^{(-i\theta/2)}\right)$$

$$P_{f\leftarrow i}(t_0) = \frac{4|V_{fi}|^2}{(E_f - E_i)^2} \sin^2\left[\left(\frac{E_f - E_i}{2\hbar}\right) t_0\right]$$




So, the leading correction is not $t$ but $t^2$, the transition is suppressed for very fast perturbations.

- Fermi Golden Rule: $(\sin x)^2 = x^2 + \ldots$



Suppose, we have degeneracy between a discrete system and continuous spectrum.

- [Probability of this state leaking out into continuum if we perturb the height of the walls suddenly?]

- How does my transition probability look like?

$$P = \sum_f P_{f\leftarrow i}(t_0)$$

$$= \int P_{f\leftarrow i}(t_0)\,\rho(E_f)\,dE_f$$

$$P = 4\int_{-\infty}^{\infty} \frac{|V_{fi}|^2 \sin^2\left(\frac{\omega_{fi}}{2} t_0\right)}{(E_f - E_i)^2}\,\rho(E_f)\,dE_f$$

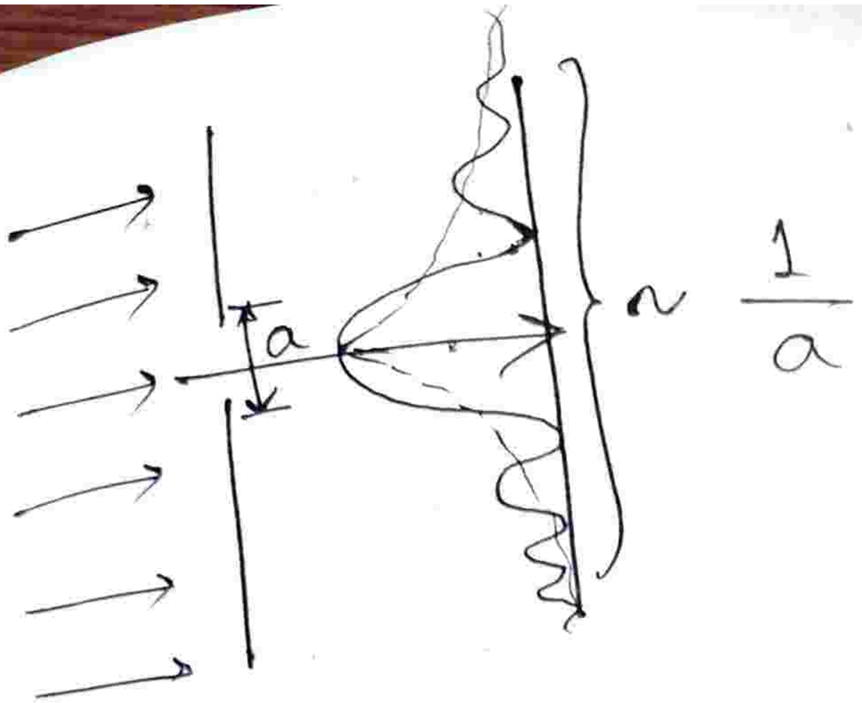
Where $\rho(E_f) \rightarrow$ Density of states

$\downarrow$

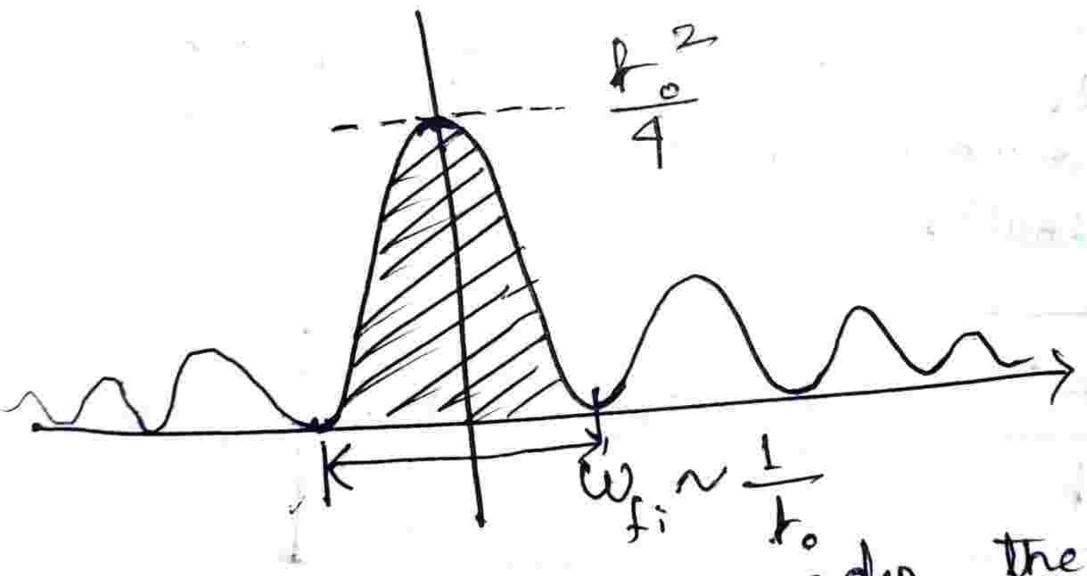
Definition depends on the particular problem, whether 1D, 2D or 3D.

$$P = \int_{-\infty}^{\infty} |V_{fi}|^2 \frac{\sin^2\left(\frac{\omega_{fi}}{2} t_0\right)\rho(E_f)}{\hbar^2\left(\frac{\omega_{fi}}{2}\right)^2}\,dE_f$$

$\hookrightarrow$ Similar to intensity of profile of single slit experiment!




The screen is the exact fourier transform of the 1-D opening!
Smaller the opening, wider the spread of the pattern.




- Dominant area under the curve will go as $O\left(\frac{t_0^2}{4}\right) O\left(\frac{1}{t_0}\right) \simeq t_0$

$\therefore$ Rate of leaking out is constant.
[In this problem, perturbation is a constant potential]

$$\int_{-\infty}^{\infty} \frac{\sin^2\left(\frac{\omega_{fi}}{2} t_0\right)}{\omega_{fi}^2} dE_f \quad \Bigg| \quad d\omega_{fi} = \frac{dE_f}{\hbar}$$

$$x = \frac{\omega_{fi}}{2} t_0 \Rightarrow d\omega_{fi} = \frac{2dx}{t_0}$$

$$= -\hbar t_0 \int_{-\infty}^{\infty} \frac{\sin^2 x}{x^2} dx = \frac{\pi}{2} \hbar t_0$$

Here, we took, density of states is thought to be constant and took it out of the integral (Long time range, narrow energy window, density of states is almost constant within this window).

$$\boxed{\frac{dP_{f \leftarrow i}}{dt} = \frac{2\pi}{\hbar} |V_{fi}|^2 \rho(E_f)}$$

↳ It is independent of time.

There should be a finite density of states, and a matrix element (perturbation) that takes you there.

# Advanced Quantum Mechanics

16/01/2026

$H(t) = H^{(0)} + \delta H(t)$,

- Harmonic drive / perturbation

$$\delta H(t) = \begin{cases} 0 , & \text{for } t \leq 0 \\ 2H' \cos(\omega t) & \text{for } t > 0 \end{cases}$$

$$c'_{f \leftarrow i} = \frac{1}{i\hbar} \int_0^{t_0} dt' \, e^{i\omega_{fi} t'} \delta H_{fi}(t')$$

Here,
$\langle f | \delta H | i \rangle$
$= (2\cos \omega t')$
$\underbrace{2 \langle f | H | i \rangle}_{H_{fi}}$

$$= \frac{H_{fi}}{i\hbar} \int_0^t e^{i\omega_{fi} t'} \left( e^{i\omega t'} + e^{-i\omega t'} \right) dt'$$

$$= \frac{H_{fi}}{i\hbar} \int_0^t \left[ e^{i(\omega + \omega_{fi}) t'} + e^{i(\omega_{fi} - \omega) t'} \right] dt'$$

$$\boxed{c'_{f \leftarrow i} = \frac{H_{fi}}{i\hbar} \left[ \frac{e^{i(\omega_{fi} + \omega)t}}{\underbrace{(i)(\omega_{fi} + \omega)}_{\simeq 0}} + \frac{e^{i(\omega_{fi} - \omega)t}}{\underbrace{(i)(\omega_{fi} - \omega)}_{\simeq 0}} \right]}$$

- Stimulated emission:

$$\omega_{fi} + \omega = 0 \Rightarrow E_f = E_i - \hbar\omega$$

- Absorption:

$$E_f = +\hbar\omega + E_i$$

$$c^{1}_{f\leftarrow i}(t_0) = -\frac{H_{fi}}{\hbar}\, \frac{e^{i(\omega_{fi}-\omega)t_0}-1}{\underbrace{\omega_{fi}-\omega}_{\simeq 0}}$$

$$= \frac{-H_{fi}}{\hbar}\, e^{i\theta/2}\, \frac{\left(e^{i\theta/2}-e^{-i\theta/2}\right)}{\theta}$$

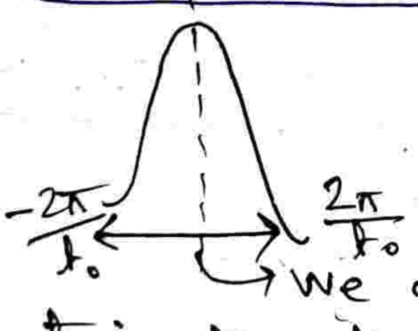
$$= \frac{-H_{fi}}{\hbar}\, e^{i\theta/2}\, \frac{2i\sin\left(\frac{\theta}{2}\right)}{\theta}$$

$$P_{f\leftarrow i}(t_0) = |c^{1}_{f\leftarrow i}(t_0)|^2$$

$$= \frac{H^2_{fi}}{\hbar^2}\, \frac{\sin^2\left(\frac{\theta}{2}\right)}{\left(\frac{\theta}{2}\right)^2}$$

$$\boxed{-\frac{2\pi}{t_0} < \omega_{fi}-\omega < \frac{2\pi}{t_0}} \rightarrow \text{estimate}$$

$$-\frac{2\pi\hbar}{t_0} + (E_i+\hbar\omega) < E_f < \left(\frac{2\pi\hbar}{t_0}\right) + (E_i+\hbar\omega)$$

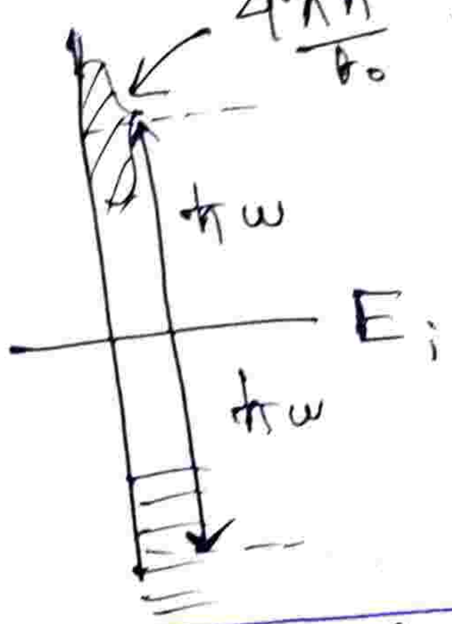



We are trying to estimate this width

Energy uncertainty gives us a window of energies within which transition takes place.

Hence, time duration of irradiation (duration of perturbation), also matters.

Infinite time limit, transition does not happen (conservation of energy). But in intermediate time durations transitions might happen.




$$P = \int P_{f \leftarrow i} \, \rho(E_f) \, dE_f$$

$$= \frac{4}{\hbar} |H_{fi}|^2 \rho(E_f) \int \frac{\sin^2(\omega_{fi} - \omega) t_0}{(\omega_{fi} - \omega)^2} \, d\omega_{fi}$$

$$\therefore \boxed{P_{\pm} = \left( \frac{2H}{\hbar} |H_{fi}|^2 \rho(E_i \pm \hbar\omega) \right) t_0}$$

This is the Fermi-Golden Rule.

- <u>Adiabatic Perturbation theory.</u>

$\tau \gg \frac{\hbar}{(\Delta E)_{min}}$ for adiabatic perturbed systems.

Let, Time varying potential, but verty very slowly. And we want to exactly solve the Hamiltonian. If we want to solve the Hamiltonian ~~at~~ instantaneously at $t = t_0$, we find the system to be in the same eigenstate as it was prepared in. But, the instantaneous eigenstate is intact "upto a phase".

Kind of like snapshots stacked together.

Let us consider the Z-spin basis:

$|\uparrow\rangle_z \ , \ |\downarrow\rangle_z$

Let, $|\psi\rangle = \alpha|\uparrow\rangle_z + \beta|\downarrow\rangle_z$

Normalization condition:

$\alpha^2 + \beta^2 = 1 \quad \left[\begin{array}{l}\alpha = x_1 + iy_1 \\ \beta = x_2 + iy_2\end{array}\right]$

$\Rightarrow x_1^2 + y_1^2 + x_2^2 + y_2^2 = 1$

$\therefore$ every state of a 2-level system (2-D of Hilbert space) lies on a 4-D sphere.

$\Rightarrow$ Geometry of Hilbert space.

Concept of Berry phase is introduced now.

21/01/2026 # Advanced Quantum Mechanics:

□ Outer product, density matrix:

$\hat{\rho} = |\Psi\rangle\langle\Psi|$

∘ $|\Psi\rangle \rightarrow e^{i\theta}|\Psi\rangle$ → There is U(1) freedom for $|\Psi\rangle$

∘ $(|\Psi\rangle)^* \rightarrow \langle\Psi| e^{-i\theta}$.

We can evolve this guy using Heisenberg equation

We can evolve this guy using Schrodinger equation

$\vec{V} \longrightarrow$ in 3.D

$\vec{V}.\vec{V} \rightarrow$ real number (+ve)

□ Inner product:

$$[a^* \; b^*]\begin{bmatrix} a \\ b \end{bmatrix} = a^* a + b^* b$$

□ Outer product [gives density matrix]

$$\begin{bmatrix} a \\ b \end{bmatrix}[a^* \; b^*] = \begin{bmatrix} |a|^2 & a b^* \\ b a^* & |b|^2 \end{bmatrix}$$

□ Diagonal elements → probability distribution
→ classical information.

□ Off diagonal elements → phase information
→ quantum information.

Because,

$$a^*b = |a||b| e^{i(\beta-\alpha)}$$ [has phase information]

where, $\begin{bmatrix} a \\ b \end{bmatrix} \equiv |\Psi\rangle = a|\uparrow\rangle + b|\downarrow\rangle$

where, $a = |a| e^{i\alpha}$, $b = |b| e^{i\beta}$

$\Rightarrow a^* = |a| e^{-i\alpha}$

$|\Psi\rangle$ is an element of vector space, but $\hat{\rho}$ is not.

$\alpha \rho_1 + \beta \rho_2 \notin$ space of density operators

even if $\alpha \Psi_1 + \beta \Psi_2 \in \mathbb{H}$ [where $\Psi_1 + \Psi_2 = \Psi \in \mathbb{H}$]

Because, pure states:

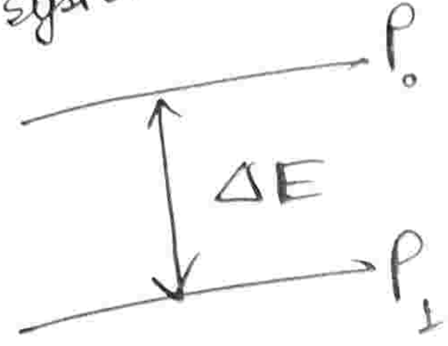
$$(|\Psi\rangle\langle\Psi|)^2 = |\Psi\rangle\langle\Psi|\Psi\rangle\langle\Psi|$$
$$= |\Psi\rangle\langle\Psi| \quad [\hat{\rho}^2 = \hat{\rho}]$$

$\alpha \rho_1 + \beta \rho_2$ gives a "mixed state", not in the same space as the pure states.

$\boxed{Z = e^{-\beta \hat{H}} = \hat{\rho}_T}$ $\boxed{\text{Partition function}}$, $\boxed{\text{where, } \beta = \frac{1}{k_B T}}$

If you give me an energy, I will

tell you probability of occupying a state with a finite temperature.
Let us say there is a two state system, with an energy gap.




at $T=0$, $P_1 = 1, P_0 = 0$
[Pure state]

at $T \to \infty$, $P_1 = \frac{1}{2}, P_0 = \frac{1}{2}$
[Counterintuitive]

at $T \to \infty$, $\dfrac{\Delta E}{k_B T} \to 0$

For infinite temperature, the energy level spacing is very small, and hence the energy levels are degenerate.
At $T \to \infty$ → very high entropy, and the 50 - 50 looks it.
at $T \to \infty$ → phase information is zero (50 - 50 means fully scrambled).

At finite temperatures → no wavef^ns, only occupancies.

Once we leave zero temperature physics, everything is in terms of density matrices.

$$|\psi\rangle = \alpha|\uparrow\rangle_z + \beta|\downarrow\rangle_z$$

$$|\alpha|^2 + |\beta|^2 = 1 \quad \Bigg| \quad \begin{aligned} \alpha &= x_1 + iy_1 \\ \beta &= x_2 + iy_2 \end{aligned}$$

$\alpha$ in 3D:

$$x_1^2 + y_1^2 + x_2^2 + y_2^2 = 1 \leftarrow$$ equation of a sphere in 4-D.

<u>Calculate and Show:</u>

general state described by:

$$|\psi\rangle = \begin{pmatrix} \cos\left(\frac{\theta}{2}\right) \\ \sin\left(\frac{\theta}{2}\right) e^{i\phi} \end{pmatrix} e^{i\alpha}$$

[Double cover, due to $\left(\frac{\theta}{2}\right)$] $\rightarrow SU(2)$

$$\theta \rightarrow [0, \pi], \quad \phi \rightarrow [0, 2\pi]$$

$$\hat{\rho}_\psi = |\psi\rangle\langle\psi| = f(\theta, \phi)$$

If we look at the parametrisation of $\theta$, $\phi$, then reside on the surface of a ~~4~~ 3-dimensional sphere. [Space of $\hat{\rho}$]

Now, equation of 3-dimensional sphere:

$x^2+y^2+z^2=1 \rightarrow (3-1)$ ~~coordinates~~, 1 dimension

[3 coordinates, 1 constraint]

↳ Now we want to write this equation without a constraint, and with cyclic coordinates, with just 2 variables.

□ $U(1) \rightarrow$ space is a circle.

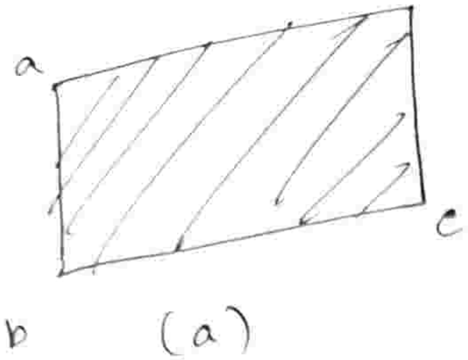
↳ collection of $\{e^{i\theta}\} \rightarrow$ obviously a circle.

$$\boxed{U(1)\times S^2 \simeq S^3}$$

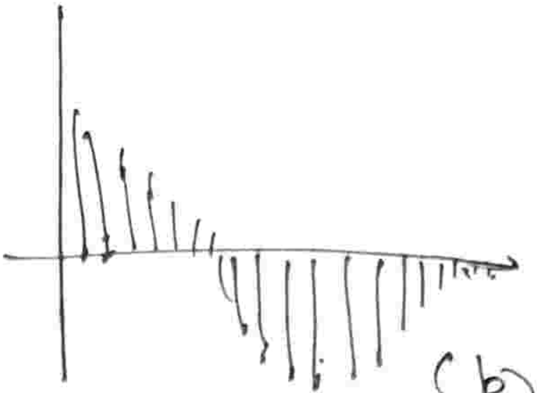
↳ space of states [sphere of 4-dimensions]

space of phases [Circle]

space of density operators: [sphere of 3-D]




(a)



(b)

Under P.B.C.'s, (a) becomes a cylinder and (b) becomes a mobius strip.

For (a), original space is same. [simple product]


(b) Gives twisted product. New space has periodicity $4\pi$.

o Photon → spin 1 particle

→ 3 projections

→ but 0-projection can not be measured as photon does not have a rest frame

→ effectively 2-level system.

## Notion of Slowness:

$H(t) \rightarrow$ is a time dependent hamiltonian.

$e^{-iHt} |n(t=t_1)\rangle \rightarrow n^{th}$ eigenstate of the instantaneous hamiltonian at $t=t_1$

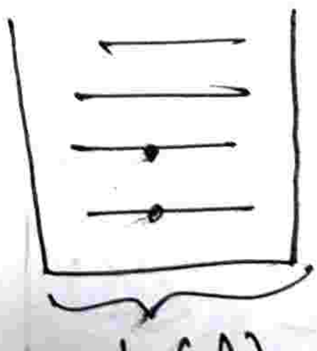
$\downarrow$

$t_1 \rightarrow t_2$

$\rightarrow |\Psi(t)\rangle \Big|_{t=t_2}$

$\rightarrow$ when we look at the state, we see that the ~~eigenstate~~ state looks like the eigenstate $|n(t=t_2)\rangle$ multiplied by a factor $e^{i\theta}$. This is under adiabatic perturbation (This is the definition).

Let us take a particle in 1-D box. But box size can increase or decrease adiabatically.



$L(t)$

$\rightarrow$ Semiclassical approach.

## Somerfield quantization condition:

Related to how only certain orbits are allowed.

$\rightarrow$ only quantized orbits allowed.

• Semiclassical approach:

- $p \sim \frac{\hbar}{L}$ ≡ momentum
- $T = \frac{L}{v} = \frac{mL}{p} \sim \frac{mL^2}{\hbar}$

≡ Time taken for a loop journey for the particle around the box.

Approximation: Over many many such loop journeys, L has not changed.

$$\frac{|\Delta L|_{\text{per cycle}}}{L} \sim \frac{\left(\frac{dL}{dt}\right) T}{L}$$

$$\sim \frac{\left(\frac{dL}{dt}\right)\frac{mL^2}{\hbar^2}}{L}$$

$$\frac{|\Delta L|_{\text{per cycle}}}{L} = \frac{\left(dL/dt\right)}{p/m}$$

$$\boxed{\frac{|\Delta L|_{\text{per cycle}}}{L} = \frac{V_{\text{wall}}}{V_{\text{particle}}} \ll 1}$$

$$T \sim \frac{\hbar}{(\Delta E)_{min}}$$

$$E_n = \frac{n^2 \hbar^2 \pi^2}{2mL^2}$$

$$(\Delta E)_m \sim \frac{\hbar^2}{mL^2}$$

$$\boxed{T \sim \frac{1}{\left(\hbar^2/mL^2\right)/\hbar} = \frac{mL^2}{\hbar}}$$

→ no $n$ dependence though.

One of the examples where quantum mechanics and semi-classical approach are consistent → adiabaticity.

$$i\hbar \frac{\partial |\Psi(t)\rangle}{\partial t} = H(\lambda(t)) |\Psi(0)\rangle$$

$H(\lambda(t))$ → Hamiltonian function of a parameter which is scalar, vector or whatever.

So, we are looking for a solution that looks like:

$$|\Psi(t)\rangle = U(t)\, |n(\lambda(t))\rangle \quad \Big| \; e^{-\frac{i}{\hbar}\int dt\, E_n(t)}$$

$$\langle \Psi | \dot{\Psi} \rangle = \dot{U} U^* + \langle n | \dot{n} \rangle = 0$$

$H|n\rangle = E_n|n\rangle$

$$e^{-\frac{i\hat{H}t}{\hbar}}|n\rangle = e^{-\frac{iE_0 t}{\hbar}}|n\rangle$$

Our eigenvalue depends on time. Hence, purely due to the dynamic, our phase is: $e^{-\frac{i}{\hbar}\int dt\, E_n(t)}$

But, we need a phase that is more than just the dynamical phase. We need to take it into account quantum geometry.

$$U^* \dot{U} = -\langle n|\dot{n}\rangle$$

$$U^* \dot{U} = -\langle n|\frac{\partial}{\partial \lambda_i}|n\rangle \dot{\lambda}^i$$

$$\boxed{\dot{U} = \left[-i\left\{-i\langle n|\frac{\partial}{\partial \lambda_i}|n\rangle\right\}\dot{\lambda}^i\right]U}$$

$$\left[\frac{d}{dt}|n(\lambda(t))\rangle = \frac{\partial n}{\partial \lambda_i}\frac{\partial \lambda_i}{\partial t}\right]$$

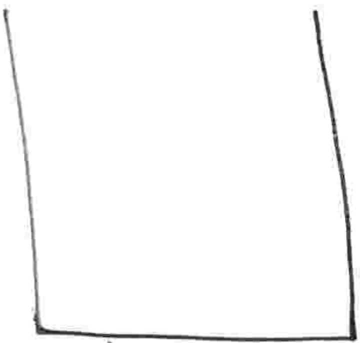
Advanced Quantum = 28/01/2026

TIPT Q.1> Solution:

(1) $H' = q\mathcal{E}L$

$H^{\circ} = -\frac{\hbar^2}{2m}\frac{d^2}{dx^2}$ | $E^{(n)} = \frac{n^2\pi^2\hbar^2}{2mL^2}$

$$(H_0' - E_n^0)\,|\psi\rangle = -(H' - E_n^{(0)})\,|n^{(0)}\rangle$$



particle in a box and weak electric field

$$\left(-\frac{\hbar^2}{2m}\frac{d^2}{dx^2} - \frac{\pi^2\hbar^2}{2mL^2}\right)\phi_1(x)$$

$$= -\left(q\mathcal{E}x - E_n^{(1)}\right)\psi_1^{(0)}(x)$$

(a) $E_1^{(1)} = \langle n_L^{(0)} | H' | n^{(0)} \rangle$

$$= \frac{2}{L}\int q\mathcal{E}x\sin\left(\frac{\pi x}{L}\right)dx$$

$$= \frac{2q\mathcal{E}^{0}}{L}\int\frac{x}{2}\left(1-\cos\left(\frac{2\pi x}{L}\right)\right)dx$$

$$\psi_1^{(1)} = \sqrt{\frac{2}{L}} \sin\left(\frac{\pi x}{L}\right)$$

(b) $$\frac{d^2\phi_1(x)}{dx^2} - \left(\frac{\pi}{L}\right)^2 \phi_1(x) = \sqrt{\frac{2}{L}} \frac{2mq\varepsilon}{\hbar^2}\left(x - \frac{L}{2}\right) \sin\left(\frac{\pi x}{L}\right)$$

$$y = y_c + y_p$$

$$\phi_1^c \rightarrow \frac{d^2\phi_1^c}{dx^2} + k^2 \phi_1^c(x) = 0$$

$$\circ\ \phi_1^c = A\cos(kx) + B\sin(kx)$$

$$g(x) = x \sin(\ \ )$$

$$\circ\ \phi_1^p(x) = (ax^2 + bx)\cos(kx)$$

$$+ (cx^2 + dx)\sin(kx)$$

$$d = 0, \quad a = \frac{mq\varepsilon L}{2\pi^2\hbar^2}$$

$$b = -\frac{mq\varepsilon L}{2\pi^2\hbar^2}\sqrt{\frac{2}{L}}, \quad c = \frac{mq\varepsilon L}{4\pi\hbar^2}\sqrt{\frac{2}{L}}$$

$$\Psi(0) = \Psi(L) = 0$$

$$B\sin(\pi) + \phi_p(L) = 0$$

$$B = -\frac{\phi_p(L)}{\sin(\pi)} = \cancel{mq\varepsilon}$$

$$= \frac{mq\varepsilon L^3}{2\pi^2\hbar^2}\sqrt{\frac{2}{L}}\left(\frac{1}{\pi^2} - \frac{1}{6}\right)$$

$$\langle \psi_1^{(0)} | \phi_1 \rangle = 0 \; ; \; E^{(2)} = \langle \psi_n^{(0)} | H' | \phi \rangle$$

$$E_1^{(2)} = 2\varepsilon \int_0^L \psi_1^{\circ}(x)\, x\, \phi_1(x)\, dx$$

$$= \frac{mq^2\varepsilon^2}{2\pi\hbar^2}\left(\frac{2}{L}\right)\int_0^L x\sin(kx)\left[\frac{L^2}{\pi}\left(\frac{1}{\pi^2} - \frac{1}{6}\right) + \frac{Lx - x^2}{\pi}\right]\cos(kx)\,dx$$

$$+ \int_0^L \frac{x^3}{2}\sin^2(kx)\,dx$$

$$\int_0^L x\sin(kx)\cos(kx)\left(Lx - x^2 - \frac{L^2}{6} - \frac{L^2}{\pi^2}\right)dx$$

$$= \frac{-L^4}{8\pi^2}$$

$$\int_0^L x^3 \sin(kx)\,dx = \frac{L^4}{8} - \frac{3L^4}{4\pi^2}$$

$$E_1^{(2)} = -\frac{mq^2\varepsilon^2 L}{24\pi^4\hbar^2}\left(1 - \frac{6}{\pi^2}\right)$$

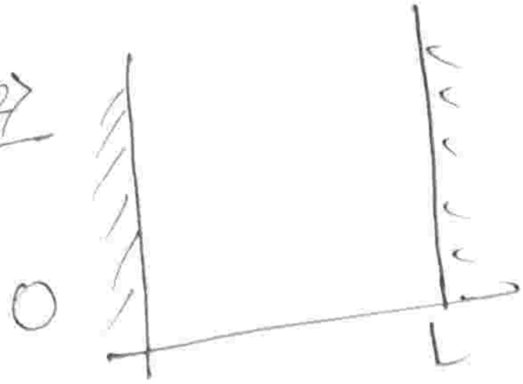
---

Now, classical polarizability:

$$E_1 = -\frac{1}{2}\alpha\varepsilon^2, \quad \vec{p} = \alpha\vec{E}$$

Here, $\alpha = \dfrac{mq^2 L}{24\pi^4\hbar^2}\left(1 - \dfrac{6}{\pi^2}\right)$

Q. 2>




$$y = \frac{x}{\left(1 + \eta/L\right)} = \frac{x}{1+y} \quad , \quad H = -\frac{\hbar^2}{2m}\frac{d^2}{dx^2}$$

$$0 < x < L+\eta$$

$$\frac{d^2}{dx^2} = -\frac{1}{(1+\epsilon)^2}\frac{d^2}{dy^2}$$

$$H_y = \frac{-\hbar^2}{2m(1+\epsilon)^2}\frac{d^2}{dy^2}$$

$$\int |\Psi(x)|^2 dx \longleftrightarrow \int |\Psi(y)|^2 dy$$

$$\frac{1}{(1+\epsilon)^2} = 1 - 2\epsilon + 3\epsilon^2 \cdots\cdots$$

$$H_y = H_0 - 2\epsilon H_0 + O(\epsilon^2)$$

$$E_n^{(1)} = \langle \psi_n^{(0)} | H' | \psi_n^{(0)} \rangle$$

$$= -2\epsilon \langle \psi^{(0)}_n | H_0 | \psi_n^{(0)} \rangle$$

$$1 = \int_0^{L+\eta} |\psi(x)|^2 dx \quad , \quad y = \frac{x}{1+\epsilon}$$

$$= \int_0^{L} |\psi(y)|^2 (1+\epsilon)\, dy$$

$$dE = \left\langle \frac{\partial H}{\partial L} \right\rangle dL$$

In adiabatic change, slow enough change of Hamiltonian, ~~final~~ initial and final Hamiltonian have same eigenstates.

$$H = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2}$$

$$H(L) = -\frac{\hbar^2}{2mL^2} \frac{d^2}{d\xi^2} = \frac{1}{L^2} H_\xi$$

$$\frac{\partial H}{\partial L} = -\frac{2}{L^3} H_\xi = -\frac{2H}{L}$$

$$dE = -\frac{2}{L} E^{(0)} \eta$$

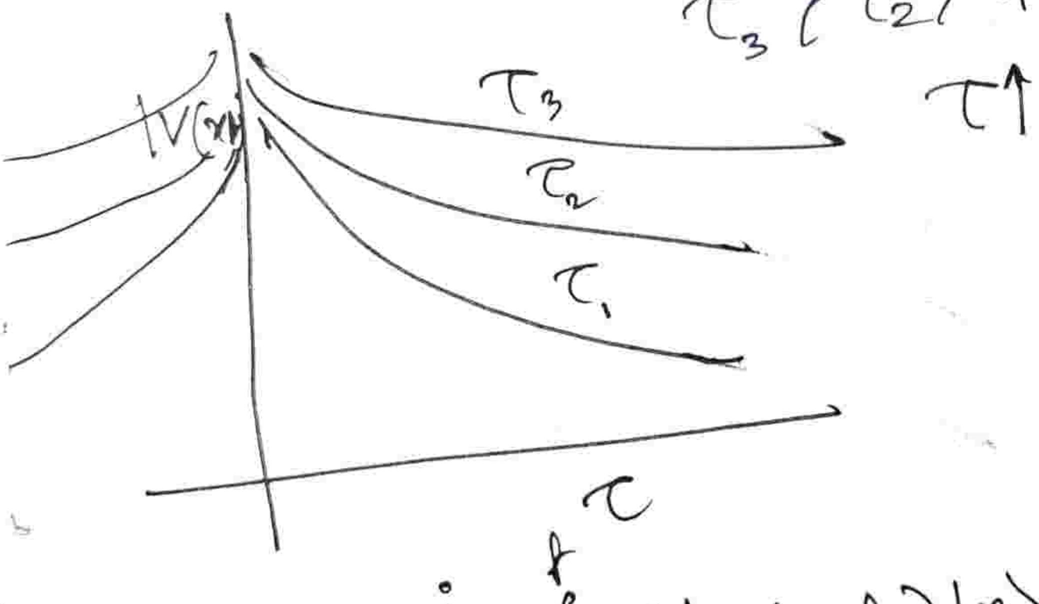
(2) $H = \frac{p^2}{2m} + \frac{1}{2} m \omega^2 x^2$ $\quad E_n = \left(n + \frac{1}{2}\right)\hbar\omega$

$$V(x,t) = -e E_0 x \exp\left(-\frac{t^2}{\tau^2}\right)$$

$$\frac{V(x,t)}{e E_0 x} = \frac{-1}{\exp\left(t^2/\tau^2\right)}$$

$\tau_3 > \tau_2 > \tau_1$

$\tau \uparrow$




$$C_n = -\frac{i}{\hbar} \int_{\alpha}^{t} \langle n | V(x,t) | n \rangle e^{i(E_m - E_n)t/\hbar} \, dt$$

$$|P_{1\leftarrow 0}(\infty)| = |c_{1\leftarrow 0}(\infty)|^2$$

$$H = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 x^2$$

$$= \hbar\omega\left(\frac{p^2}{2m\omega\hbar} + \frac{m\omega}{2\hbar}x^2\right)$$

$$= \hbar\omega\left[\left(\sqrt{\frac{m\omega}{2\hbar}}x + i\frac{p}{\sqrt{2m\hbar\omega}}\right)\left(\sqrt{\frac{m\omega}{2\hbar}}x - i\frac{p}{\sqrt{2m\hbar\omega}}\right)\right.$$

$$\left. - i(px - xp)\sqrt{\frac{m\omega}{2\hbar}}\frac{1}{\sqrt{2m\omega\hbar}}\right]$$

$$= \hbar\omega\left(\hat{a}^{+}\hat{a} + \frac{1}{2}\right)$$

$$\hat{a}^{+} = \sqrt{\frac{m\omega}{2\hbar}}\hat{x} + i\frac{\hat{p}}{\sqrt{2m\omega\hbar}}$$

$$\hat{a} = \sqrt{\frac{m\omega}{2\hbar}}\hat{x} - i\frac{\hat{p}}{\sqrt{2m\hbar\omega}}$$

$$\hat{x} = \sqrt{\frac{\hbar}{2m\omega}}(\hat{a} + \hat{a}^{+})$$

$1, \; 3/2 \, \hbar\omega$

$0, \; \frac{1}{2} \hbar\omega$

$$C_{1\leftarrow 0}(t) = \frac{i}{\hbar} \int_{-\infty}^{t} \langle 1 | \hat{x} | 0 \rangle e E_0 e^{-\frac{t^2}{\tau^2}} e^{i\omega t} \, dt \; (eE_0)$$

$$= \frac{-i}{\hbar} (eE_0) \sqrt{\frac{\hbar}{2m\omega}} \int_{-\infty}^{t} \langle 1 | a^{\dagger} + a | 0 \rangle e^{\frac{-t^2}{\tau^2} + i\omega t} \, dt$$

$$= -\frac{i}{\hbar} (eE_0) \sqrt{\frac{\hbar}{2m\omega}} \int_{-\infty}^{\infty} e^{-t^2/\tau^2 + i\omega t} \, dt$$

$$= -\frac{i}{\hbar} (eE_0) \sqrt{\frac{\hbar}{2m\omega}} \sqrt{\pi} \, \tau e^{\left(\frac{-\omega^2 \tau^2}{a}\right)}$$

$$P_{1\leftarrow 0}(\infty) = |C_{1\leftarrow 0}(\infty)|^2$$

$$p \propto \tau^2 \exp\left(-\frac{\omega^2 \tau^2}{2}\right)$$

$$\boxed{\int_{-\infty}^{\infty} e^{-\alpha x^2 + \beta x} \, dx = \sqrt{\frac{\pi}{a}} \, e^{\beta^2/4\alpha}}$$

$H_0$, $H_0|n\rangle = E_n|n\rangle$

$\psi(t) = e^{-iH_0 t/\hbar}$ ; now, $P_{10} \propto \tau^2$

Now, $\omega\tau \ll 1$

~~$\psi(t)$~~

$\Rightarrow P_{10} \to 0$ [In adiabatic case, no transition]

$$V(x,t) = V(x) f(t)$$

$$C_{m\leftarrow 0}(t) = -\frac{i}{\hbar}\int_{-\infty}^{\alpha} \langle m|V(x)|0\rangle e^{-i(E_m - E_n)t/\hbar} f(t)\,dt$$

$$= -\frac{i}{\hbar}\langle m|V(x)|0\rangle \int_{-\infty}^{\infty} e^{i\omega_{m0}t} f(t)\,dt$$

$$= -\frac{i}{\hbar}\langle m|V(x)|0\rangle F(\omega_{m0})$$

$$\boxed{P \propto |F(\omega_{m0})|^2}$$

$$f(t) = e^{i\omega t}$$

$$\int_{-\infty}^{\infty} e^{i\omega_{m0}t + i\omega t}\,dt = \delta(\omega_{m0} - \omega)$$

$\boxed{\omega_{m0} \to 0}$

$$L(t) = L\left(1 + \frac{t}{T}\right)$$

$$\varepsilon = \frac{x}{L(t)}, \quad x \in [0, L(t)]$$

$$\varepsilon \in [0, 1]$$

$$\psi(x,t) = \sqrt{\frac{2}{L(t)}} \sin\left(\frac{n\pi x}{L(t)}\right)$$

$$= \sqrt{\frac{2}{L(t)}} \sin(n\pi\varepsilon)$$

$$\boxed{\psi(x,t) = \frac{1}{\sqrt{L(t)}} \phi(\varepsilon)}$$

$$i\hbar \frac{\partial \psi(x,t)}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2}$$

$$\frac{\partial \psi(x,t)}{\partial t} = \frac{\partial}{\partial t}\left[\{L(t)\}^{-1/2} \phi(\varepsilon)\right]$$

$$= -\frac{1}{2} L(t)^{-3/2} \dot{L}(t) \phi(\varepsilon)$$

$$+ L(t)^{-1/2} \partial\phi/\partial t + L(t)^{-1/2} \left(\frac{\partial \varepsilon}{\partial t}\right)$$

$$= -\frac{1}{2} L(t)^{-3/2} \dot{L}(t)\, \phi(\varepsilon)$$

$$+ L(t)^{-1/2} \frac{\partial \phi}{\partial t} - x \varepsilon\, \cancel{L}(t)^{-3/2} \frac{\partial \phi}{\partial \varepsilon}$$

$$\frac{\partial \varepsilon}{\partial t} = - \frac{x}{L(t)^2} \dot{L}(t)$$

$$\boxed{\frac{\partial \varepsilon}{\partial t} = - \frac{x \varepsilon}{L(t)}}$$

$$i\hbar \frac{\partial \psi(x,t)}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2}$$

$$\frac{\partial \psi}{\partial x} = \frac{\partial}{\partial x}\left( \frac{\phi(\xi)}{\sqrt{L(t)}} \right) = \frac{1}{\{L(t)\}^{3/2}} \frac{\partial \phi}{\partial \xi}$$

$$i\hbar \frac{\partial \phi(\xi)}{\partial t} = -\frac{\hbar^2}{2mL^2} \frac{\partial^2 \phi}{\partial \varepsilon^2} + i\hbar \frac{\dot{L}(t)}{L(t)} \left(1 + \varepsilon \frac{\partial}{\partial \varepsilon}\right) \phi(\xi)$$

$$= H_{eqn}\, \phi(\xi)$$

$$\boxed{H_{eff} = -\frac{\hbar^2}{2mL^2} \frac{\partial^2}{\partial \xi^2} + i\hbar \frac{\dot{L}(t)}{L(t)} \left(1 + \varepsilon \frac{\partial}{\partial \varepsilon}\right)}$$

$$\Psi_{gs} = \sqrt{\frac{2}{L}} \sin\left(\frac{\pi x}{L}\right)$$

$$\Psi = \sqrt{\frac{1}{L}} \sin\left(\frac{\pi x}{2L}\right)$$

$$\frac{\partial \psi(\xi, t)}{\partial t} = \frac{\partial \phi}{\partial t} + \frac{\partial \alpha}{\partial \xi} \cancelto{0}{\left(\frac{\partial \xi}{\partial t}\right)}$$

Advanced quantum : 03/02/2026

Adiabatic

Slowly waning Hamiltonian

→ same state in which it was prepare

$P_{0\to i}$ , where 'i' is $i^{th}$ excited state.

$$\hat{H}(t)\,|\psi(t)\rangle = E(t)\,|\psi(t)\rangle$$

↳ This is not the solution of the Schrodinger equation. This is "instantaneous eigenstate".

$$i\hbar \frac{\partial}{\partial t}|\Psi\rangle = \hat{H}(t)|\Psi\rangle$$

Space of states → sphere

Sol$^n$ of S.E. in terms of $|\psi(t)\rangle$ times something.

Let the trial solution be

$$|\Psi(t)\rangle = C(t)\,\underbrace{\exp\left[\frac{1}{i\hbar}\int^{t} E(t')\,dt'\right]}\times|\psi(t)\rangle$$

[Dynamical phase has nothing to do with Geometry !]

Now if we use this ansatz to find its eq$^n$ of motion:

$$i\hbar\frac{\partial}{\partial t}|\bar{\Psi}(t)\rangle = i\hbar\frac{\partial}{\partial t}\left[c(t)\exp\left[\frac{1}{i\hbar}\int_0^t E(t')dt'\right] \times|\Psi(t)\rangle\right.$$

[Simply, apply chain rule]

$$\frac{\partial}{\partial t}e^{f(t)} = e^{f(t)}\frac{\partial f}{\partial t}, \quad f = \int_0^t dt' f'(t')$$

$$\frac{\partial}{\partial t}e^{f(t)} = f(t)\, e^{f(t)}$$

$$\therefore \dot{c}(t)\exp\left[\frac{1}{i\hbar}\int_0^t E(t')dt'\right]|\Psi(t)\rangle$$

$$+c(t)\exp\left[\frac{1}{i\hbar}\int_0^t E(t')dt'\right]|\dot{\Psi}(t)\rangle = 0$$

$$\Rightarrow \dot{c}(t)|\Psi(t)\rangle = -c(t)|\dot{\Psi}(t)\rangle$$

$\hookrightarrow$ Differential equation for c

$$\dot{c}(t) = -c(t)\langle\Psi(t)|\dot{\Psi}(t)\rangle$$

$$\Rightarrow c(t) = \exp\left[-\int_0^t \langle\Psi(t')|\dot{\Psi}(t')\rangle dt'\right]$$

$$c(t) = e^{-(x+iy)} = \underbrace{e^{+ix}\, e^{\mp iy}}$$

$e^{+y}$ is a factor

$$\mathrm{Re}\left[\langle \Psi(t) | \dot{\Psi}(t) \rangle\right] = 0$$

$$\langle \Psi(t) | \dot{\Psi}(t) \rangle = 1$$

$$\underbrace{\langle \dot{\Psi}(t) | \Psi(t) \rangle}\ \underbrace{\langle \Psi(t) | \dot{\Psi}(t) \rangle} = 0$$

Complex conjugate

$$Q_B = -i \int_0^t \langle \Psi(t') | \dot{\Psi}(t') \rangle \, dt'$$

$$= -i \sum_i \int_{\lambda_0}^{\lambda(t)} \langle \Psi \lambda_i(t) | \frac{\partial}{\partial \lambda_i} | \Psi(\lambda_i(t)) \rangle \frac{\partial \lambda_i}{\partial t'} dt'$$

The number $\frac{\partial \lambda_i}{\partial t'}$ gives us an estimate of How fast a Hamiltonian $H(\lambda_i)$ is changing in the parameter space.

$$\oint_{\lambda_i^0}^{\lambda(t)} \langle \Psi(\lambda_i(t)) | \frac{\partial}{\lambda_i} \Psi(\lambda_i(t)) \rangle \, d\lambda$$

$$= \int \langle \Psi(\lambda) | \nabla_\lambda | \Psi(\lambda) \rangle \, d\lambda$$

↳ Now, in a closed loop:

$$\oint \langle \Psi(\lambda) | \nabla_\lambda | \Psi(\lambda) \rangle \, d\lambda$$

⇒ <u>Berry phase</u>:

The phase the wavefunction had at $t=0$ and that it accumulated during its evolution and comes back to initial position.

Let initial state be:

$$|\Psi\rangle = C_1 |1\rangle + C_2 |1\rangle$$

↓ evolution in time, picks up a phase which is not part of phase freedom.

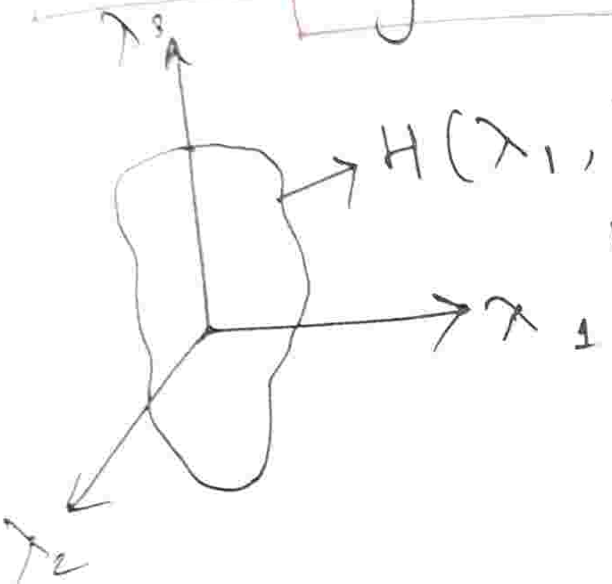
Shows up as interference pattern.

$\ulcorner$ independent of $\frac{d\lambda}{dt}$

$$\oint \{ \langle \Psi(\lambda) | \vec{\nabla}_\lambda | \Psi(\vec{\lambda}) \rangle \} d\lambda$$

$\Rightarrow$ can change to a surface integral by stokes Theorem.

$$\boxed{\Theta = -i \oint \langle \Psi(\lambda) | \nabla_\lambda | \Psi(\lambda) \rangle d\lambda}$$




$$|\Psi(t)\rangle = e^{-if(\lambda)} |\Psi(\lambda)\rangle$$

## Quantum Mechanics 04/03/2026

Instantaneous eigenstate :

(i) $\hat{H}(t) |\Psi(t)\rangle_n = E(t) |\Psi(t)\rangle$ — (1)

Schrodinger equation:

(ii) $i\hbar \frac{\partial}{\partial t} |\bar{\Psi}(t)\rangle = \hat{H}(t) |\bar{\Psi}(t)\rangle$ — (2)

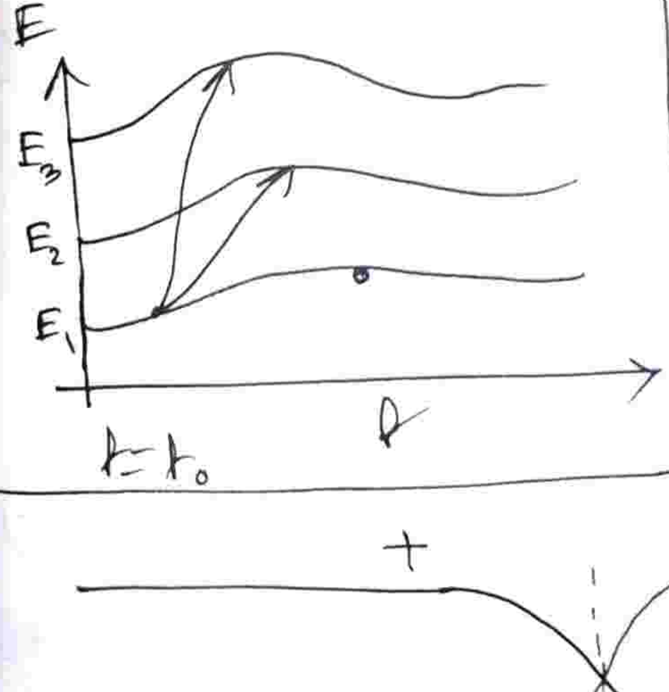
Ansatz solution:

(iii) $|\tilde{\Psi}(t)\rangle = \sum_n C(t) \exp\left[-\frac{i}{\hbar} \int_0^t E(t') dt'\right] |\Psi(t)\rangle_n$

A general schrodinger equation will not preserve the $n$ - label.

$\{|\Psi(t)\rangle_n\}$ is a complete set. So, if initial eigenstate is $|\Psi(t)\rangle_n$, one should expect it to span out to the whole basis as a sum. But, we make an assumption that no such transition occurs.

$\therefore$ This ansatz already encodes adiabaticity.

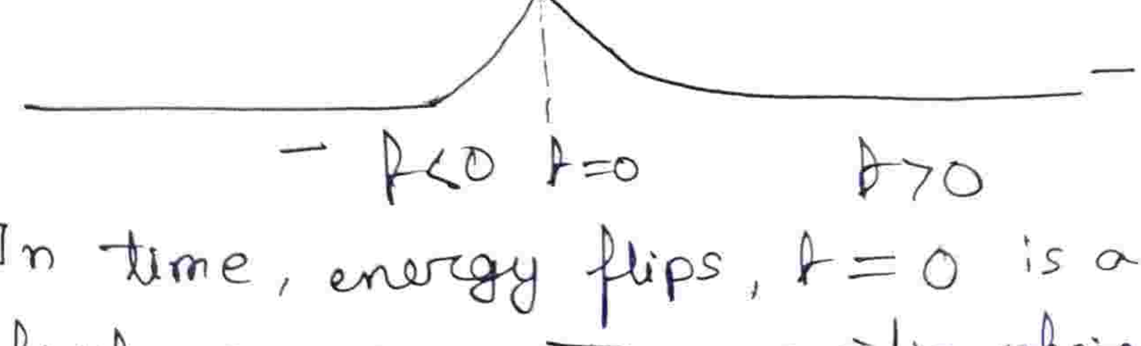



Eg: Spin-$\frac{1}{2}$ in magnetic field.

$\hat{H} = \vec{B}.\vec{S}$,

$|\vec{B}| = \alpha t$ [No]

$\vec{B} = \alpha t \hat{z}$ [Yes]




In time, energy flips, $t=0$ is a level crossing. This specific choice of $t=0$ has broken time reversal symmetry.

Adiabaticity ensures a finite distance between two energy levels.

Because once there is level crossing, we can not tell which is which because it can split and go both ways.

$$LHS = i\hbar\Big[\dot{c}(t)\exp\Big[-\frac{i}{\hbar}\int_0^t E(t')dt'\Big]|\psi(t)\rangle$$

$$+\, c(t)\times\Big(-\frac{i}{\hbar}\frac{\partial}{\partial t}\Big[\int_0^t E(t')dt'\Big)\exp\Big[-\frac{i}{\hbar}\int_0^t E(t')dt'\Big]|\psi(t)\rangle$$

$$+\, c(t)\exp\Big[-\frac{i}{\hbar}\int_0^t E(t')dt'\Big]|\dot{\psi}(t)\rangle\Big]$$

$$= i\hbar\Big[\dot{c}(t)\exp\Big[-\frac{i}{\hbar}\int_0^t E(t')dt'\Big]|\psi(t)\rangle$$

$$+\Big(-\frac{i}{\hbar}\Big)E(t)\exp\Big[-\frac{i}{\hbar}\int_0^t E(t')dt'\Big]|\psi(t)\rangle + (\dots)$$

$|\overline{\Psi}(t)\rangle \longleftarrow$

We multiply (i) by $z = c(t)\exp\Big[-\frac{i}{\hbar}\int_0^t E(t')dt'\Big]$

We obtain:

$$H(t)|\overline{\Psi}(t)\rangle = E(t)|\overline{\Psi}(t)\rangle$$

RHS : $H(t)|\overline{\Psi}(t)\rangle$

Hence we can rewrite the equation:

$$~~|\dot{c}(t)\rangle~~ \quad \dot{c}(t)|\psi(t)\rangle = -c(t)|\dot{\psi}(t)\rangle$$

[Cancelling exponentials on both sides]

$$\Rightarrow \dot{c}(t)\langle\psi(t)|\psi(t)\rangle = -c(t)\langle\psi(t)|\dot{\psi}(t)\rangle$$

$$\Rightarrow \frac{dc(t)}{c(t)} = -\langle\psi|\dot{\psi}(t')\rangle dt$$

$$\Rightarrow \ln c(t)\Big|_{c(t=0)}^{c(t)} = -\int_{t_0}^{t}\langle\psi(t')|\dot{\psi}(t')\rangle dt'$$

$$\Rightarrow \ln\frac{c(t)}{c(t_0)} = -\int_{t_0}^{t}\langle\psi(t')|\dot{\psi}(t')\rangle dt'$$

$$\Rightarrow \boxed{c(t) = c(t_0)\exp\left[-\int_{t_0}^{t}\langle\psi(t')|\dot{\psi}(t')\rangle dt'\right]}$$

Now we have to qualify that it is a phase (pure complex number).

- Is $\langle\psi|\dot{\psi}\rangle$ pure complex number?

$$\langle\psi(t)|\psi(t)\rangle = 1 \rightarrow \boxed{\text{Statement of unitarity}}$$

$$\Rightarrow \langle\psi(t)|\dot{\psi}(t)\rangle + \langle\dot{\psi}(t)|\psi(t)\rangle = 0$$

Solution 2

One trivial solution is, we know that these two terms are related by complex conjugation.

□ Soln 1:

$$\Rightarrow Re\left[\langle\psi(t)|\dot{\psi}(t)\rangle\right]=0$$ → S.E. ensures this, in general

□ Soln 2:

$$\langle\psi(t)|\dot{\psi}(t)\rangle=0$$ (Trivial, not the case)

Looking at the instaneous eigenstates of the system:

$$\hat{H}(t)|\psi(t)\rangle = E(t)|\psi(t)\rangle$$

$$\dot{H}(t)|\psi(t)\rangle + H(t)|\dot{\psi}(t)\rangle = \dot{E}(t)|\psi(t)\rangle + E(t)|\dot{\psi}(t)\rangle$$

$$\Rightarrow \langle\psi|\dot{H}|\psi\rangle + \langle\psi(t)|H(t)|\dot{\psi}(t)\rangle$$

$$= \dot{E}(t)\langle\psi(t)|\psi(t)\rangle + E(t)\langle\psi(t)|\dot{\psi}(t)\rangle$$

We are looking for a solution for which $\langle\psi(t)|\dot{\psi}(t)\rangle = 0$

$$\boxed{\dot{E}(t) = \langle\psi|\dot{H}|\psi\rangle + \langle\psi|H|\dot{\psi}\rangle}$$

↳ Condition for solution 2 to occur.

Curvature of phase space through which the states are being dragged
→ source of geometric phase.
Now, for higher dimensional space where Hilbert ~~ph~~ space is flat, no geometric phase arises.

When perturbation is large, ~~we~~ the ansatz solution is wrong in itself!

Instaneous eigenstates:

$$H(t)|\Psi_n(t)\rangle = E_n|\Psi_n(t)\rangle$$

The Schrodinger equation:

$$H(t)\,|\overline{\Psi}(t)\rangle = i\hbar\frac{\partial}{\partial t}|\overline{\Psi}(t)\rangle$$

Sub-stitute

$$|\overline{\Psi}(t)\rangle = \sum_n c_n(t)|\Psi_n(t)\rangle$$

$$i\hbar\sum_n\left[\dot{c}_n|\Psi_n\rangle + c_n|\dot{\Psi}_n\rangle\right] = \sum_n c_n E_n|\Psi_n\rangle$$

Multiplying both sides by $\langle\Psi_k|$

$$i\hbar\sum_n\left[\dot{c}_n\langle\Psi_k|\Psi_n\rangle + c_n\langle\Psi_k|\dot{\Psi}_n\rangle\right] = \sum_n c_n E_n\langle\Psi_k|\Psi_n\rangle$$

[P.T.O]

$$\Rightarrow \boxed{i\hbar \sum_n [\dot{c}_n \delta_{kn} + c_n \langle \Psi_k | \dot{\Psi}_n \rangle] = \sum_n c_n E_n \delta_{kn}}$$

- Case: $k = n$ (We want a differential eq$^n$ for k)

$$i\hbar \dot{c}_k + i\hbar \sum_n c_n \langle \Psi_k | \dot{\Psi}_n \rangle = c_k E_k$$

$$\Rightarrow i\hbar \dot{c}_k = c_k E_k - i\hbar \sum_n c_n \langle \Psi_k | \dot{\Psi}_n \rangle$$

$$= \boxed{c_k (E_k - i\hbar \langle \Psi_k | \dot{\Psi}_k \rangle)}$$

$$\boxed{- i\hbar \sum_{n \neq k} c_n \langle \Psi_k | \dot{\Psi}_n \rangle}$$

Correction

Rate of leakage

Adiabatic approximation is setting $\langle \Psi_k | \dot{\Psi}_n \rangle = 0$

- Now question is, what do we mean when we say $\langle \Psi_k | \dot{\Psi}_n \rangle$ is "small"?

$$H(t) | \Psi_n(t) \rangle = E_n(t) | \Psi_n(t) \rangle$$

$$\Big\downarrow \text{time derivative}$$

$$\dot{H} | \Psi_n \rangle + H | \dot{\Psi}_n \rangle = \dot{E}_n | \Psi_n \rangle + E_n | \dot{\Psi}_n \rangle$$

Taking inner product with $\langle \Psi_k |$ $[k \neq n]$

$$\langle \Psi_k | \dot{H} | \Psi_n \rangle + \langle \Psi_k | H | \dot{\Psi}_k \rangle = \dot{E}_n \langle \Psi_k | \Psi_n \rangle + E_n \langle \Psi_k | \dot{\Psi}_n \rangle$$

Here, $\langle \Psi_k | H = E_k \langle \Psi_k | \dot{\Psi}_n \rangle$

$$\langle \Psi_k | \Psi_n \rangle = 0$$

$$\Rightarrow \langle \Psi_k | \dot{\Psi}_n \rangle (E_n - E_k) = \langle \Psi_k | \dot{H} | \Psi_n \rangle$$

$$\boxed{\langle \Psi_k | \dot{\Psi}_n \rangle \Big|_{n \neq k} = \frac{\langle \Psi_k | \dot{H} | \Psi_n \rangle}{E_n - E_k}} \ll 1$$

→ expectation value of rate of change of Hamiltonian, is it capable of taking a state $|\Psi_n\rangle$ to $|\Psi_k\rangle$?

If $\langle \Psi_k | \dot{\Psi}_n \rangle = 0$, always adiabatic.

If energy cost is very high, $(E_n - E_k \to \infty)$, $\langle \Psi_k | \dot{\Psi}_n \rangle = 0$.

This is the small dimensionless parameter which determines how "small".

∘ What is "geometric" about this phase?

$$C_k(t) = C_k(0)\exp\left[\frac{1}{i\hbar}\int_0^t \left(E_k(t') - i\hbar\langle\psi_k|\dot{\psi}_k\rangle\right)dt'\right]$$

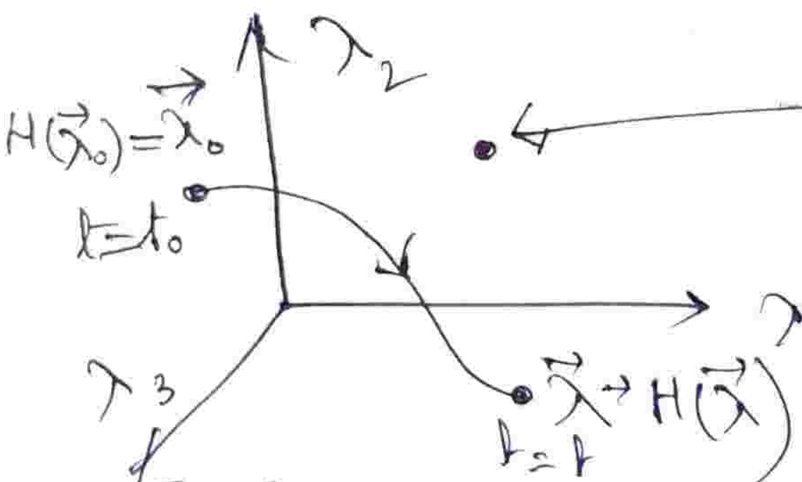
$$= C_k(0)\, e^{i\theta_k}\, e^{i\gamma_k}$$

$$\boxed{\begin{aligned} \theta_k &= -\frac{1}{\hbar}\int_0^t E_k(t')\,dt' \rightarrow \text{Dynamical Phase} \\ \gamma_k &= \int_0^t \nu_k(t')\,dt' \rightarrow \text{geometric phase} \\ & \qquad \nu_k = i\langle\psi_k|\dot{\psi}_k\rangle \end{aligned}}$$

▸ What is parameter space?

Hamiltonian is a function of $n$ nos. of parameters : $H(\lambda_1, \lambda_2, \ldots, \lambda_n)$

real no $\in \mathbb{R}$




Each point is a specific Hamiltonian,

In every point on the trajectory

from $H(\vec{\chi}_0)$ to $H(\vec{\chi})$, we change the potential, in order to trace this path. We also introduce time evolution.

Now we recalculate the geometric phase in the parameter space:

$$\langle \Psi_k | \dot{\Psi}_k \rangle = \langle \Psi_k(t) | \frac{\partial}{\partial t} | \Psi_k(t) \rangle$$

$$\boxed{\Psi_k(t) = \Psi_k(\vec{\lambda}(t))}$$

In case of 1-D SHO, $\omega = \sqrt{\frac{k}{m}}$ inculcates $k \equiv k(t)$, $m \equiv m(t)$ (although changing mass is a bit weird) and is a fundamental time dependent parameter of this problem.

$$\langle \Psi_k | \dot{\Psi}_k \rangle = \langle \Psi_k(t) | \frac{\partial}{\partial t} | \Psi_k(t) \rangle$$

$$\boxed{\langle \Psi_k | \dot{\Psi}_k \rangle = \sum_i \underbrace{\langle \Psi_k | \frac{\partial}{\partial \lambda_i} | \Psi_k \rangle}_{\text{$i$th component of vector potential.}} \dot{\lambda}_i}$$

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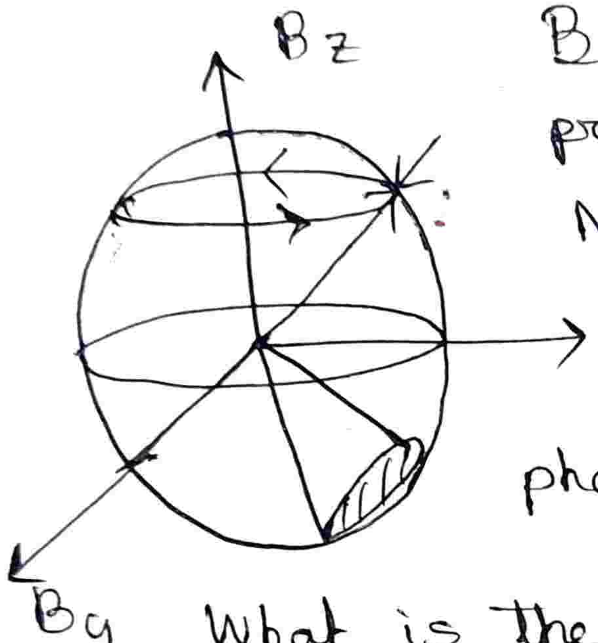
In case of the spin-$\frac{1}{2}$ problem.




Berry phase for a spin-$\frac{1}{2}$ problem, is not unique. Many geometric paths that correspond to the same geometric phase.

What is the berry phase accumulated by the spin-$\frac{1}{2}$ particle in its loop journey?

In this case, if berry phase $\theta_0 = \frac{\Omega_0}{2}$, many possible cones, many possible solid angles, not unique Berry phase.

- Actual Classnotes Begin

(1) Space $\rightarrow$ (i) Space of Hamiltonians (Parameter space) $R^n \rightarrow$ every point in the space correspond to Hamiltonian.

Let, $\vec{H} = \vec{B} \cdot \vec{S}$ [$\vec{S} \equiv$ spin half]

$H = \vec{B} \cdot \vec{S}$ , $\vec{B} = (B_x, B_y, B_z)$

where, $B_x, B_y, B_z \in \mathbb{R}^3$

Every distinct $B_x$, $B_y$, $B_z$

$\rightarrow$ Unique Hamiltonian.

(ii) Space in question is a vector space.

Hermitian matrices form a vector space: $\underbrace{\vec{B}_1 + \vec{B}_2 = \vec{B}}$

$\hookrightarrow$ closure under addition.

Hermitian matrices form a vector space over field of $\mathbb{R}$.

[For a two dimensional problem, vector space is 4-D]

## ② Hilbert space (Space of States)

~~Space~~ Vector space defined over the field of complex number:

$$z_1|\Psi_1\rangle + z_2|\Psi_2\rangle = |\Psi\rangle$$

$e^{-i\theta\hat{n}\cdot\vec{S}}$ $\rightarrow$ This rotates

$\hookrightarrow$ intrinsic spin $\rightarrow$ lives in $\mathbb{R}^3$.

actually a matrix!

If we take spin-$\frac{1}{2}$, $\vec{S} = \hbar\frac{\vec{\sigma}}{2}$

$$\begin{array}{ccc} {}_z\langle\uparrow|\uparrow\rangle_z & & {}_z\langle\uparrow|\downarrow\rangle_z \\ & \sigma_z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} & \\ {}_z\langle\downarrow|\uparrow\rangle_z & & {}_z\langle\downarrow|\downarrow\rangle_z \end{array}$$

${}_z\langle\uparrow|\downarrow\rangle_z \rightarrow$ Cross terms: Off diagonal elements.

${}_z\langle\downarrow|\downarrow\rangle_z \rightarrow$ Expectation values: Diagonal elements

$|\uparrow\rangle_z$

$|\downarrow\rangle_z$ $\leftarrow$ Hilbert space

## Adiabaticity :

"Dragging" of a state in such way which only allows a phase freedom.

In physical space, $\vec{S}$ is an operator, but in its internal structure, its matrix ~~sd~~ components, is governed by the State-space details.

Now we previously saw:

$$\boxed{C_k(t) = C_k(0)\, e^{i\theta_k}\, e^{i\gamma_k}}$$

Where $C_k(0)$ is initial probability amplitude of a process.

$$\boxed{\gamma_k = \int_0^t \gamma_k(t')\,dt' = i\langle\Psi_k|\dot{\Psi}_k\rangle}$$

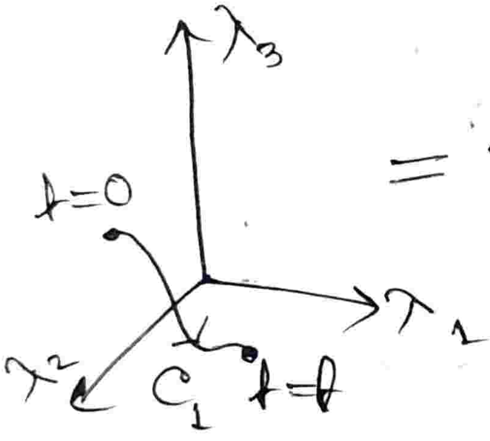
Now in parameter space:

$$\langle \psi(\vec{\lambda}(t)) | \frac{\partial}{\partial t} | \psi(\vec{\lambda}(t)) \rangle$$

$$= \sum_j \langle \psi_k | \frac{\partial}{\partial \lambda_j} | \psi(\vec{\lambda}) \rangle \frac{\partial \lambda_j}{\partial t}$$

$$\gamma_k = i \sum_j \int_0^t \langle \psi_k | \frac{\partial}{\partial \lambda_j} | \psi_k \rangle \dot{\lambda}_j \, dt'$$

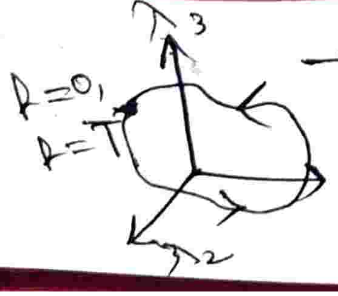
$$= i \sum_j \int_{C_1} \underbrace{\langle \psi_k | \frac{\partial}{\partial \lambda_j} | \psi_k \rangle}_{A_j} d\lambda_j$$




$$= i \sum_j \int_{C_1} A_j \, d\lambda_j = i \int_{C_1} \vec{A} \cdot d\vec{\lambda}.$$

Here, $\vec{A}$ is a vector defined on $\mathbb{R}^n$, not in Hilbert space.

$$\boxed{\gamma_k = i \int_{C_1} \vec{A} \cdot d\vec{\lambda}}$$




→ Now we consider a closed loop journey in parameter space.

$$\gamma_k = i \oint \vec{A} \cdot d\vec{\lambda}$$

[P.T.O]

Now, we interchange this ~~to~~ with Stokes theorem [make it an area integral!]

$$\gamma_k = i\oint \vec{A}\cdot d\vec{\lambda} = i\iint_S \underbrace{(\vec{\nabla}\times\vec{A})}_{\vec{B}_{eff}}\cdot d\vec{s}$$

Here: $\vec{\nabla}\times\vec{A} \equiv$ Berry Curvature

> The nomenclature comes from the fact that the Berry phase arises due to curvature (non-trivial geometry) of the Hilbert space!

$A_i = \langle \Psi_k | \frac{\partial}{\partial \lambda_i} | \Psi_k \rangle \rightarrow$ This vector is defined upto a phase.

$$\therefore |\Psi_k\rangle = \underbrace{e^{i\theta}}_{} |\Psi_k(\vec{\lambda})\rangle$$

$\hookrightarrow$ Global Gauge Transformation

$\theta$ is independent of $\lambda$.

$\therefore \langle \Psi_k |$ and $| \Psi_k \rangle$ will come with conjugate phases and exponentials will cancel, leaving $\langle \psi | \frac{\partial}{\partial \lambda_i} | \psi \rangle$ invariant.

$$|\Psi_k(\lambda_i)\rangle \rightarrow e^{i\theta}|\Psi_k(\lambda_i)\rangle$$

$$\langle\Psi_k(\lambda_i)| \rightarrow e^{-i\theta}\langle\Psi_k(\lambda_i)|$$

$\therefore A_i$ is invariant under this transformation.

The question is, is $\gamma_k$ gauge-invariant? If yes, then $\gamma_k$ is a physical quantity.

$$A_i' \rightarrow \langle\Psi_k|e^{-i\phi(\lambda)}\frac{\partial}{\partial\lambda_i}e^{i\phi(\lambda)}|\Psi_k\rangle$$

$$\Rightarrow A_i' = \left\{i\frac{\partial}{\partial\lambda}\phi(\vec{\lambda})\right\}\langle\Psi_k|\Psi_k\rangle$$

$$+ \langle\Psi_k|\frac{\partial}{\partial\lambda_j}|\Psi_k\rangle$$

$$= \left\{i\frac{\partial}{\partial\lambda}\phi(\vec{\lambda})\right\} + \langle\Psi_k|\frac{\partial}{\partial\lambda_j}|\Psi_k\rangle$$

$$\Rightarrow A_i' = \left\{i\frac{\partial}{\partial\lambda}\phi(\vec{\lambda})\right\} + A_i$$

$$\Rightarrow \boxed{A_i' = i\nabla_\lambda\phi(\lambda) + A_i}$$

↳ gradient in parameter space.

Now we look at $\gamma_k \rightarrow \gamma_k'$

$$\gamma_k' = i \iint_S \left(\vec{\nabla} \times \left\{ i\vec{\nabla}_\lambda \phi(\vec{\lambda}) + A_i \right\}\right) \cdot \vec{ds}$$

$$\boxed{\gamma_k' = \gamma_k = i \iint_S (\vec{\nabla}_\lambda \times \vec{A}) \cdot \vec{ds}} \rightarrow \text{Gauge invariance}$$

$\therefore$ Berry phase ;

(i) Time independent.

(ii) Gauge independent.

(iii) Dependent on Geometric path.

- For degenerate points like $\vec{B} = (B_x, B_y, B_z) = (0, 0, 0)$, no adiabaticity.

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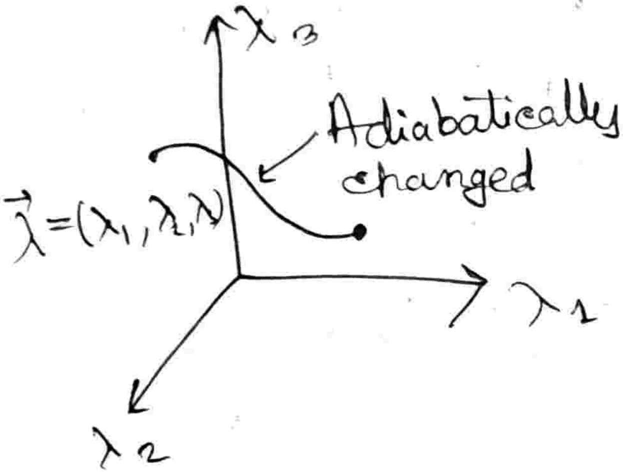
$$\gamma_k = i \int_0^t \langle \psi_k | \dot{\psi}_k \rangle \, dt'$$

- Quantity of interest




$$\langle \psi | \dot{\psi}_k \rangle$$
$$= \langle \psi(\vec{\lambda}(t)) | \dot{\psi}(\vec{\lambda}(t)) \rangle$$
$$= \sum_i \langle \psi_k | \frac{\partial}{\partial \lambda_i} | \psi_k \rangle \dot{\lambda}_i$$

$$\langle \psi_k | \frac{\partial}{\partial \lambda} | \psi_k \rangle = A_i$$

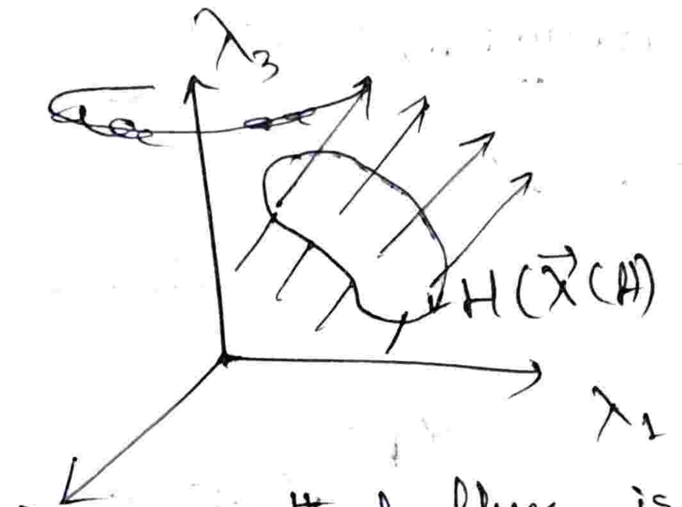
$$\gamma_k = i \int_S (\vec{\nabla} \times \vec{A}) \cdot \vec{ds}$$

According to Gauge freedom (Phase freedom) $|\psi(\vec{\lambda})\rangle \Rightarrow e^{i\phi(\vec{\lambda})} |\psi(\vec{\lambda})\rangle$

$$\vec{A} \to \vec{A} \pm i \nabla \phi(\vec{\lambda})$$

$\gamma_k \to (\vec{\nabla}_\lambda \times \vec{A})$ invariant as $\vec{\nabla} \times (\vec{\nabla} \phi(\vec{\lambda})) = 0$

Hence, $\gamma_k = i \int_S (\vec{\nabla} \times \vec{A}) \cdot \vec{ds}$ is a Gauge invariant quantity, and hence an experimentally measurable.




Statement:

If Hamiltonian traces a path that is a closed curve in the parameter space, there is a flux passing through it and that flux is proportional to the $\vec{B}$.

$$\vec{A} = \mathrm{Im} \langle \Psi_k(\vec{\lambda}) | \vec{\nabla}_\lambda \Psi_k(\vec{\lambda}) \rangle \equiv \text{Vector potential}$$

[$\Psi_k(\vec{\lambda}) \equiv k^{th}$ eigenstate at $\lambda^{th}$ point]

Let
$H = \vec{B} \cdot \vec{\sigma}$ [What might be a possible basis?]
assuming $\vec{B} = |\vec{B}|\,\hat{z}$, a nice choice of basis is $|\uparrow\rangle_z$, $|\downarrow\rangle_z$

Suppose we have a rotating $\vec{B}$,

$$\vec{B} = (B_x \sigma_x \cos(\omega t) + B_y \sigma_y \sin(\omega t))$$

$$H = \vec{B} \cdot \vec{\sigma} = |\vec{B}|\,(\hat{n} \cdot \vec{\sigma})$$

→ This operator part determines the eigenstate

$\hat{n}$ = instantaneous direction of $\vec{B}$.

We go for a co-moving basis:

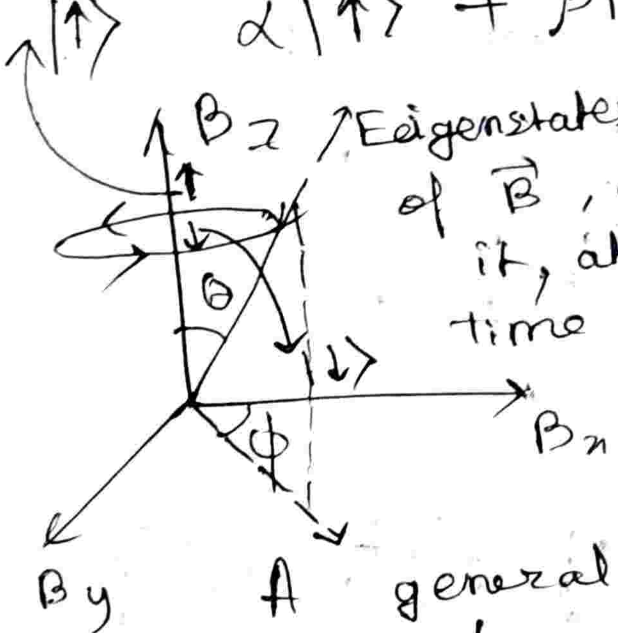
$|\uparrow\rangle_z$ and $|\downarrow\rangle_z$ along direction $\hat{n}$.

$(\vec{\sigma}\cdot\hat{n}) \equiv |\uparrow\rangle_{\hat{n}}, \; |\downarrow\rangle_{\hat{n}} \Rightarrow$ $\boxed{\text{Instantaneous eigenstates.}}$

In parameter space, $\hat{n} \equiv n(\vec{\lambda})$

A possible eigenstate is:

$\alpha|\uparrow\rangle + \beta|\downarrow\rangle$




Eigenstates ~~for~~ in direction of $\vec{B}$, another opposing it, at each point in time for the rotating $\vec{B}$. These are the instantaneous eigenstates.

A general spinor looks like:

$$|\psi\rangle = \begin{pmatrix} \cos\frac{\theta}{2} \\ \sin\frac{\theta}{2}\, e^{i\phi} \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}\cos\left(\frac{\theta}{2}\right) + \sin\left(\frac{\theta}{2}\right)e^{i\phi}\begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

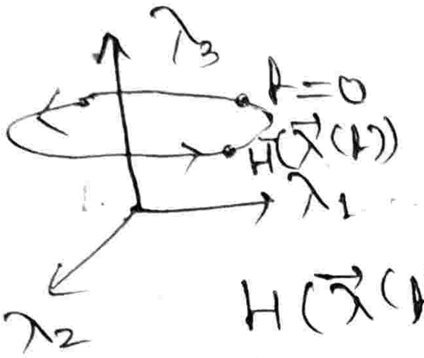
$$= |\uparrow\rangle_z \cos\left(\frac{\theta}{2}\right) + |\downarrow\rangle_z \sin\frac{\theta}{2}e^{i\phi}$$

$[\theta \equiv \theta(\vec{\lambda})]$

$[\phi \equiv \phi(\vec{\lambda})]$

$\vec{A}(t) = \mathrm{Im}\langle \Psi_k | \vec{\nabla} \Psi_k \rangle$

$\hookrightarrow$ Instantaneous eigenstate




[We going to expand it in fixed basis that is frozen in time]

$H(\vec{\lambda}(t))$ has complete set of eigenbasis $\{|n(\vec{\lambda})\rangle\}$

Suppose we take $\vec{\lambda}_0$ out of the trajectory, define Hamiltonian $H(\vec{\lambda}_0)$. Let it have a complete set of basis, i.e. reference basis $\{|R\rangle_n\}$. This is NOT the eigenbasis of $H(\vec{\lambda}(t))$.

$$\vec{A}(\vec{\lambda}) = \mathrm{Im}\langle \Psi_k | \vec{\nabla} \Psi_k \rangle$$

$$= \mathrm{Im}\left[\sum_i \sum_j \{c_j^* \langle j|\}\{\vec{\nabla} c_i |i\rangle\}\right]$$

$$= \mathrm{Im} \sum_{ij} (c_j^* \vec{\nabla} c_i) \langle j | i \rangle$$

$$\boxed{\vec{A}(\vec{\lambda}) = \mathrm{Im} \sum_i c_i^* \vec{\nabla} c_i}$$

$$\vec{\nabla} \times \vec{A} = \vec{B}_{eff} = \vec{\nabla} \times \left[\mathrm{Im} \sum_i (c_i^* \vec{\nabla} c_i)\right]$$

Using identity:

$$\vec{\nabla} \times (f\vec{V}) = f\vec{\nabla} \times \vec{V} + (\vec{\nabla} f) \times \vec{V}$$

Now, $\vec{V} = \vec{\nabla} C_i$

$$\vec{\nabla} \times (\vec{\nabla} C_i) = 0$$

$$\vec{\nabla} \times \vec{A} = \mathrm{Im} \sum_i \vec{\nabla} C_i^* \times (\vec{\nabla} C_i)$$

$$\vec{B}_{eff} = \mathrm{Im} \langle \vec{\nabla} \Psi_k | \times | \vec{\nabla} \Psi_k \rangle \qquad \left| \sum_n |n\rangle\langle n| = \mathbb{1} \right.$$

$$= \mathrm{Im} \sum_{n \neq k} \langle \nabla k | n \rangle \times \langle n | \nabla k \rangle$$

[$\langle \nabla k | n \rangle \rightarrow$ vector in $\lambda$-space, scalar in H-space]

- The condition of adiabaticity has to be:

$$\langle n' | \nabla n \rangle = \frac{\langle n' | \vec{\nabla} H | n \rangle}{E_n - E_{n'}}$$

$$\vec{B}_{eff} = \mathrm{Im} \sum_{n \neq n'} \frac{\langle n | \vec{\nabla}_\lambda H | n' \rangle \times \langle n' | \vec{\nabla}_\lambda H | n \rangle}{(E_n - E_{n'})^2} \quad \text{— (i)}$$

- Let us look at how spin-$\frac{1}{2}$ in $\vec{B}$ field behaves like;

$$H = \frac{1}{2} \vec{B} \cdot \vec{\sigma} = \frac{1}{2} [B_x \sigma_x + B_y \sigma_y + B_z \sigma_z]$$

$$\vec{B} = \vec{B}(t) \Rightarrow H = \frac{1}{2} \begin{bmatrix} B_z & B_x - iB_y \\ B_x + iy & -B_z \end{bmatrix}$$

$$\vec{\nabla} H = \left(\frac{\partial}{\partial B_x}\hat{i} + \frac{\partial}{\partial B_y}\hat{j} + \frac{\partial}{\partial B_z}\hat{k}\right)\left[\frac{1}{2}\{B_x\sigma_x + B_y\sigma_y + B_z\sigma_z\}\right]$$

$$= \frac{1}{2}\left[\sigma_x\hat{i} + \sigma_y\hat{j} + \sigma_z\hat{k}\right]$$

Let there be two eigenstates $|+\rangle$, $|-\rangle$ with eigenvalues $E_+$, $E_-$.

$$E_+ = \frac{1}{2}\sqrt{B_x^2 + B_y^2 + B_z^2}$$

$$E_- = -\frac{1}{2}\sqrt{B_x^2 + B_y^2 + B_z^2}$$

$$E_+ - E_- = |\vec{B}| = \sqrt{B_x^2 + B_y^2 + B_z^2}$$

For a two-level system, no "summation" involved in (1), i.e., the formula of $\vec{B}$ field. This two, is a two level system with eigenstates $|+\rangle$ and $|-\rangle$. So we have to calculate the matrix elements $\langle +|\vec{\nabla} H|-\rangle$, $\langle -|\vec{\nabla} H|+\rangle$, to evaluate the $\vec{B}_{eff}$ (exercise).

Geometric phase is not a property of the adiabatic Hamiltonian, rather a property of the Hilbert's space.

We will define a notion of parallelism. For vectors, cross product is zero. How do we define it for Hilbert space vectors?

- Let us ask a question: Let $\vec{A} \parallel \vec{B}$, $\vec{B} \parallel \vec{C}$, does it imply $\vec{A} \parallel \vec{C}$.
  ↳ Only implies in flat space.

- Pancharatnam's Notion: If $\langle A|B\rangle$ is real and positive, then they are parallel.

∘ Why?

Phase freedom says: $|A\rangle \rightarrow e^{i\theta_A}|A\rangle$

$$|B\rangle \rightarrow e^{i\theta_B}|B\rangle$$

$$\langle A|B\rangle = z = re^{i\theta}$$

$$\Rightarrow z' = re^{i(\theta_A - \theta_B)}e^{i\theta}$$

We can always make a choice $\theta_A - \theta_B = -\theta$

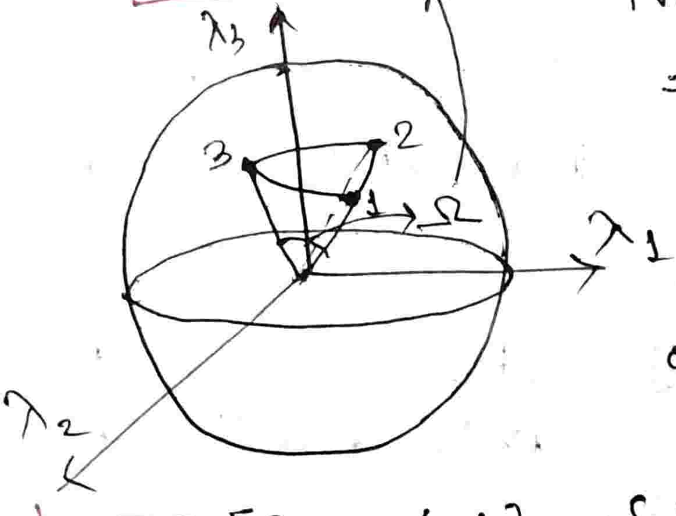
[Phase choice that make $\langle A|B\rangle$ real and +ve] again, $|C\rangle = e^{i\theta_c} |C\rangle$

$\underbrace{\langle B|C\rangle}_{\text{Real phase by choice}} = r' e^{i\theta'}$

$\langle C|A\rangle$ = not real. There is some geometric phase between $|A\rangle$ and $|C\rangle$.

To have a non-trivial loop, we need at least three points [three vectors in Hilbert space].

$\boxed{\text{Geometric phase picked} = \frac{1}{2}\Omega}$

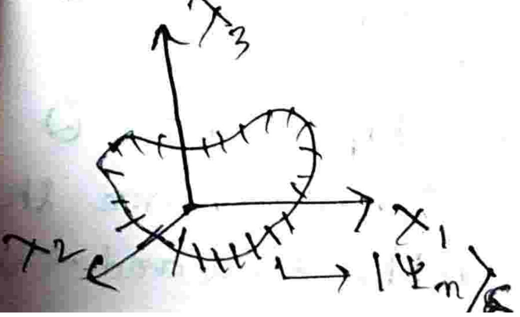



Phase of $\langle 1|2\rangle\langle 2|3\rangle\langle 3|1\rangle$

$= z = re^{i\theta}$

$\langle 1|2\rangle\langle 2|3\rangle\langle 3|1\rangle$ is a series of projection of one state on another.

$Tr\left[\{|1\rangle\langle 1|\}, \{|2\rangle\langle 2|\}, \{|3\rangle\langle 3|\}\right]$

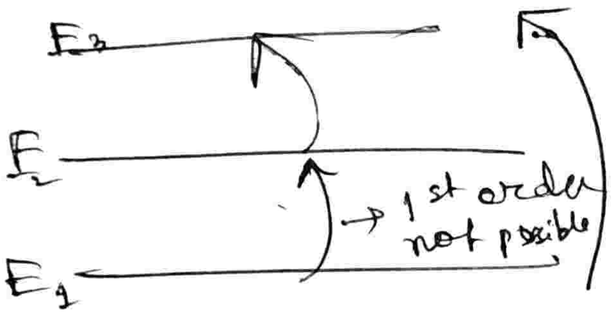



$\Rightarrow \langle n|\hat{O}|n\rangle = Tr[\langle n|n\rangle \hat{O}]$

$\langle \Psi_n(\delta_1 = 1) | \Psi_n(\delta = 2)\rangle \cdot$
$\langle \Psi_n(\delta = 2) | \Psi_n(\delta = 3)\rangle \ldots$

Quantum Tutorial: TDPT

4. Three Level Atomic System, monochromatic laser of frequency $\hbar\omega \simeq \frac{E_3 - E_1}{2}$




$C^{(1)} = \langle 3|\hat{x}|1\rangle \cancel{\neq} 0$

1st order transition does not work here.

$$\vec{E} = \mathcal{E}\cos(\omega t)\hat{\epsilon}, \quad \vec{p} = p\hat{\epsilon}$$

$$V(t) = -\vec{p}\cdot\vec{E} = -p\mathcal{E}\cos(\omega t)$$

$$= Ve^{-i\omega t} + Ve^{i\omega t}$$

$$C_3^{(2)}(t) = \left(-\frac{i}{\hbar}\right)^2 \sum_m \int_0^t dt' \int_0^{t'} dt'' \langle 3|V_I(t')|m\rangle\langle m|V_I(t'')|1\rangle$$

$$V_I(t) = e^{\frac{iH_0 t}{\hbar}} V(t) e^{-iH_0 t/\hbar}$$

$$\langle 3|V_I(t')|2\rangle = \langle 3|e^{iH_0 t/\hbar} \underbrace{V(t')}_{V_{32}} e^{-iH_0 t/\hbar}|2\rangle$$

$$= e^{i\omega_{32} t'} \langle 3|V(t')|2\rangle$$

Quantum Tutorial: $|\Psi\rangle = (\hat{P}+\hat{Q})|\Psi\rangle$

Q.8) $\mathcal{H}$ $\left.\begin{matrix} \text{---} \\ \text{---} \\ \text{---} \end{matrix}\right\} M^{\perp}$ $\mathbb{I} = \sum_{k \in \mathcal{H}} |k\rangle\langle k|$

$\left.\begin{matrix} \text{---} \\ \text{---} \\ \text{---} \end{matrix}\right\} M$ $= \sum_{k \in M} |k\rangle\langle k|$

$\Rightarrow \mathbb{I} = \hat{P} + \hat{Q}$ $+ \sum_{k \in M} |k\rangle\langle k|$

$$(\hat{P})^2 = \left(\sum_{k \in M} |k_1\rangle\langle k_1|\right)\left(\sum_{k \in M} |k_2\rangle\langle k_2|\right)$$

$$= \sum_{k_1, k_2} |k_1\rangle\langle k_2|k_2\rangle\langle k_2|$$

$$\mathbb{I} = \sum_{k_1, k_2} |k_1\rangle \, \delta_{k_1, k_2} \langle k_2|$$

$$= \sum_{k} |k\rangle\langle k| = \hat{P}$$

$$\therefore \hat{P}\hat{Q} = \left(\sum_{k\in M} |k\rangle\langle k|\right)\left(\sum_{k'\in M^\perp} |k'\rangle\langle k'|\right)$$

$$= \sum_{\substack{k\in M \\ k'\in M^\perp}} |k\rangle\langle k|k'\rangle\langle k'| = 0$$

(b) $H_{eff}$ : $(\hat{P}+\hat{Q})|\psi\rangle = \mathbb{I}\,|\Psi\rangle = |\Psi\rangle$

$$\Rightarrow \hat{Q}\hat{H}(\hat{P}+\hat{Q})|\psi\rangle = \hat{Q}\hat{H}\,|\Psi\rangle$$

$$\Rightarrow (\hat{Q}\hat{H}\hat{P}+\hat{Q}\hat{H}\hat{Q})|\psi\rangle = E\hat{Q}|\Psi\rangle$$

$$\Rightarrow \hat{Q}\hat{H}\,\hat{P}|\Psi\rangle = (E-\hat{Q}H\hat{Q})\,\hat{Q}|\Psi\rangle$$

$$\hat{Q}|\Psi\rangle = (E-\hat{Q}\hat{H}\hat{Q})^{-1}\hat{Q}\hat{H}\hat{P}|\Psi\rangle$$

$$\hat{P}\hat{H}\,(P+Q)|\Psi\rangle = \hat{P}\hat{H}|\Psi\rangle$$

$$\Rightarrow \hat{P}\hat{H}\hat{P}|\Psi\rangle + PHQ|\Psi\rangle = EP|\Psi\rangle$$

$$\Rightarrow PHP|\psi\rangle + PHQ(E-QHQ)^{-1}QHP|\psi\rangle$$

$$= EP|\psi\rangle$$

$$\Rightarrow \left(PHP + PHQ(E-QHQ)^{-1}QHP\right)(P|\psi\rangle)$$

$$= E\,(P|\psi\rangle)\,|\psi'\rangle$$

$$\therefore H_{eff} = PHP + PHQ\,(E-QHQ)^{-1}QHP$$

$$\delta E \propto \langle\psi|H'|\psi\rangle$$

$$H_{eff} = H_0 + H' \quad \rightarrow \{|k\rangle\} \rightarrow H_0$$

$$\langle m|H_{eff}|n\rangle = \langle m|P\,H_0P|n\rangle + \langle m|PH'P|n\rangle$$

$$+ \langle m|PHQ\,(E-QHQ)^{-1}\,QHP|n\rangle$$

$$\langle m|PH_0P|n\rangle$$

$$= \sum_{k_1,k_2} \langle m|k_1\rangle\langle k_1|H_0|k_2\rangle\langle k_2|n\rangle$$

$$= \delta_{m,k_1}\langle k_1|H_0|k_2\rangle\,\delta_{k_2,n}$$

$$= \langle m|H_0|n\rangle$$

$$= \sum_{k,k'} \langle m|H|k\rangle \langle k|(E - \Theta H \Theta)^{-1}|k'\rangle \langle k'|H|n\rangle$$

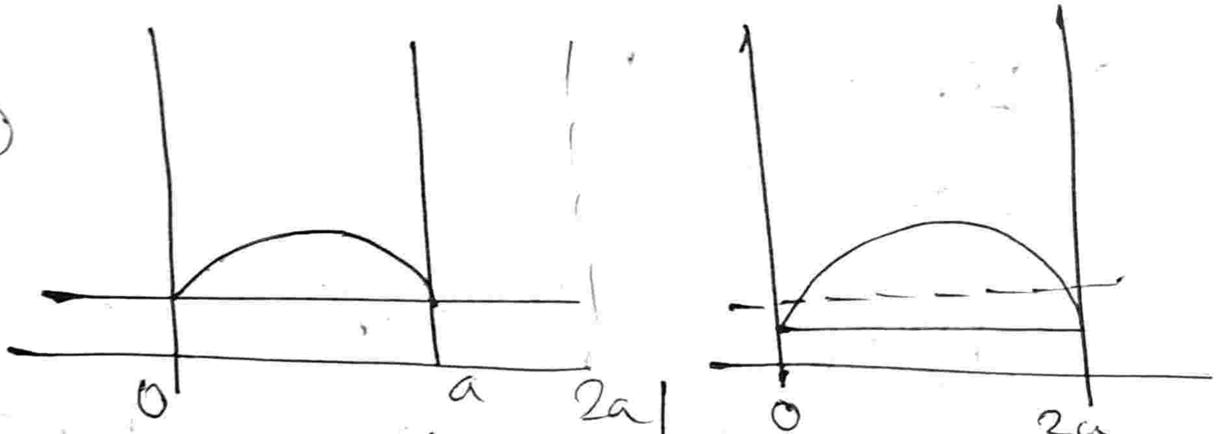
$$= \delta_{m,k_1} \langle k_1|H_0|k_2\rangle \delta_{k_2,n}$$

$$= \langle m|H_0|n\rangle$$

$$= \sum_{k,k'} \langle m|H|k\rangle \langle k|(E - \Theta H \Theta)$$

$$= \sum_{n} \langle m|H|k\rangle \{E - E(P)\}^{-1} \langle k|H|n\rangle$$

(2)



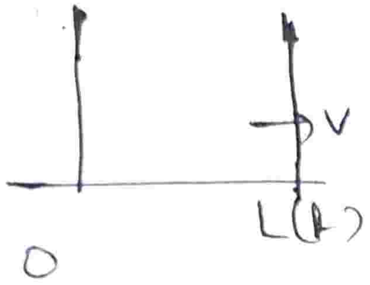

$\psi_1(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{\pi x}{a}\right)$, | $\psi_2(x) = \sqrt{\frac{2}{2a}} \sin\left(\frac{\pi x}{2a}\right)$

$E = \frac{\hbar^2}{2m}\left(\frac{\pi}{a}\right)^2$

$= |\langle \psi_i(x) | \psi'(x)\rangle|^2 = \left| \int_0^a \psi_1(x)\psi'(x)dx + 0\right|^2$

$= \left|\int_0^{2a} \psi_i(x \in (a,a])\, \psi'(x)dx\right|^2 = \left|\sqrt{\frac{2}{a}}\sqrt{\frac{2}{2a}} \int_0^a \sin\left(\frac{\pi x}{a}\right) \sin\left(\frac{\pi x}{2a}\right)\right|^2 = \frac{32}{9\pi^2}$

(b)




$$L(t) = a + \left(\frac{t}{T}\right)a$$

$$v \frac{dL}{dt} = \left(\frac{a}{T}\right)$$

$$\xi = \frac{x}{L(t)} \quad \Bigg| \quad -\frac{\hbar^2}{2m}\frac{\partial^2 \Psi(x,t)}{\partial x^2} = \frac{\partial \Psi(x,t)}{\partial t}$$

$$\Psi(x,t) = \frac{1}{\sqrt{L(t)}}\phi(\xi,t)$$

$$\frac{\partial \Psi}{\partial x}, \frac{\partial^2 \Psi}{\partial x^2}, \frac{\partial \Psi}{\partial t}$$

(c) $L(t) = a + \frac{t}{T}a$

$T \to \infty$, $v = \frac{a}{T} \to 0$

$$|\Psi(x,t)\rangle = e^{i(\theta+\gamma)}|\psi(x,t)\rangle$$

Dynamical phase: $\theta(t) = -\frac{1}{\hbar}\int_0^t E(t)\,dt$

Berry phase: $\gamma = i\langle\psi(x,t)|\frac{\partial}{\partial x}\psi(x,t)\rangle$

$$\theta(t) = -\frac{1}{\hbar}\int_0^T \frac{\hbar^2\pi^2}{2mL(t)^2}\,dt$$

$$= -\frac{1}{\hbar}\frac{\hbar^2\pi^2}{2ma^2}\int_0^T \frac{dt}{\left(1+\frac{t}{T}\right)^2}$$

$$= -\frac{1}{\hbar}\frac{\hbar^2\pi^2}{2ma^2}(-T)\left[\frac{1}{1+\frac{t}{T}}\right]_0^T$$

$$= -\frac{\hbar\pi^2 T}{4ma^2}$$

$$\gamma = i\int_0^t \left\langle \Psi(x,t) \middle| \frac{\partial \Psi(x,t)}{\partial t}\right\rangle dt$$

$$\left[\frac{\partial \Psi(x,t)}{\partial t} = \frac{\partial \Psi(R(t))}{\partial R}\cdot\frac{\partial R}{\partial t}\right]$$

$$= \int_{R(0)}^{R(t)} \left\langle \Psi(R) \middle| \frac{\partial \Psi}{\partial R}\right\rangle \frac{\partial R}{\partial t}\, dt$$

$$= i\int_{R(0)}^{R(t)} \left\langle \Psi(R) \middle| \frac{\partial \Psi}{\partial R}\right\rangle dR$$

$$\Psi_{gs}(x,t) = \sqrt{\frac{2}{L(t)}}\sin\left(\frac{\pi x}{L(t)}\right)$$

$$\frac{\partial |\Psi_{gs}\rangle}{\partial t} = \sqrt{2}\{L(t)\}^{-3/2}\left(-\frac{1}{2}\right)\sin\left(\frac{\pi x}{L(t)}\right)\dot{L}(t) +$$ [P.T.O]

$$+\sqrt{\frac{2}{L(t)}}\sin\left(\frac{\pi x}{L(t)}\right)\left\{-\frac{\pi x}{\{L(t)\}^2}\dot{L}(t)\right\}$$

$$\left\langle \psi_{gs} \middle| \frac{\partial \psi_{gs}}{\partial t} \right\rangle = \int_0^{L(t)} dx\, \psi^*_{gs} \frac{\partial \Psi_{gs}}{\partial t} = 0$$

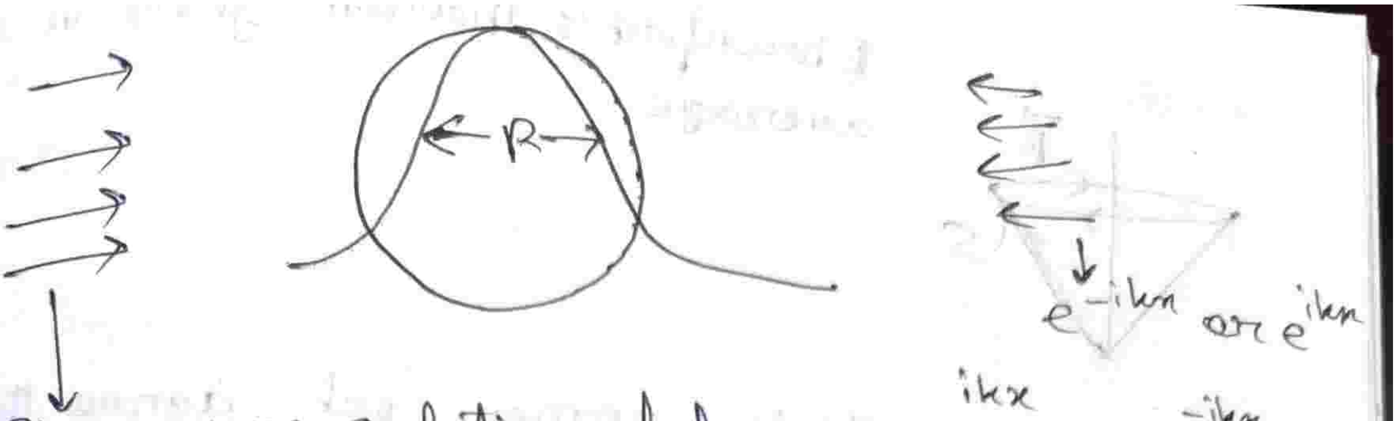
Advanced Quantum Mechanics 25/02/2026

## Scattering Theory :

Delta potential is an extremely ~~ex~~ localised potential in space.

We have to develop a notion of "farness" from this potential — where "free particles" form good basis.

Scattering states are "far away" from the potential → free states, not bound.



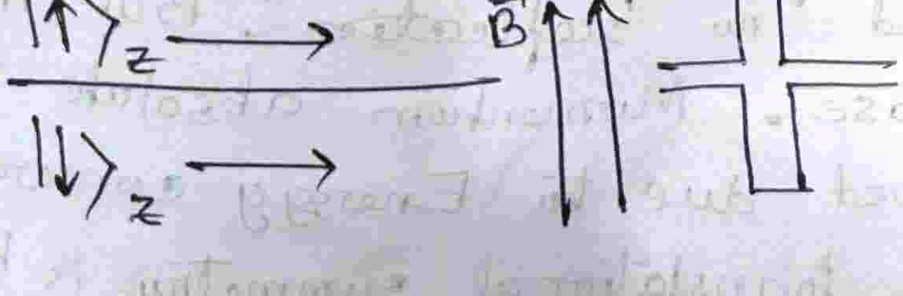

Plane wave solution of form $e^{ikx}$ or $e^{-ikx}$. And in between them, some "unitary" operation happens $\rightarrow$ Intensity of beam is conserved. Elastic collisions ensure energy conservation.

So, $\frac{(\hbar k)^2}{2m}$ remains conserved (for elastic collisions). We will consider only elastic situations for now.

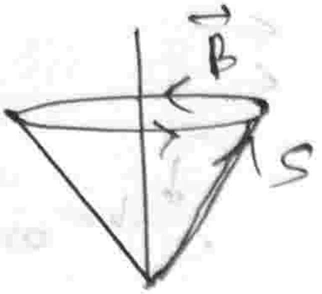
There can be additional quantum numbers we could associate with the incident wave (due to properties like intrinsic angular momentum).

$\frac{1}{\sqrt{2}}(|\uparrow\rangle_z + |\downarrow\rangle_z)$

One spin will see a well, another a barrier (Zeeman split)



Ehrenfest's theorem gives us the average




Now the particle comes out across the potential, the spin quantum number on the outcoming wavefunction tells us the ~~shap~~ form of the potential barrier.

So : $\underline{\text{incoming}}$ $\qquad\qquad$ $\underline{\text{Outcoming}}$

$$\frac{1}{\sqrt{2}}\left(|\uparrow\rangle_z + |\downarrow\rangle_z\right) \longrightarrow \cancel{\alpha|\alpha} \; \alpha|\uparrow\rangle_z + \beta|\downarrow\rangle_z$$

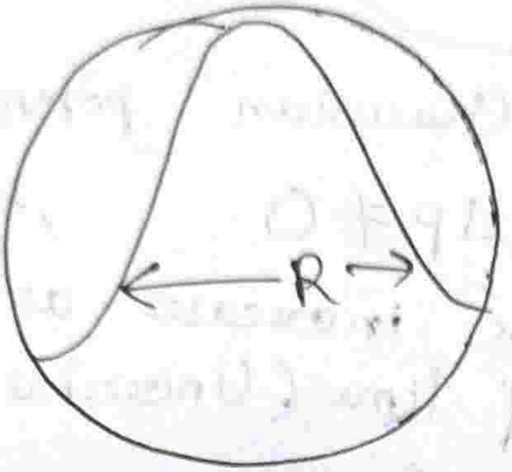
Exactly 50% probability of either state.

$[\alpha, \beta$ arbitrary$]$

But, $|\alpha|^2 + |\beta|^2 = 1$

One information we can extract from the scattering problem, one is intensity

And another is $\left|\frac{\alpha}{\beta}\right| = z = re^{i\theta}$. Here, we assumed "no reflection". But this is not the case. Momentum absolute value is conserved due to Energy conservation. But ~~space~~ translational symmetry is broken, hence $k$ ~~is~~ can go to $-k$ (reflection happens).

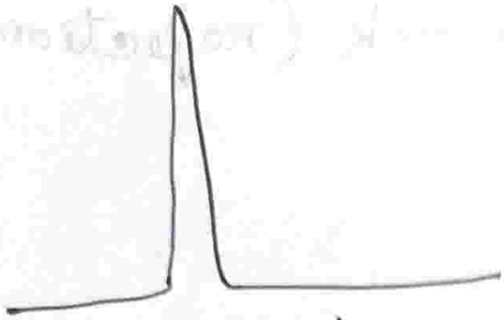


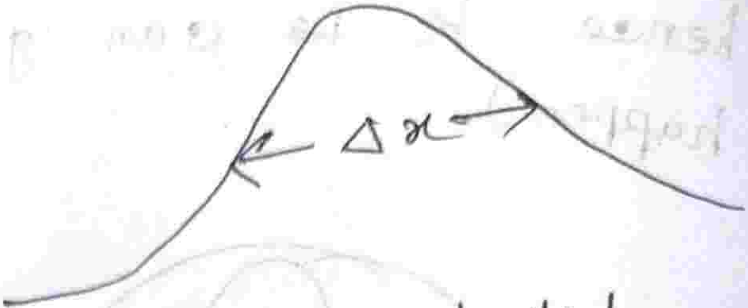

Now, possible plane wave form ~~equation~~ solution: $te^{ikx}$, $re^{-ikx} + e^{ikx}$

[Intuition: $e^{-ikx}$ "reflects" off, goes back in reverse direction, $e^{ikx}$ "transmitted" along same direction as incident wave].

We do we not talk about time here?
Because we are dealing with time averaged quantities (i.e., intensities) here. So, we are basically dealing with currents here.

A particle does not go backward and forward in a single particle. Fraction of experiment that gives a forward/backward outgoing particle, tells us the corresponding probability.






δ- potential

$\langle p \rangle = 0$

[Particles accelarated by electric and magnetic fields]

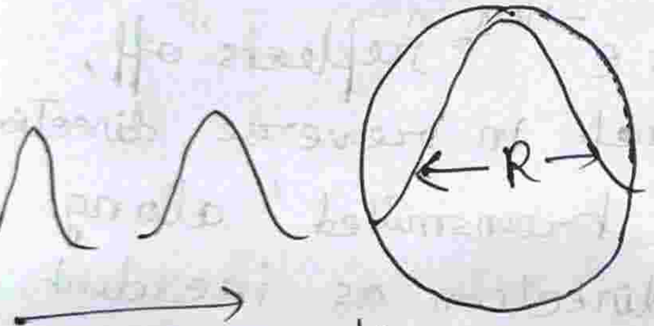
Gaussian potential

$\Delta p \neq 0$ , $\Delta x \neq 0$

$\Delta x$ increases as a function of time (Uncertainty principle)

$e^{-i\hat{H}t/\hbar}|x_0\rangle \rightarrow \delta(x-x_0)$

(We take momentum basis, position basis gives weird situation, i.e. being at both sides of potential with equal probability)

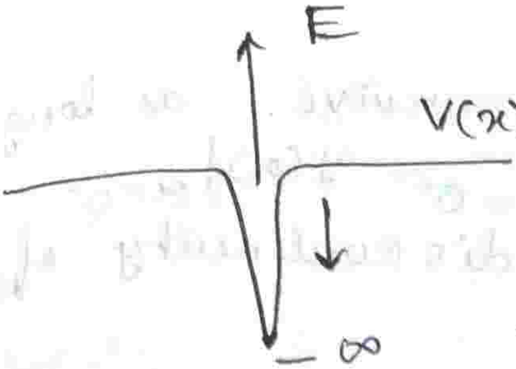



Width expands

What makes scattering still valid?

$\Rightarrow \lambda << R$ (Range of potential).

Entire scattering scenario happens at +ve energies, because -ve energy particles are completely localised.




$V(x) = -a\delta(x)$

The S.E.

$$-\frac{\hbar^2}{2m}\frac{d^2}{dx^2}\psi(x) - a\delta(x)\psi(x) = -|E|\psi(x)$$

$\Rightarrow$ This S.E. has two solutions, bound state and scattering state solutions.

- $-$ve Energy Solutions: $(e^{-ikx})$

$$\psi(x) = Ae^{-kx} + Be^{+kx} \quad , x<0$$

$$= Ce^{-kx} + De^{+kx} \quad , x>0$$

$$-E = \frac{p^2}{2m} \Rightarrow p = i\sqrt{2Em}$$

Here, as $-$ve energy solutions, at, $x = \pm\infty$, $\psi(x) = 0$. This gives us: $A = D = 0$.

$$-\frac{\hbar^2}{2m}\int_{-\varepsilon}^{+\varepsilon}\frac{d}{dx}\left(\frac{d\psi}{dx}\right)dx - a\int_{-\varepsilon}^{+\varepsilon}\delta(x)\psi(x)dx = -|E|\int_{-\varepsilon}^{+\varepsilon}\psi(x)dx$$

$$\Rightarrow -\frac{\hbar^2}{2m}\left[\frac{d\psi(x)}{dx}\bigg|_{x=+\varepsilon} - \frac{\partial\psi(x)}{\partial x}\bigg|_{x=-\varepsilon}\right) - a\psi(x=0) = -|E|\underbrace{\int_{-\varepsilon}^{+\varepsilon}\psi(x)dx}_{0}$$

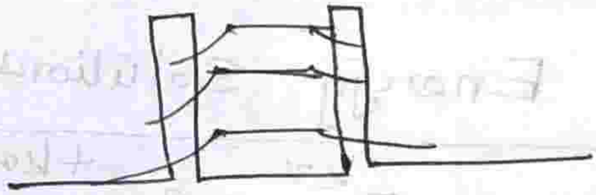
$$\int_{-\epsilon}^{+\epsilon} \psi(x)\, dx$$ does not survive as long

[**Because:** $\psi(x)|_{x=0^+} = \psi(x)|_{x=0^-}$]

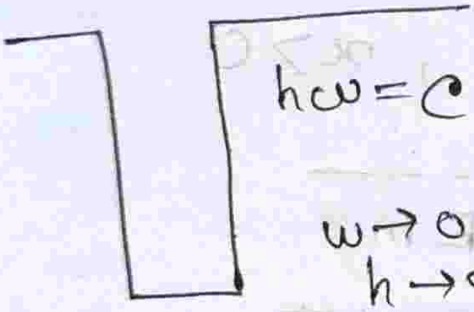
as there is not a discontinuity of $\infty$ strength.

Dirac equation puts space and time in equal footing by using 1st derivative of space, unlike Schrodinger equation.

$$E = -\frac{ma^2}{2\hbar^2}$$



Quasi-bound state



$h\omega = C$

$\omega \to 0$

$h \to \infty$

$\boxed{\text{1-}\overset{ve}{d}\text{ Delta potential has 1 bound state.}}$

→ Levinson's Theorem

$$\psi_R(x) = \begin{cases} e^{ikx} + r e^{-ikx}, & x \to -\infty \\ t e^{ikx}, & x \to \infty \end{cases}$$

$$|t|^2 + |r|^2 = 1, \quad |t(a,k)| = |t(-a,k)|$$

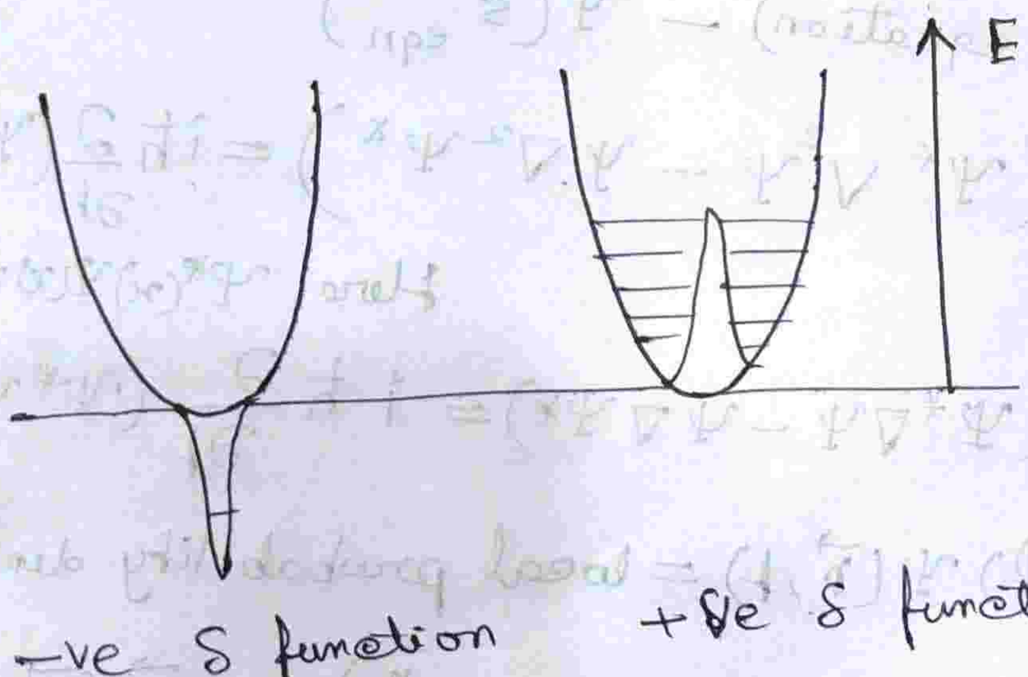
⇒ It is not possible to infer from transmission coefficient the sign of potential.

But the difference comes in with the argument : $\text{Arg}\left[f(a,k)\right] \neq \text{Arg}\left[f(-a,k)\right]$

Density of states [distribution of states in energy] changes from +ve to −ve delta function.

Exercise: 1

$$\int_0^{\infty}\left(\rho(\varepsilon)\Big|_{a\to +ve} - \rho(\varepsilon)\Big|_{a\to -ve}\right) d\varepsilon = 1$$




−ve $\delta$ function +ve $\delta$ function

Exercise 2 : Show phase shift between $f(a,k)$, $f(a,-k)$ as $k\to 0$, is $\pi$.

Continuity equation for probability

$$\int_{-\infty}^{\infty} |\Psi(x,t)|^2 dx = 1 \rightarrow \text{independent of time}$$

$$-\frac{\hbar^2}{2m}\nabla^2\Psi + V(\vec{r})\,\Psi = i\hbar\frac{\partial \Psi}{\partial t} \rightarrow \text{Complex conjugation}$$

$$-\frac{\hbar^2}{2m}\nabla^2\Psi^* + \underbrace{V(\vec{r})}_{\text{Real}}\Psi^* = -i\hbar\frac{\partial \Psi^*}{\partial t}$$

* Real / imaginary depends on momentum / position representation.

$$\Psi^*(S \text{ equation}) - \Psi(S \text{ eqn})^*$$

$$= -\frac{\hbar^2}{2m}\left(\Psi^*\nabla^2\Psi - \Psi\nabla^2\Psi^*\right) = i\hbar\frac{\partial}{\partial t}\left(\Psi^*_{(x)}\Psi_{(x)}\right)$$

Here, $\Psi^*(x)\Psi(x) = P(x)$

$$-\frac{\hbar}{2mi}\vec{\nabla}\left(\Psi^*\nabla\Psi - \Psi\nabla\Psi^*\right) = i\hbar\frac{\partial}{\partial t}(\Psi^*\Psi)$$

$\Psi^*(\vec{r},t)\,\Psi(\vec{r},t)$ = local probability density

$$-\frac{i\hbar}{2m}\left(\Psi^*\nabla\Psi - \Psi\nabla\Psi^*\right) = \vec{J}(\vec{r},t)$$

Advanced Quantum Mechanics : 26/02/2026

The actual transmission/ reflection probability is the ratio of transmitted / reflected current to the incident current.

Plane waves are solutions of free particle problems. But for delta function potential, plane waves are not eigenstates! If they were, they had no need of evolving!

$$\int_{-\infty}^{\infty} \underbrace{|\Psi(x,t)|^2}_{P(x,t)} dx = \text{const. in time} = 1$$

$P(x,t) \to$ probability density.

Probability is like an incompressible fluid, currents possible.

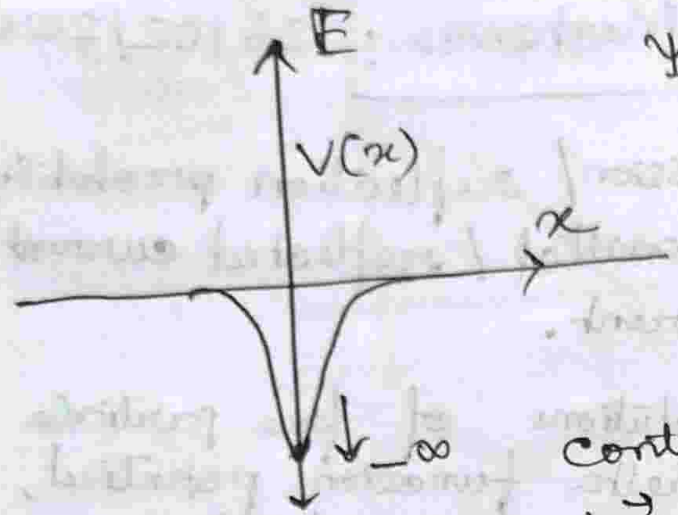
The probability being a function of time is telling us it is NOT an eigenstate. It is a "scattering state".

Probability current:

$$-\frac{i\hbar}{2m}(\Psi^* \nabla \Psi - \Psi \nabla \Psi^*) = \vec{J}(x,t)$$

$$\frac{\partial \rho}{\partial t} + \vec{\nabla}\cdot\vec{J} = 0 \leftarrow \text{Continuity equation.}$$

Exercise: R. Shankar. 5.3.1 → Discusses imaginary potential




$$\Psi(x) = \begin{cases} e^{ikx} + r e^{-ikx}, & x \to -\infty \\ t e^{ikx}, & x \to +\infty \end{cases}$$

As there is no time dependence, in 1-D, continuity equation becomes:

$$\frac{d\vec{J}}{dx} = 0. \Rightarrow \vec{J} \text{ stationary.}$$

$$\vec{J}(x) = \frac{-i\hbar}{2m}\Big[\big(e^{-ikx} + r^* e^{ikx}\big)\big((ik)e^{ikx} - r(ik)e^{-ikx}\big) - \big(e^{ikx} + e^{-ikx}\big)\big((-ik)e^{-ikx} + r^*(ik)e^{ikx}\big)\Big]$$

$$\Rightarrow \vec{J}(x) = \underbrace{\frac{\hbar k}{m}}_{\text{Dimension of velocity [group velocity]}}(1 - |r|^2) \rightarrow \text{independent of } x.$$

→ Dimension of velocity [group velocity ~~velocity~~]

$\Rightarrow$ $\frac{\hbar k}{m}$ going forward, $|r|^2 \frac{\hbar k}{m}$ going backward. $\therefore$ depending on resistance strength, $|r|$, current drops! Naturally.

~~If~~ Is $\rho$ a function of $x$? Yes!

$$\rho(x) = \Psi^*(x)\Psi(x) = \big(e^{ikx} + r e^{-ikx}\big)\big(e^{-ikx} + r^* e^{ikx}\big)$$

$$\boxed{\rho(x) = 1 + |r|^2 + 2|r|\cos(kx + \delta)}$$

$$\Rightarrow \delta = \text{Arg}(r)$$

If we pass through an impurity, what phase do we accumulate? ~~phase~~
extra phase accumulated:

- By reflection: $Z_1 e^{i\theta_1}$
- By transmission: $Z_2 e^{i\theta_2}$

Where does the oscillation term $2|r|\cos(kx+\delta)$ come in in the density formula? $\Rightarrow$ Interference of incident and reflected waves.

$\Rightarrow$ Friedel oscillations.

By looking at the extra-phase accumulated, we can classify the symmetry of system and hence, classify the impurity that broke the said symmetry.

For delta potential, it is perfectly symmetric about 0.

$$V(1-|r|^2) = V(|t|^2) \Rightarrow |r|^2 + |t|^2 = 1$$

But if $V_L \neq V_R$, $|r|^2 + |t|^2 \neq 1$!

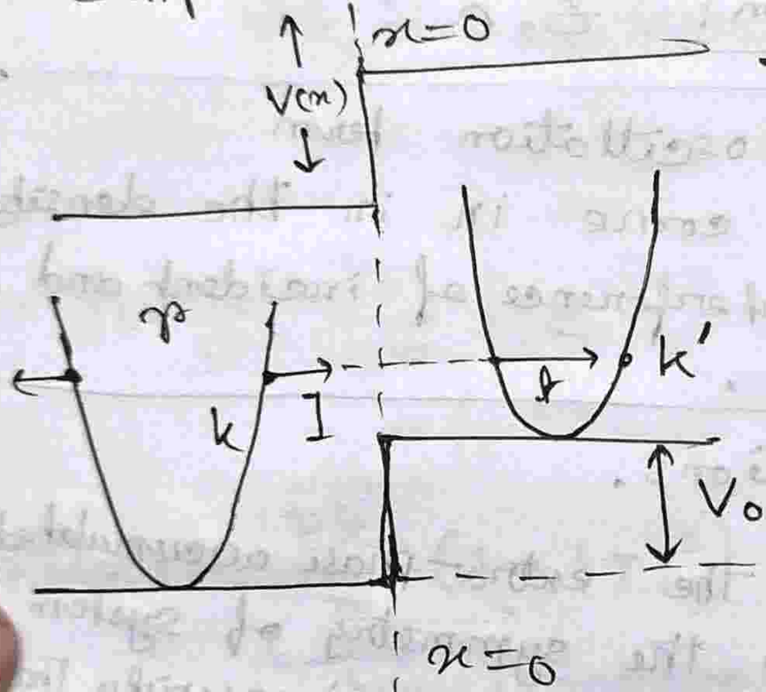
Because, $|r|^2$, $|t|^2$ are NOT probabilities.

Group velocity is : $V_g = \frac{dE}{dk}$

Density of states : $\rho = \frac{dk}{dE}$

$\therefore$ V is equivalent to D.O.S. mismatch.

Simplest example: $(V_L \neq V_R)$



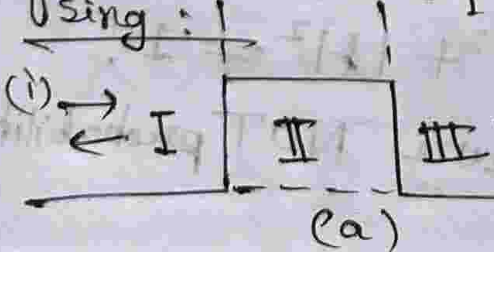
Reflection probability will still ~~have~~ be interpreted by $|r|^2$, but $|t|^2$ will not be interpreted as transmission probability.

Because, tangents have equal slopes for both incident and reflected wave, not for transmitted one.

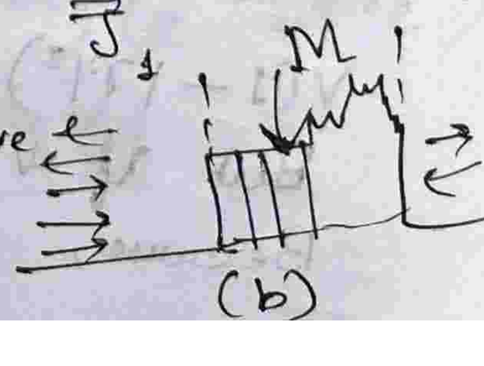
$$\therefore T = \frac{\vec{J}_T}{\vec{J}_I} \quad ; \quad R = \frac{\vec{J}_R}{\vec{J}_I}$$

Using :

(i)



How do we solve



Here we use the scattering matrix!
We split the irregular potential into small patches $\xrightarrow{(b)}$ each patch represented by a matrix M [Transform matrix].

If we multiply all these matrices sequentially, we obtain the ~~fin~~ outcoming planewaves, if we know the incoming plane waves.

$$\Psi(x) = \begin{cases} Ae^{ikx} + Be^{-ikx} & \to \text{I} \\ Ce^{-kx} + De^{kx} & \to \text{II} \text{ in Fig(a)} \\ Fe^{ikx} + Ge^{-ikx} & \to \text{III} \end{cases}$$

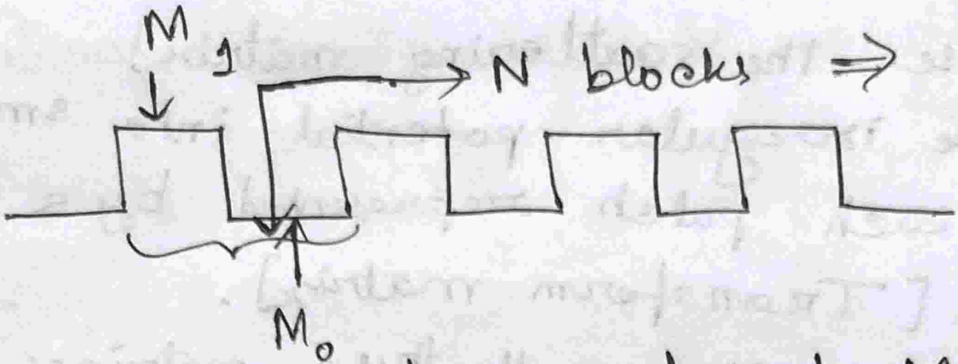
↳ 2nd order ODE, 2 b.c.'s required, we are provided with 2 b.c's for I, III.

- Scattering matrix:

$$\begin{pmatrix} A \\ B \end{pmatrix} = \begin{pmatrix} M_{11} & M_{12} \\ M_{21} & M_{22} \end{pmatrix} \begin{pmatrix} F \\ G \end{pmatrix} \leftarrow$$

This is not scattering matrix. This simply connects amplitude

We are looking for general solutions devoid of preferred direction, i.e. waves can come and go out through either left or right. This matrix connects each incoming-outgoing amplitude pair.




N blocks $\Rightarrow (M_1 M_0)^N$ is the transform matrix.

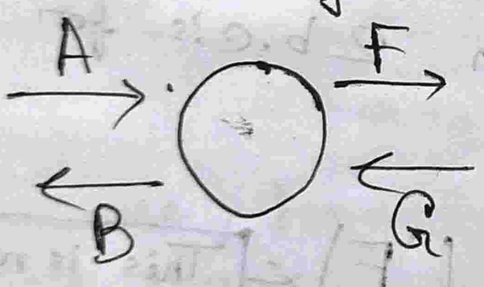
Free space has a trivial M-matrix, $r = 0$, $t = 1$.

## Actual Scattering matrix:

$$\begin{pmatrix} B \\ F \end{pmatrix} = \begin{pmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{pmatrix} \begin{pmatrix} A \\ G \end{pmatrix}$$

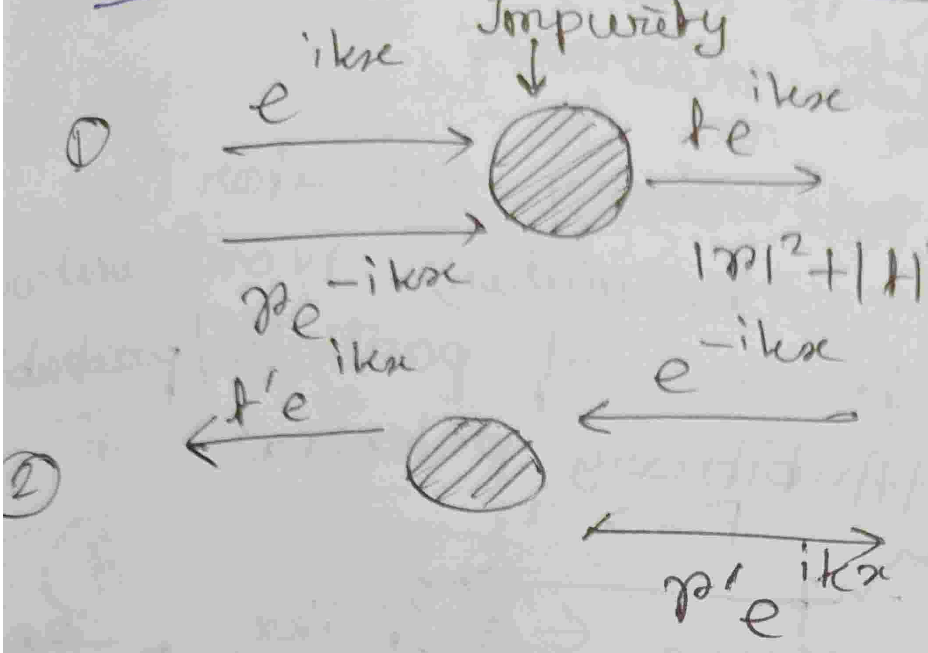
outgoing ~~outgoin~~ ~~incoming~~

incoming




This matrix carries symmetry information of system

# Advanced Quantum Mechanics 11/03/26




$$|r|^2 + |t|^2 = 1$$

$$|r'|^2 + |t|^2 = 1$$

∴ S-matrix :

$$S = \begin{bmatrix} r & t \\ t' & r' \end{bmatrix}$$

↓

Unitary matrix! → Entropy not generate

→ Acts on incoming amplitude gives outgoing amplitude.

## S-Matrix Transmissions

- Left to Right
- Right to Left
-

$t \rightarrow |t| e^{i\theta}$

$t' \rightarrow |t'| e^{i\theta'}$

$r \rightarrow |r| e^{i\phi}$

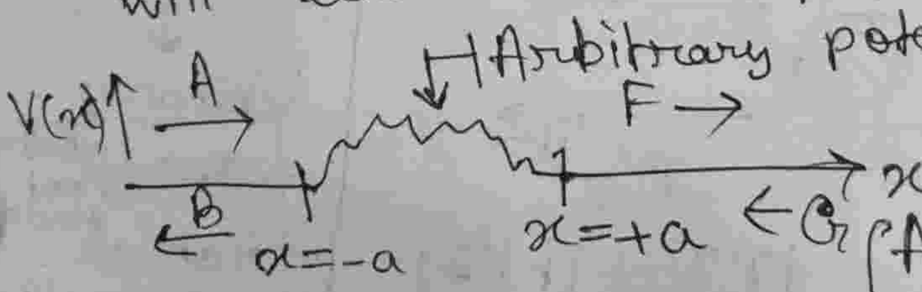
$r' \rightarrow |r'| e^{i\phi'}$

<u>Recall:</u> If an operator is Unitary :

$$U^{+} = U^{-1}$$

A possible representation of this is: $e^{i\alpha M}$ where $\alpha$ is real, M is a Hermitian matrix.

$\therefore e^{-iHt}$ is an unitary operator (as long as H is hermitian). Non-unitarity will lead to loss of particles/probability.




Solution to this will roughly look like:

$$\Psi(x) = \begin{cases} Ae^{ikx} + Be^{-ikx}, & x < -a \\ Ce^{-kx} + De^{kx}, & -a < x < a \\ Fe^{ikx} + Ge^{-ikx}, & a < x \end{cases}$$

The idea is to take the incoming modes of amplitudes A and G, and to act on them with scattering matrix.

$$\begin{pmatrix} B \\ F \end{pmatrix}^{out} = \begin{pmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{pmatrix} \begin{pmatrix} A \\ G \end{pmatrix}^{in}$$

Current in region $x < -a$ ; $\underbrace{-a < x < a}_{\text{not considered now}}$ ; $x > a$

$$J = v \{ |A|^2 - |B|^2 \}$$

$$= v \{ |F|^2 - |G|^2 \} \Rightarrow |A|^2 - |B|^2 = |F|^2 - |G|^2$$

$$\Rightarrow |A|^2 + |G|^2 = |F|^2 + |B|^2$$

The $v$'s cancel out only if potential goes to zero at infinities. Otherwise, as in case of an $\infty$ - extending potential such as step potential, $v$'s are not equal even at $\pm \infty$.

There is a row vector version to this.

The row vector version is:

$$(B \quad F) = (A \quad G)\begin{pmatrix} S_{11} & S_{21} \\ S_{12} & S_{22} \end{pmatrix}$$

↳ S-transpose.

↓ Take complex conjugate

$$(B^* \quad F^*) = (A^* \quad G^*)\, S^{\dagger} \quad \text{—} \quad (2)$$

$$\begin{pmatrix} B \\ F \end{pmatrix} = S \begin{pmatrix} A \\ G \end{pmatrix} \quad \text{—} \quad (1)$$

Multiplying (1) and (2) sidewise:

$$(B^* \quad F^*)\begin{pmatrix} B \\ F \end{pmatrix} = (A^* \quad G^*)\, S^{\dagger} S \begin{pmatrix} A \\ G \end{pmatrix}$$

$$\Rightarrow |B|^2 + |F|^2 = (A^* \quad G^*)\, S^{\dagger} S \begin{pmatrix} A \\ G \end{pmatrix}$$

Current conservation gives LHS is $|A|^2 + |G|^2$

$$\Rightarrow |A|^2 + |G|^2 = (A^* \quad G^*)\, S^{\dagger} S \begin{pmatrix} A \\ G \end{pmatrix}$$

$$\Rightarrow S^{\dagger} S = \mathbb{I} \rightarrow \text{Unitary S-matrix.}$$

Lack of unitarity means entropy generation — ~~lack~~ loss of information. Decoherence can not be written as

$e^{ikx}$ because this is invertible in nature, but loss of information to nature is mathematically not invertible.

$$S^\dagger S = \mathbb{I} \quad \left( S = \begin{pmatrix} r & t \\ t' & r' \end{pmatrix}, \; S^\dagger = \begin{pmatrix} r^* & (t')^* \\ t^* & (r')^* \end{pmatrix} \right)$$

$$\Rightarrow \begin{pmatrix} |r|^2 + |t|^2 & t r^* + t'^* r' \\ & |r|^2 + |t|^2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Only two phases, are independent, rest of the phases depend on each other. We now look at the symmetry.

- <u>Parity</u> $(x \to -x)$

$$\Psi(x) = \begin{cases} Ae^{ikx} + Be^{-ikx}, & x < -a \\ Ce^{-kx} + De^{kx}, & -a < x < a \\ Fe^{ikx} + Ge^{-ikx}, & x > a \end{cases}$$

$\downarrow x \to -x$

$$\Psi(-x) = \begin{cases} Ae^{-ikx} + Be^{ikx}, & x > a \\ Ce^{kx} + De^{-kx}, & -a < x < a \\ Fe^{-ikx} + Ge^{ikx} & (-a > x) \end{cases}$$ | Both sol$^n$ and domains flip.

$$\begin{pmatrix} F \\ B \end{pmatrix} = \begin{pmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{pmatrix} \begin{pmatrix} G \\ A \end{pmatrix} \to \begin{pmatrix} B \\ F \end{pmatrix} = \begin{pmatrix} S_{22} & S_{21} \\ S_{12} & S_{11} \end{pmatrix} \begin{pmatrix} A \\ G \end{pmatrix} \Rightarrow S_{11} = S_{22}$$

$\therefore$ Probability of reflection from both sides is equal, so is the phase picked up while being reflected in either direction. $S_{11} = S_{22}$

Time Reversal (Potential is real, unlike $\sigma_y$ when $\vec{B}$

$$\psi(x) = \begin{cases} Ae^{ikx} + Be^{-ikx} & (x < -a) \\ Ce^{-kx} + De^{kx} & (-a < x < a) \\ Fe^{ikx} + Ge^{-ikx} & a < x \end{cases}$$

$\Big\downarrow$ Time reversal

$$\psi_1(x) = \begin{cases} A^* e^{-ikx} + B^* e^{ikx} & (-a > x) \\ C^* e^{-kx} + D^* e^{kx} & (-a < x < a) \\ F^* e^{-ikx} + G^* e^{ikx} & (a < x) \end{cases}$$

$$\begin{pmatrix} A^* \\ G^* \end{pmatrix} = \begin{pmatrix} S_{11}^{\equiv r} & S_{12}^{\equiv t} \\ S_{21}^{\equiv t'} & S_{22}^{\equiv r'} \end{pmatrix} \begin{pmatrix} B^* \\ F^* \end{pmatrix}$$

$$\begin{pmatrix} B \\ F \end{pmatrix} = \begin{pmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{pmatrix} \begin{pmatrix} A \\ G \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} B \\ F \end{pmatrix} = SS^* \begin{pmatrix} B \\ F \end{pmatrix} \Rightarrow S_{12} = S_{21}^*$$

Time reversal symmetry does not constrain reflection, it constrains transmission. All information about symmetry is included in phase. Phase picked up from left to right is same as the one picked up from right to left.

- $|r|^2 + |t|^2 = 1$ does not carry any symmetry information, it simply arises from current conservation.

$$\boxed{\begin{aligned} t &\Rightarrow S_{12} = S_{21} \rightarrow \text{time reversal} \\ r &\Rightarrow S_{11} = S_{22} \rightarrow \text{Parity} \end{aligned}}$$

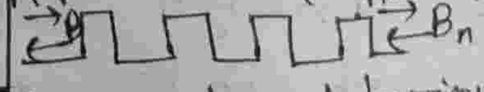
$\hookrightarrow \begin{pmatrix} r & t \\ t & r \end{pmatrix} \equiv$ Scattering matrix of a system with both Parity and time reversal symmetries.

**Extra**

CPT → Charge, parity, time-reversal symmetry together → System invariant under any transformation.

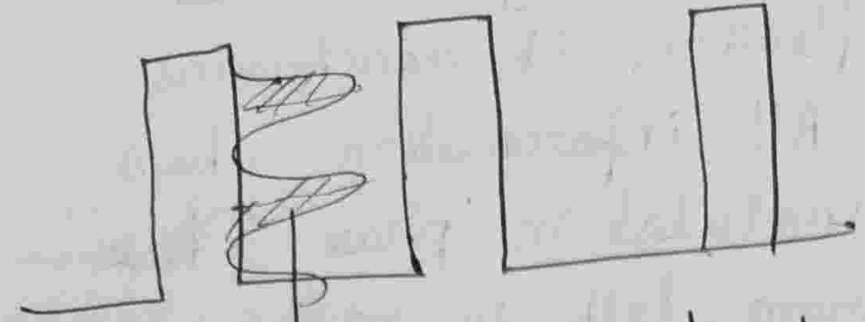
Transform matrix (M)

$$\begin{pmatrix} F \\ G \end{pmatrix} = \begin{pmatrix} M_{11} & M_{12} \\ M_{21} & M_{22} \end{pmatrix} \begin{pmatrix} A \\ B \end{pmatrix}$$



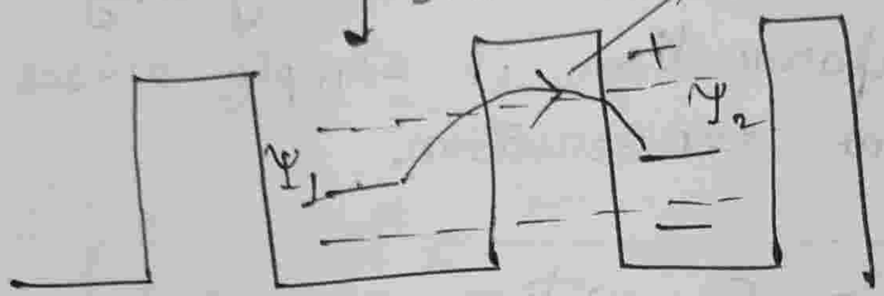

$$\begin{pmatrix} A_n \\ B_n \end{pmatrix} = M^n \begin{pmatrix} A \\ B \end{pmatrix}$$

n = number of barriers.



→ DOS Localised here

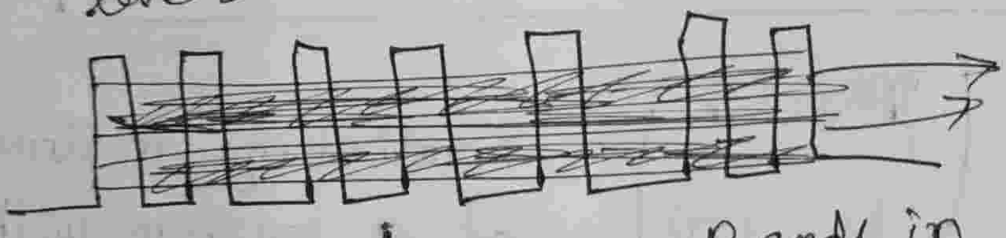
↓ Instead, Tunnelling




$$\Psi_+ = \Psi_1 + \Psi_2$$

$$\Psi_- = \Psi_1 - \Psi_2$$

Level splitting delocalised levels → non-degenerate.



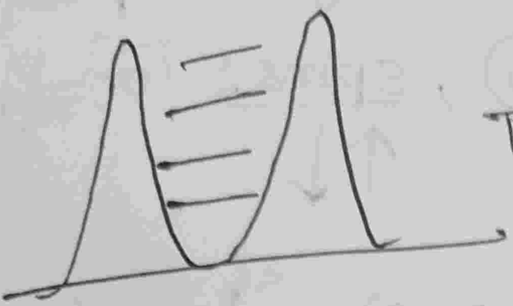
→ Bandwidth

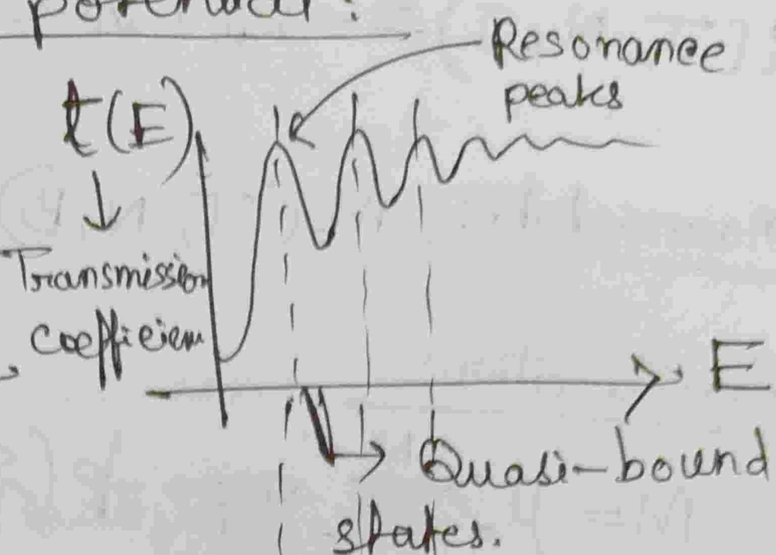
Bands in solids

- Three overlapping orbitals → Three such bands!

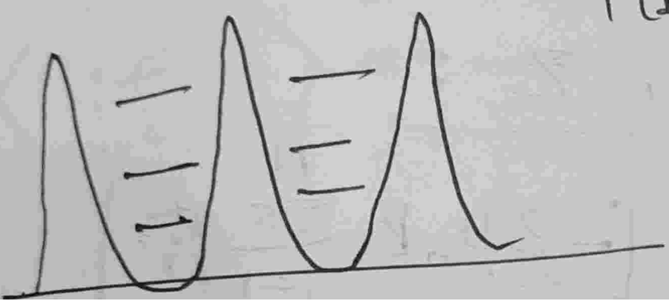
Now → scattering approach. If I shoot a particle from left, will it come out through left? If yes, it is metal. If not, it is insulator.

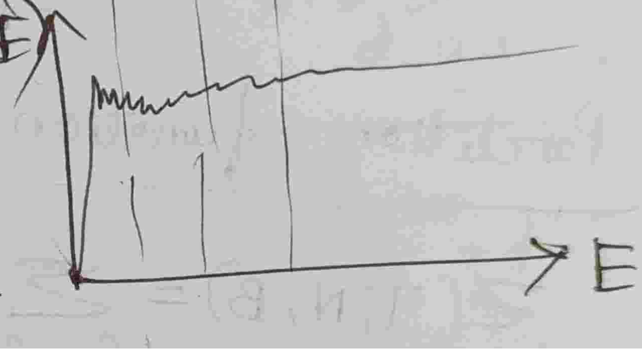
## Double delta potential:





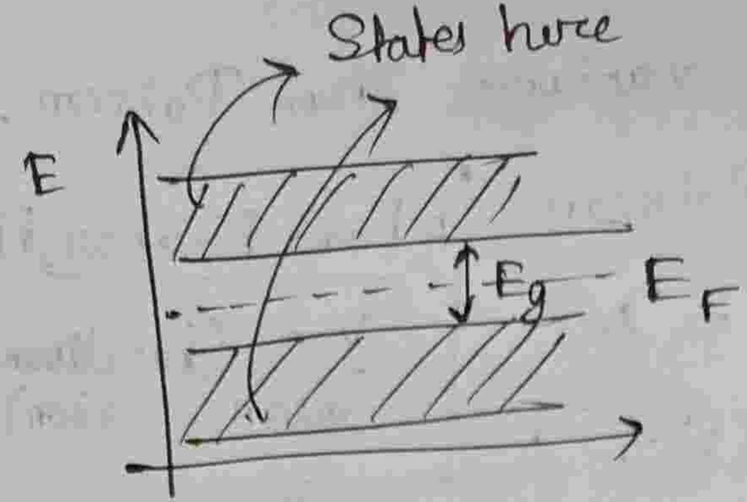

when we add such barriers,



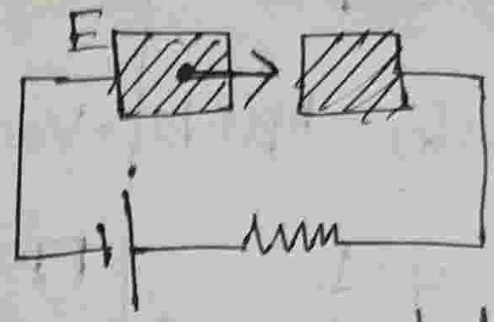


Advanced Quantum Mechanics : 12/03/26

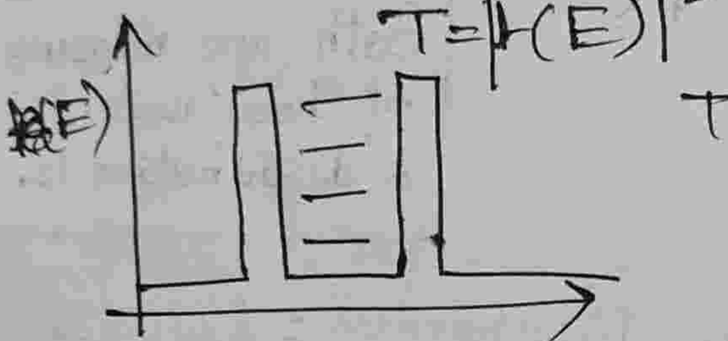
Suppose a Bulk Solid → piece of metal. We are interested in transmission coefficient, but that requires guarantee of phase coherence. Scattering Theory gives us how much energy of "shooting" gives us a finite transmission coefficient and which gives us infinite.

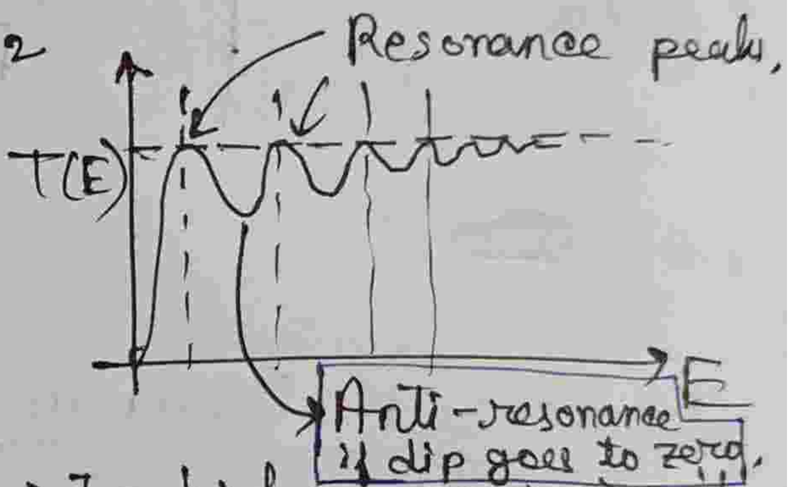



Question is, if we shoot an $e^-$ through a gap:




So, looking at $|t|^2$ tells us what amount of current passes.

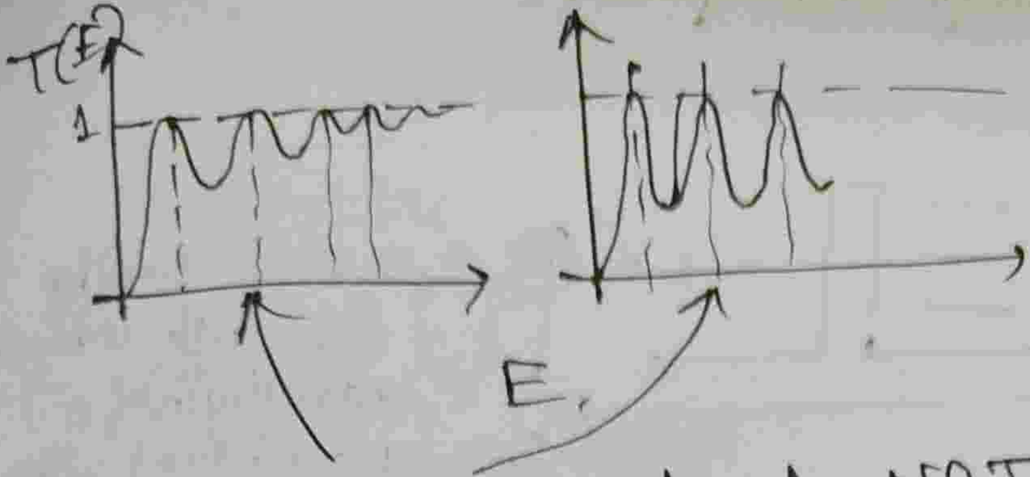



↳ In higher energies, it is classical behavior, because the electron being shot does not even feel the impurity, i.e. height of barrier is way less than $e^-$ energy.

So, the modes are defined (because boundary conditions are fixed), barrier height does not affect energy location of resonance peak.

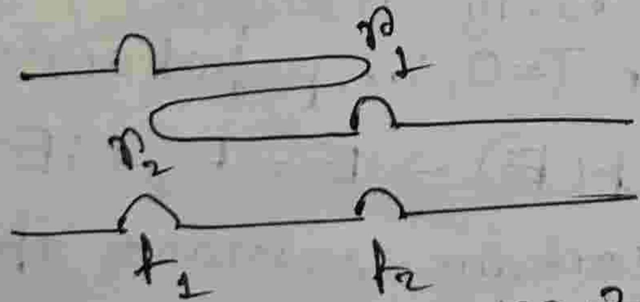
But if $h_2 >> h_1$




Resonance peaks do NOT change E-location. However, peaks become sharper. Lifetime of particle goes

$$\text{Line width} \propto \frac{1}{\text{Life time}}$$

Why do we get a situation where transmission is perfect? Due to interference. There are infinite no. of paths in which particle can pass through the gap.

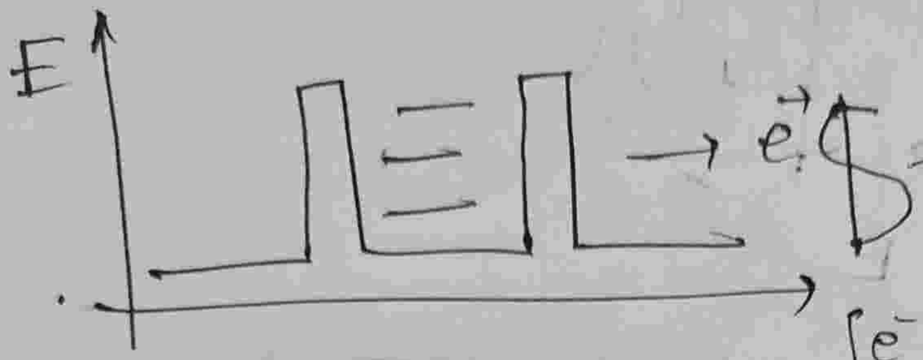



← Reflected twice, shot through twice

← Shot through twice

$|r_1|^2+|r_2|^2+|t_1|^2+|t_2|^2 \nRightarrow$ total probab.

All these paths either constructively or destructively interfere.




→ side cavity (1 full de-broglie wavelength) of $e^-$)
[$e^-$ quasi bound within cavity] → Finite lifetime within cavity

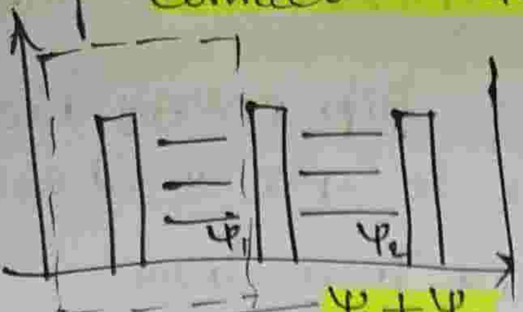
□ Problem Statement

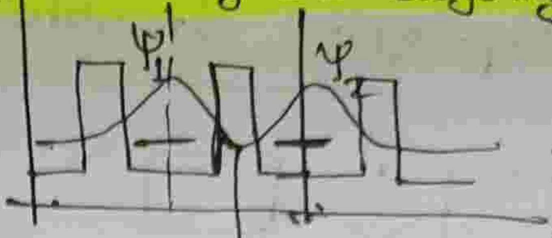
What do we need to introduce into the system such that there is perfect anti-resonance in 1-D?

Ans. We introduce a cavity off the electron channel such that $e^-$ has a lifetime there, and couple it with a channel so it imposes a boundary condition by its phase and kills off transmission amplitude. Now, if amplitudes of all paths (direct / indirect) from channel through cavity interfere destructively, we get $T=0$, i.e. perfect anti-resonance.

We use complex $t(E) \Rightarrow E = E_R + iE_I$, and poles of $t(E)$ contain resonance information, ∴ Taking $|t|^2$ is not necessary anymore.

The M-matrix from this unit gives wavefunctions connection between incoming and outgoing.



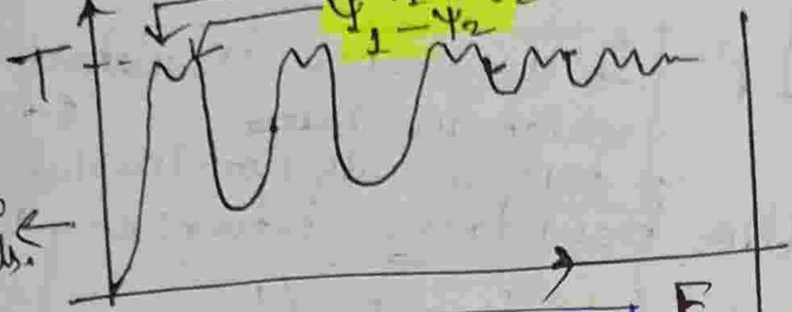

→ overlap of $\psi_1$, $\psi_2$ given by height of middle barrier.

Overlap gives energy splitting:

$\psi_1 \pm \psi_2 = \psi_{\pm}$ ⇒ extent of hybridization

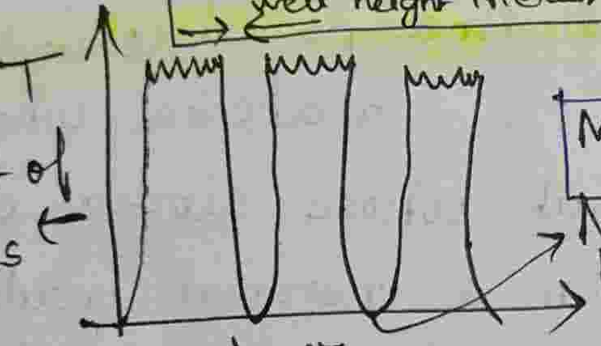
$\psi_1 + \psi_2$

$\psi_1 - \psi_2$




Two wells

decreases as middle well height increases




Lot of wells

More number of walls → more possibility of destructive interference.

No states here.

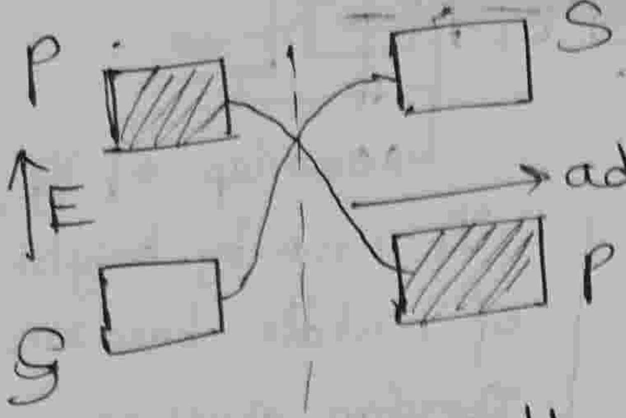
The lower the height of middle well, distance between each peak is lessened.

No radiation idea included because we are dealing with stationary states.

Atoms are wells in 3-D. Lot of such atoms are brought together to form a real solid. Their s-orbitals overlap to form s-band (energy splits), p-orbitals form p-bands

Topological Insulators: (Metal on surface)

(S, p flip energy hierarchy due to spin orbit coupling)




→ adiabatic transitions

It generates insulators which are ~~trivia~~ topologically non-trivial

↳ metallic boundary between

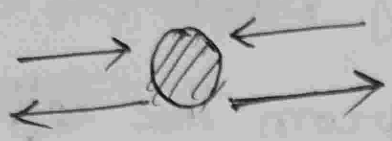
Quantized conductance/channel ← $\boxed{e^2/h}$

Summary: (Quantized conductance per edge channel $e^2/h$)

Topological insulator is a material whose bulk is insulating but whose surface or edge is metallic. In a normal insulator, the conduction and valence bands have a certain ordering (s ~~above~~ (maybe p over s). With spin-orbit coupling present, ordering inverts as parameters change adiabatically. Band gap closes temporarily, produces metallic level crossing. Gap reopens with inverted bands → material becomes topological insulator. At the boundary between this material and air/vacuum (trivial insulator) → band structures must connect → forces metallic edge states (channels). Quantized conductance $= e^2/h$.

## Advanced Quantum 13/03/2026

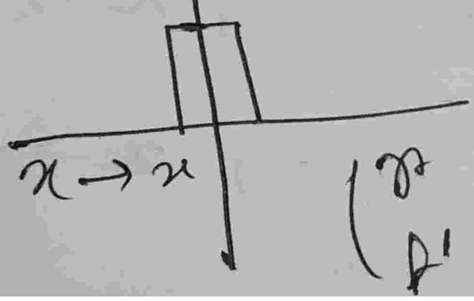
Scattering problem with parity symmetry.



→ $n \times n$ scattering matrix for $n$ incoming and outgoing modes.

Scattering matrix is unitary, hence if we diagonalize it, all my eigenvalues will be of the form $e^{i\theta}$

Simple example → square well → reflection symmetry




→ Scattering matrix should look diagonal.

$$\begin{pmatrix} r & t \\ t' & r' \end{pmatrix} \xrightarrow{\text{Parity}} \begin{pmatrix} r & t \\ t' & r \end{pmatrix}$$

Beam incident from both sides with 50% probability → then it scatters off.

↳ symmetric linear combination → parity state.

If parity is a good quantum number, parity channels do not talk to each other. Positive parity states do not scatter into negative parity states and vice-versa.

We have to prepare parity eigenstate as scattering eigenstate. If the impurity maintains parity, scattering matrix for the system will look diagonal in the parity basis.

For particle incident from the left:

$$\Psi_L = \begin{cases} e^{ikx} + r e^{-ikx}, & x \to -\infty \\ t e^{ikx}, & x \to +\infty \end{cases}$$

Incident from the right:

$$\Psi_R = \begin{cases} e^{-ikx} + r e^{-ikx}, & x \to +\infty \\ t e^{-ikx}, & x \to -\infty \end{cases}$$

□ <u>Notation</u>

Ingoing $\begin{cases} \text{Right-moving};\ I_R(x) = e^{+ikx}, & x \to -\infty \\ \text{Left-moving};\ I_L(x) = e^{-ikx}, & x \to +\infty \end{cases}$

Outgoing $\begin{cases} \text{Right-moving};\ O_R(x) = e^{ikx}, & x \to +\infty \\ \text{Left-moving};\ O_L(x) = e^{-ikx}, & x \to -\infty \end{cases}$

$$\begin{pmatrix} \Psi_L \\ \Psi_R \end{pmatrix} = \begin{pmatrix} I_R \\ I_L \end{pmatrix} + S \begin{pmatrix} O_R \\ O_L \end{pmatrix} \quad \Big| \quad \underset{\to S(k)}{S} = \begin{pmatrix} t' & r' \\ r & t \end{pmatrix}$$

$\boxed{\Psi(x,k) = \Psi^*(x,-k)}$ ⇒ scattering matrix as a function of momenta follows: $\boxed{S^*(k) = S(-k)}$

↳ Due to time-reversal symmetry.

↳ simply take complex conjugate of S.E., put $k \to -k$, it maps back to itself.

→ We can expect the Scattering matrix to be invariant.

□ <u>Parity symmetric Barrier:</u>

$\hat{V}(x) = \hat{V}(-x)$, $\hat{P}f(x) = f(-x)$

$\Rightarrow \hat{P}^2 f(x) = f(x) \Rightarrow \hat{P}^2 = \mathbb{1}$

$\hat{P}$ has 2 eigenvalues, $\pm 1$

$\Rightarrow \hat{P}^2 = \mathbb{1}$

$\Rightarrow (\hat{P} - \mathbb{1})(\hat{P} + \mathbb{1}) = 0 \Rightarrow \hat{P}$ has eigenvalues $\pm 1$

- $+1 \to$ even parity state
- $-1 \to$ odd parity state

A natural construction of eigenstate will be:

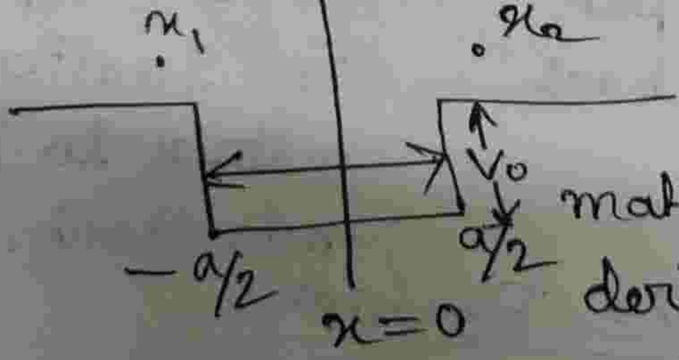
$$\Psi_+(x) = \Psi_R(x) + \Psi_L(x) = \Psi_R(x) + \Psi_R(-x)$$

$$\Psi_-(x) = -\Psi_R(x) + \Psi_L(x) = -\Psi_R(x) + \Psi_R(-x)$$

$$\therefore \hat{P}\Psi_+ = +\Psi_+ \; ; \; \hat{P}\Psi_- = -\Psi_-$$

We now do the scattering problem in the scattering eigenbasis, as $-1$ is nothing but $e^{i\pi}$, $\Psi_R(x)$ and $\Psi_L(x)$ has phase shift of $\pi$ (quite intuitive ng!).

Let us look at the problem:




When we use parity basis, we need to match solution and derivative only on one side.

$$q^2 = k^2 + \frac{2mV_0}{\hbar^2}$$

Positive parity state: $\Psi_+(x) = A(e^{iqx} + e^{-iqx})$

$x \in \left[-\frac{a}{2}, \frac{a}{2}\right]$

We notice: $\Psi_\pm(x) = A(e^{iqx} \pm e^{-iqx})$

- Matching of wavefunctions (at point in the right side);

$$\underbrace{e^{-ika/2} + r e^{ika/2}}_{\boxed{\Psi_R \text{ (Incident wave from right contributes)}}} + \underbrace{t e^{ika/2}}_{\boxed{\Psi_L \text{ (from left)}}} = A(e^{iqa/2} + e^{-iqa/2})$$

(We can similarly solve for $\Psi_-$, $(r-t)$)

- Matching derivatives;

$$\Rightarrow -e^{-ika/2} + (r+t)e^{ika/2} = \frac{q}{k}A(e^{iqa/2} - e^{-iqa/2})$$

(Solve for variables $(r+t)$ and $A$)

After diagonalizing scattering matrix, we should get eigenvalues $(r \pm t)$.

We should check if $(r+t)$ is just a phase.

$$r+t = (-e^{-ka})\,\frac{q\tan\left(\frac{qa}{2}\right) - ik}{q\tan\left(\frac{qa}{2}\right) + ik}$$

$$\frac{R - iR'}{R + iR'} = \frac{r' e^{i\theta'}}{r e^{i\theta}} \text{ ; where, } r' = r = \sqrt{R^2 + R'^2}$$

↳ RHS is just a phase. Scattering

→ ~~Only trans~~ problem is reduced to transmission problem.

$\Psi_- \xrightarrow{i\theta_-}$

$\Psi_-$ and $\Psi_+$ are parity channels that do not transmit onto one another

$\Psi_+ \xrightarrow{i\theta_+}$

means, $\langle \Psi_+ | S | \Psi_- \rangle = 0$, $\langle \Psi_- | S | \Psi_+ \rangle = 0$

Extra: (Reason)

$$\langle \Psi_+ | S | \Psi_- \rangle$$

$$= \langle \Psi_+ | P S P | \Psi_- \rangle \quad [P^2 = \mathbb{I}]$$

$$= (+1)(-1) \langle \Psi_+ | S | \Psi_- \rangle$$

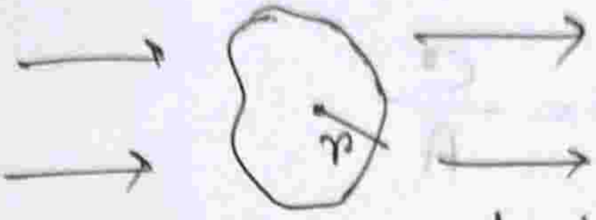
$$= - \langle \Psi_+ | S | \Psi_- \rangle$$

$$\Rightarrow \langle \Psi_+ | S | \Psi_- \rangle = 0$$

# Advanced Quantum Mechanics:

18/03/2026

## Tutorial:




Incoming $\qquad$ outgoing

$e^{-}$s $\quad e^{-}e^{+} \to 2\pi \quad e, \pi$

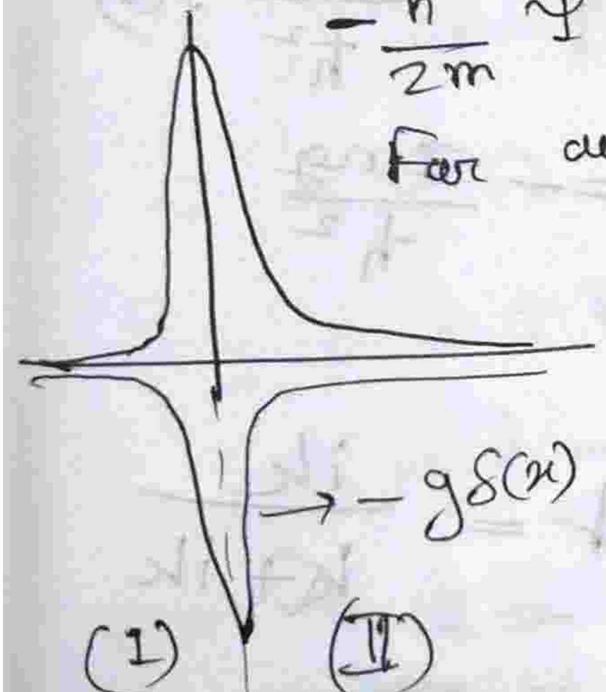
$\quad \hookrightarrow$ interaction

If range of interaction is "r", free particle at $x \gg r$.

### (ii) Attractive Delta Potential

$$V(x) = -g\delta(x)$$

$$-\frac{\hbar^2}{2m}\Psi'' - g\delta(x)\Psi = E\Psi(x)$$




Far away; $(x \gg r)$, $V(x) = 0$

$$\Rightarrow -g\delta(x) = 0$$

$$\therefore \Psi'' + k^2\Psi = 0; \; k = \sqrt{\frac{2mE}{\hbar^2}}$$

$$\Psi = \begin{cases} Ae^{ikx} + Be^{-ikx}, & \text{in I} \\ Ce^{ikx} + De^{-ikx}, & \text{in II}. \end{cases}$$

We write it in terms of:

$$\Psi = \begin{cases} e^{ikx} + r e^{-ikx} \\ t e^{ikx} \end{cases}$$

$$r = \frac{B}{A}, \quad t = \frac{C}{A}$$

• Boundary Conditions:

(i) $\Psi(0^+) = \Psi(0^-)$

$$\Rightarrow 1 + r = t$$

(ii) $$\int_{-\epsilon}^{\epsilon} -\frac{\hbar^2}{2m}\frac{d^2\Psi}{dx^2}\,dx - g\int_{-\epsilon}^{\epsilon}\delta(x)\,dx = \cancel{\int_{-\epsilon}^{\epsilon} E\,dx}$$

$$\Psi'(0^+) - \Psi'(0^-) = -\frac{2mg}{\hbar^2}\Psi(0)$$

$$\Rightarrow ikt - ik(1-r) = -\frac{2mg}{\hbar^2}$$

Let, $K = \frac{2m}{\hbar^2}$

$$r = \frac{-K}{K+ik}, \quad t = \frac{ik}{K+ik}$$

$$-\frac{\hbar^2}{2m}\left(\frac{d\Psi}{dx}\bigg|_{0+} - \frac{d\Psi}{dx}\bigg|_{0-}\right) = g\Psi(0)$$

$$\Rightarrow \frac{d\psi}{dx}\bigg|_{0+} - \frac{d\psi}{dx}\bigg|_{0-} = -\frac{2mg}{\hbar^2}\psi(0)$$

$$\begin{pmatrix} A_L^{out} \\ A_R^{in} \end{pmatrix} = \begin{pmatrix} r & t \\ t & r \end{pmatrix} \begin{pmatrix} A_L^{in} \\ A_R^{in} \end{pmatrix}$$

$\hookrightarrow$ Parity.

Here $x \to -x$ and $V(x)$ is symmetric,
$\Rightarrow$ parity basis $\rightarrow$ no off diagonal forms.

$$S^p = \begin{pmatrix} t-r & 0 \\ 0 & t+r \end{pmatrix}$$

$$S_\pm = t \pm r$$

$$S = t - r = \frac{iK + k}{ik - K}$$

pole at $k = -iK$ $\rightarrow$ scattering matrix not analytic. $\rightarrow$ outgoing particle has to go to zero.

S matrix is unitary

$$S(k) = e^{2i\delta(k)}$$

$$\delta(k) = \tan^{-1}\left(\frac{K}{k}\right)$$

$k \to 0$, $\delta(0) = \frac{\pi}{2}$ — Low energy

$k \to \infty$, $\delta(0) = 0$ — High energy

$\delta(k)$ is enough to show where wave is distorted.

$$t = \frac{ik}{ik + K} = \frac{k}{k - iK}$$

Poles of the transmission (or scattering) amplitude correspond to bound states

$$E = -\frac{\hbar^2 K^2}{2m}, \quad \psi \propto e^{ikx} = e^{-Kx}$$

Levinson's Theorem:

$$\delta(0) - \delta(\infty) = n\pi$$

$$\delta(0) - \delta(EE) \Rightarrow \frac{3\pi}{2} - \frac{\pi}{2}$$

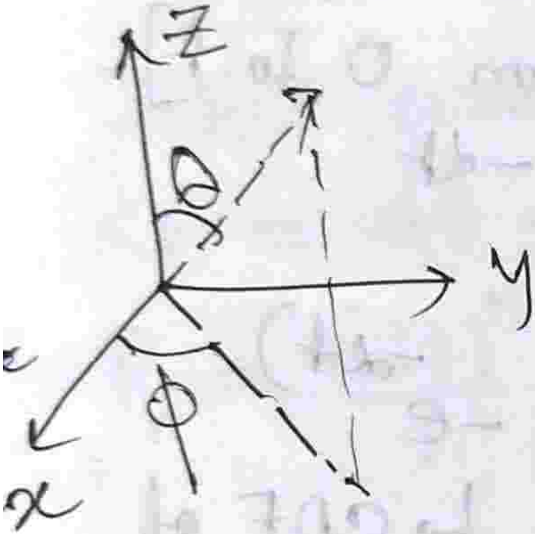
# Advanced Quantum Mechanics

## 19/03/26 : Scattering in 3D :

$$\frac{\hat{p}^2}{2m}|\psi\rangle = E|\psi\rangle \quad \text{[Free particle]}$$

↳ Free part of SE.

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial Z^2}$$




$$x = r \sin\theta \cos\phi$$
$$y = r \sin\theta \sin\phi$$
$$z = r \cos\theta$$

$$\nabla^2 = \frac{1}{r^2} = \frac{\partial}{\partial r}\left(r^2 \frac{\partial}{\partial r}\right) - \frac{\hat{L}^2}{\hbar^2 r^2}$$

$$\hat{L} = \vec{r} \times \vec{P}, \quad \vec{P} \simeq \vec{\nabla}$$

$$\hat{L} \sim (\vec{r} \times \vec{\nabla}) = (\vec{r} \times \vec{\nabla})$$

$$\hat{L}^2 = -\hbar^2 \left[\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta \frac{\partial}{\partial\theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\phi^2}\right]$$

$[\hat{L}^2, L_z] = 0$ , $[\hat{L}^2, \hat{L}_x] = 0$

$[\hat{L}^2, \hat{L}_y] = 0$

We usually work with $\hat{L}_z$ basis. We want the largest set of mutually commuting operators.

$$\vec{L} \times \vec{L} = i\hbar \vec{L}$$

$$\Rightarrow [\hat{L}_x, \hat{L}_y] = i\hbar \hat{L}_z \rightarrow 2 \text{ such equation.}$$

It is algebraically easy to take the $[\hat{L}^2, \hat{L}_z] = 0$ , and we obtain the simultaneous eigenbasis of these two.

→ Set of mutually commuting operators:

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If two operators commute with Hamiltonian, do they mutually commute? Yes, And the eigenbasis (simultaneous) of such mutually commutators is the best basis to work on.

$|n\,l\,m\rangle$ is the simultaneous eigenbasis.

$$\hat{L}^2|l,m\rangle = \hbar^2 l(l+1)|l,m\rangle$$

$l = 0, 1, 2 \ldots$

$l=0 \rightarrow$ s-wave $\rightarrow$ spherically symmetric

Impurities in system not spherically symmetric. But for low energy incidence on impurity (low momentum), (hence less de-broglie wavelength), $\Rightarrow$ the impurity can be approximated to be spherically symmetric.

$$m_l = -l, (-l+1), \ldots, (l-1), l$$

$$\langle x|P\rangle = e^{ip.x/\hbar}$$

Coordinate representation of angular momentum eigenstates in coordinate space is $Y_{l,m}(\theta,\phi)$

$$-\frac{\hbar^2}{~~2m~~}\left[\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right)+\frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\phi^2}\right]Y_{l,m}(\theta,\phi)$$

$$= \hbar^2 l(l+1)\,Y_{l,m}(\theta,\phi)$$

In continuum, orthonormality relation

$$\int e^{ipx}e^{-ip'x}dx = \#\,\delta(p-p')$$

Going back to the Schrodinger equation:

$$-\frac{\hbar^2}{2m}\left[\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial \theta}\right)-\frac{\hat{L}^2}{\hbar^2 r^2}\right]\Psi(\vec{r})=E\Psi(\vec{r})$$

- separation of variables:

$$\Psi(\vec{r}) = R(r)\,Y_{l,m_l}(\theta,\phi)$$

$$-\frac{\hbar^2}{2m}\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{d}{dr}R(r)\right)+\frac{\hbar^2 l(l+1)}{2m^2}Y_{l,m}(\theta,\phi)$$

$$=ER(r)$$

Separating;

$$-\frac{\hbar^2}{2m}\frac{d}{dr}\left(r^2\frac{d}{dr}R(r)\right)+\frac{\hbar^2}{2m}l(l+1)R(r)$$

$$=Er^2R(r)$$

$$r^2\frac{d^2}{dr^2}R_l(r)+2r\frac{d}{dr}R_l(r)$$

$$+\left[k^2r^2-l(l+1)\right]R_l(r)=0$$

Solution (general) to This function are Bessel function of 1st and 2nd kind (linearly combined). We know Bessel function of 2nd kind are not defined at origin.

Solution is of form:

$$R_\ell(r,k) = \sqrt{\frac{2}{\pi}}\, k\, j_\ell(kr)$$

$$\int_0^\infty R_\ell(r,k) R_{\ell'}(r,k') r^2 dr = \delta(k-k')\,\delta_{\ell\ell'}$$

$$R_\ell(r) = A_\ell j_\ell(kr) + B_\ell n_\ell(kr)$$

$$\frac{\hat{p}^2}{2m}|\psi\rangle = E|\psi\rangle$$

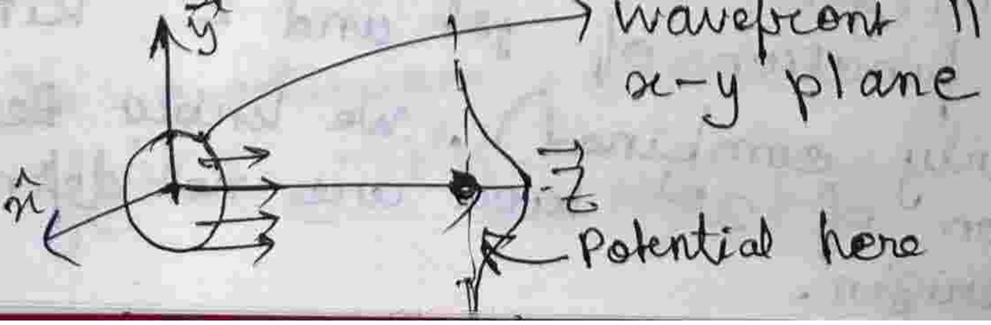
$$\Rightarrow \left[\underbrace{-\frac{\hbar^2}{2m}\frac{1}{r}\frac{\partial^2}{\partial r^2}}_{} + \frac{\hbar^2}{2m}\frac{\ell(\ell+1)}{r^2}\right]\psi = E\psi$$

↳ free of radial motion.

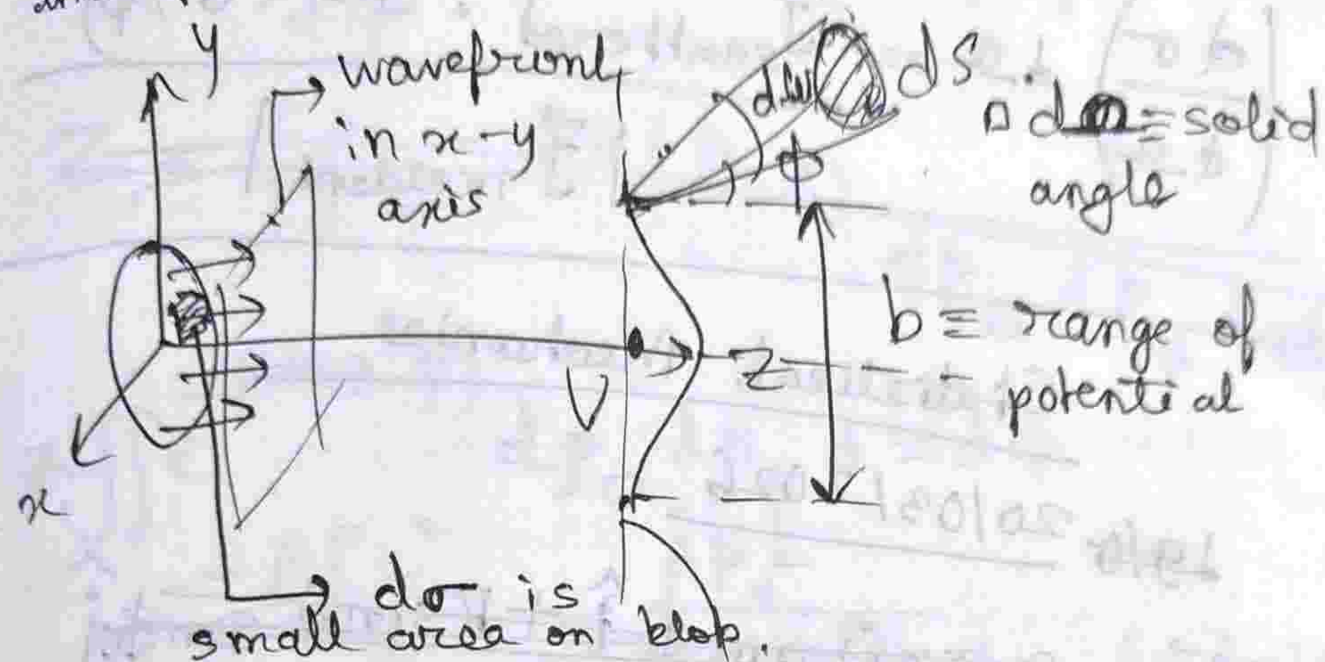
- Why does the angular part of "free SE" suffer from a potential which goes to $\infty$ at $r=0$?

$\Rightarrow$ Centrifugal Potential.

- Setting of the scattering problem



The "blob" of flux needs to be at least as big as the range of the potential (b).




Q: Given an incoming flux, we have detector (dS), how many particles we recieve on area dS per unit time.

Differential scattering cross-section:

$\rightarrow$ dependence of $\left(\frac{d\sigma}{d\Omega}\right)$

$$\left(\frac{d\sigma}{d\Omega}\right)(\theta, \phi)\, d\Omega$$

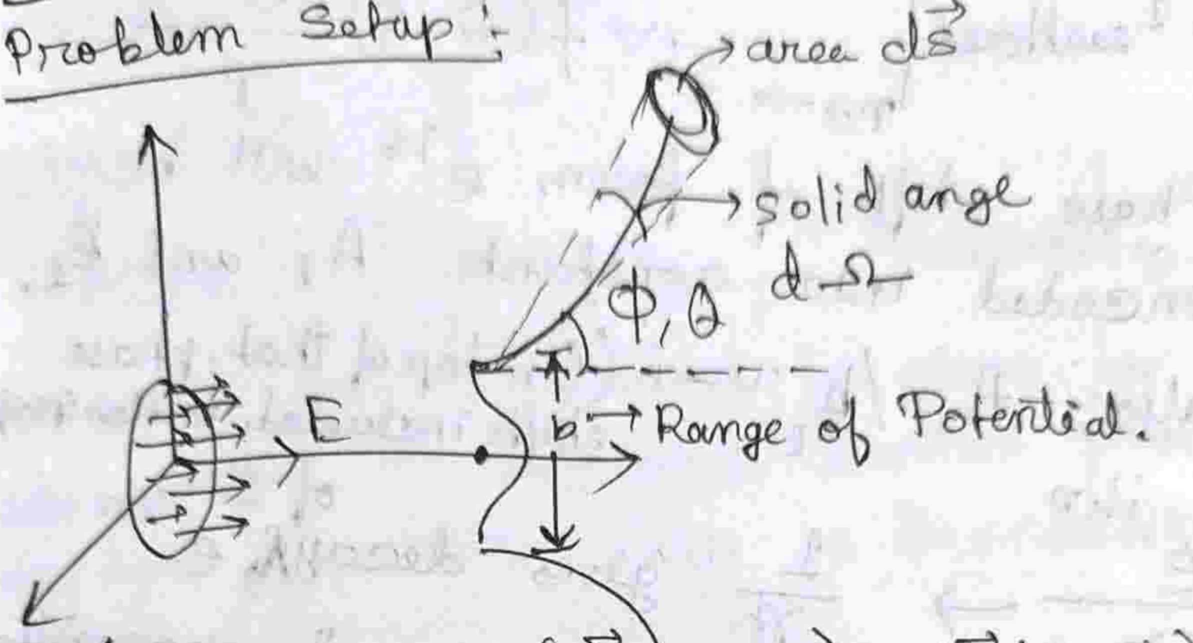
$$= \frac{\text{no. of particles scattered into } d\Omega \text{ per second}}{\text{no. of incident particles / second / } d\sigma \text{ area in the x-y plane.}}$$

$$\frac{d\sigma}{d\Omega} = \frac{\text{scattered flux in } (\theta, \phi) \text{ direction}}{\text{incident flux}}$$

$$\left(\frac{d\sigma}{d\Omega}\right) d\Omega = \frac{\vec{J}_{scattered} \cdot d\vec{S}(\theta, \phi)}{|\vec{J}_{incident}|}$$

# Advanced Quantum Mechanics : 20/03/2026

## Problem Setup :-




$$\left(\frac{d\sigma}{d\Omega}\right) d\Omega = \frac{(\vec{J}_{scattered}) \cdot d\vec{S}(\theta, \phi)}{|\vec{J}_{incident}|}$$

Now let us introduce a radially dependent (spherically symmetric) potential. The TISE becomes:

$$\left[-\frac{\hbar^2}{2m}\vec{\nabla}^2 + V(|\vec{r}|)\right]\Psi(r, \theta, \phi) = E\Psi(r, \theta, \phi)$$

$$\Psi_{incident}(\vec{r}) = e^{i\vec{k}\cdot\vec{r}} = e^{i|\vec{k}|(\hat{k}, \hat{r})} = e^{ikz}$$

we can convert it into:

$$E = \frac{\hbar^2 k^2}{2m}$$

$$\Psi_{scattered} = \sum_{l,m}\left(A_l\, j_l(kr) + B_l\, n_l(kr)\right) Y_{l,m}(\theta, \phi)$$

$r \to \infty$ ; $j_l(kr) \xrightarrow{r\to\infty} \sin\left(kr - \frac{l\pi}{2}\right)/kr$ ; $n_l(kr) \xrightarrow{r\to\infty} -\cos\left(kr - \frac{l\pi}{2}\right)/kr$

$$\Rightarrow \Psi_{scattered}\Big|_{r \to -\infty} \sim f(\theta,\phi)\,\frac{e^{ikr}}{r}$$

Phase shifts of form $e^{i\delta}$ will be encoded into constants $A_l$ and $B_l$.
also; $A_l / B_l \sim -i$ (On top of that, phase shifts included, problem specific)

$\frac{e^{ikr}}{r} \to \frac{1}{r}$ gives decay of $\Psi$, $e^{ikr}$ means "outgoing" propagation.

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$A_l / B_l = -i$ does not rule out the presence of phase shifts encoded within $A_l$ and $B_l$.

$$\Psi_{scattered} \simeq \sum_{l, m_l} \frac{1}{(kr)}\left(-i B_l \sin\left(kr - l\frac{\pi}{2}\right) - B_l \cos\left(kr - l\frac{\pi}{2}\right)\right) Y_{l,m_l}(\theta,\phi)$$

$$= \sum_{l, m_l} \frac{-B_l}{(kr)} e^{i(kr - l\pi/2)} Y_{l,m_l}(\theta,\phi)$$

$$\therefore e^{-i\frac{\pi}{2}l} = (-i)^l$$

$$\Psi_{scattered} = \left(\underbrace{\sum_{l, m_l} \cdots}_{f(\theta,\phi) \to \text{absorbs all } \theta, \phi \text{ term.}}\right) \frac{e^{ikr}}{r}$$

□ Total wavefn:

$$\Psi_k \Big|_{r\to\infty} = \Psi_{incident}\Big|_{r\to\infty} + \Psi_{scattered}\Big|_{r\to\infty}$$

In 1-D, total current is given by subtraction of reflected current from incident current. This is transmitted current, forward moving.

In 3-D, interference terms between incident and reflected current interfere and cross terms that are neither forward nor backward.

□ Friedel Oscillations:

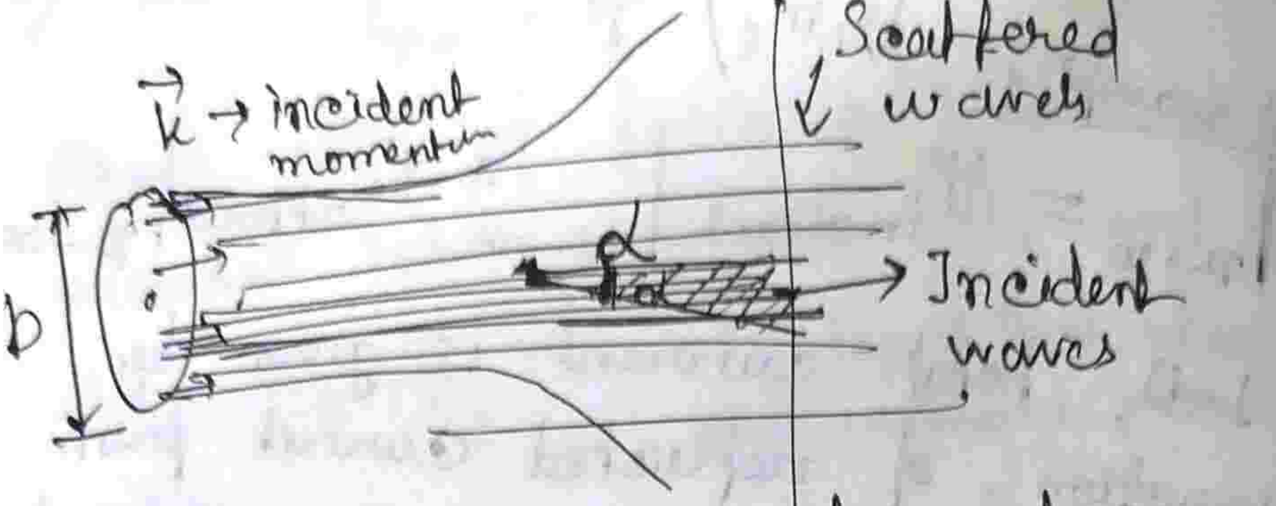
$$|\Psi|^2 = 1 + |r|^2 + 2[\operatorname{Re} r]\cos(2kx)$$

$$j(r) = v - v|\Psi(r)|^2 \longleftrightarrow \text{might have prefactors, sir did not write.}$$

$$\Psi_k\big|_{r\to\infty} = e^{ikz} + f(\theta,\phi)\frac{e^{ikr}}{r}$$

($z = r\cos\theta$)

The $A_l / B_l \sim -i$ in asymptotic region is given by symmetry imposed boundary condition.




if scattering waves do not come into the regime where incident waves are being detected, no interference for them.

$$L = \vec{r} \times \vec{p} \quad , \quad b \times k_z = L_{max}$$

- Finite b truncates the $\infty$ sum

$$\sum_{l, m_l} \frac{B_l}{(kr)} e^{i\left(kr - l\frac{\pi}{2}\right)} Y_{l, m_l}(\theta, \phi)$$

Plane waves $\to$ superposition of spherical waves (they form a complete basis).

- When a plane wave scatters off a localized, spherically symmetric potential, it is expanded in the spherical (partial wave) basis as a superposition of angular momentum modes "$l$". Rotational symmetry implies conservation of angular momentum, so different $l$ sectors do not mix. In matrix terms, the Hamiltonian is block diagonal in This basis, $\langle l'm'|H|lm\rangle \propto \delta_{ll'}\delta_{mm'}$; with no coupling between sectors. Hence, the scattering problem decomposes into independent channels for each $l$.

○ Current Calculation:

□ $\vec{J}_{incident} = -\frac{i\hbar}{2m}(\Psi^* \nabla\Psi - \Psi\nabla\Psi^*)$

This is current expression in general

$= \left(\frac{\hbar k}{m}\right)\hat{z} = v\hat{z}$

□ $\vec{\nabla}\Psi_{scattered}\Big|_{r\to+\infty}$ $\left(\cong \vec{\nabla} = \frac{\partial}{\partial r}\hat{r} + \frac{1}{r}\frac{\partial}{\partial\theta}\hat{\theta} + \frac{1}{r\sin\theta}\frac{\partial}{\partial\phi}\hat{\phi}\right]$

□ $\vec{\nabla}\Psi_{scattered}$

$\simeq \frac{\partial}{\partial r}\left(f(\theta,\phi)\frac{e^{ikr}}{r}\right)\hat{r}$

$\simeq f(\theta,\phi)(ik)\frac{e^{ikr}}{r}$ → We plug it in current equation:

$\vec{J}_{scattered} = \frac{v|f(\theta,\phi)|^2}{r^2}\hat{r}$

$\vec{J}_{scattered} \cdot d\vec{S}(\theta,\phi) = \vec{J}_{scatt.}\hat{r}\,\underbrace{dA}_{r^2 d\Omega}$

$= \frac{v|f|^2}{r^2}\cdot r^2 d\Omega$

$$\vec{J}_{\text{scattered}} \cdot d\vec{S}(\theta,\phi) = v|f(\theta,\phi)|^2 d\Omega$$

$$\frac{d\sigma}{d\Omega} d\Omega = \frac{\vec{J}_{\text{scattered}} \cdot d\vec{S}}{|\vec{J}_{\text{enc}}|} = |f(\theta,\phi)|^2 d\Omega$$

For spherical symmetry, $\Phi$ drops out because it is NOT an angle with respect to $z$-axis, along which the impurity lies.

<u>R. Shankar</u>: 19.56, page 555

2nd edition.

Advanced Quantum Mechanics : 25/03/2026

Differential cross section is equal to squared scattering amplitude.

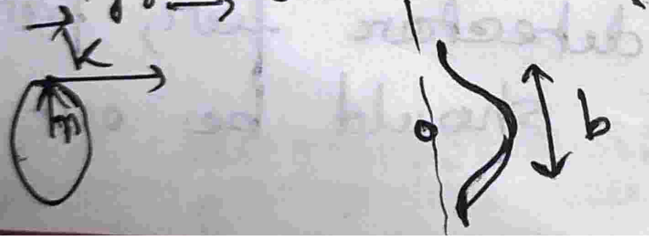
$$\frac{d\sigma}{d\Omega} = |f(\theta, r)|^2$$

The $r$ dependence maybe parametric ($kr$)

$$\Psi_{scattered}(r, \theta) = f(\theta, r)\frac{e^{ikr}}{r}$$

$$\Psi_{incident} = e^{ikz} = e^{ikr\cos\theta}$$

$r \to \infty$

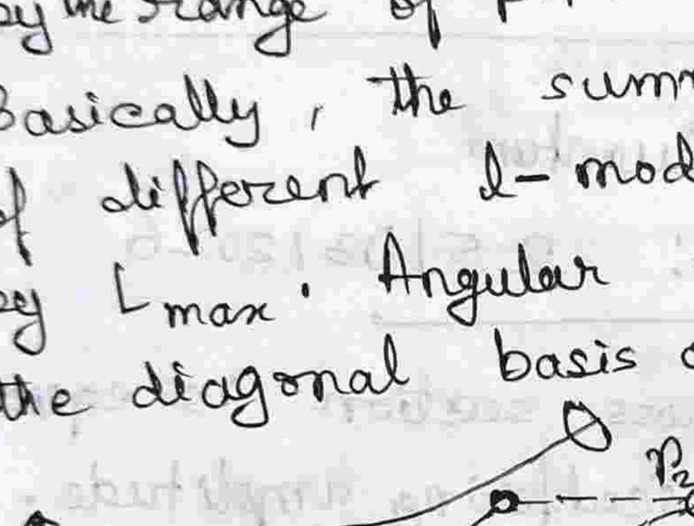



$\therefore$ Maximum angular momentum the beam is carrying is:

$\vec{L} = \vec{b} \times \vec{k} \equiv \vec{L}_{max}$

For partial wave analysis, one problem is split into different problems (angular momentum modes), which do not mix with one another. We deal with the collection of phase shifts acquired by the various angular momentum modes, but is this collection infinite? No! It is constrained in the upper bounds by the range of potential "b", $L_{max} = \vec{b} \times \vec{k}$

Basically, the summation of series of different $l$-modes is truncated by $L_{max}$. Angular momentum basis is the diagonal basis of the problem.




Total current at detector:

$j(r, \theta, \phi)$

$= j_{scattered} + j_{incident} + j_{interference}$

$\vec{j}_{total} \propto \#(\Psi^* \nabla\Psi - h.c)$

0 in 1-D

We don't want the background term introduced by $j_{interference}$. One way to do it is to put the detector far, far away. Also detector should be out of the beam radius. If the detector is very far away, it gets access to smaller and smaller angles while having minimum disturbances from the beam radius region.

$$\Psi_{total} = \Psi_{incident} + \Psi_{scattered}$$

$$\Rightarrow \Psi_{total} = e^{ikr\cos\theta} + f(\theta)\frac{e^{ikr}}{r}$$

$$\Psi_{incident} = e^{ikr\cos\theta}\Big|_{r\to\infty}$$

$$\Rightarrow \Psi_{inc} = \sum_{l=0}^{\infty} (i)^l (2l+1) \underbrace{\frac{(\sin(kr - l\pi/2))}{kr}}_{j_l(kr)|_{r\to\infty}} \underbrace{P_l(\cos\theta)}_{\text{Legendre Polynomial}}$$

- Partial wave Amplitude:

$$f(\theta, k) = \sum_{l=0}^{\infty} (2l+1) \frac{a_l}{k} P_l(\cos\theta)$$

$$\Psi_{total} = \sum_{l=0}^{\infty} \frac{(2l+1)}{2ik}\left[-\frac{e^{-i(kr - l\pi/2)}}{r} + (1+2ia_l)\frac{e^{ikr}}{r}\right] \times P_l(\cos\theta)$$

→ $\sin kr$ decomposed into incoming and outgoing waves, $e^{\pm ikr}$.

From statement of unitarity, $a_l$ is a phase shift.

$$1 + 2i a_l = e^{2i\delta_l}, \quad l = 0, 1, 2, \ldots$$

$$\Rightarrow a_l = \frac{1}{2i}\left(e^{2i\delta_l} - 1\right)$$

$$= e^{i\delta_l}\left(\frac{e^{i\delta_l} - e^{-i\delta_l}}{2i}\right)$$

$$\Rightarrow a_l = e^{i\delta_l} \sin(\delta_l)$$

$$f(\theta) = \sum_{l=0}^{\infty} \frac{(2l+1)}{k}\left(e^{i\delta_l} \sin\delta_l\right) P_l(\cos\theta)$$

$$\frac{d\sigma}{d\Omega} = |f(\theta)|^2$$

$$= \frac{1}{k^2} \sum_{l,l'}^{\infty} (2l+1)(2l'+1) e^{i(\delta_l - \delta_{l'})} \sin\delta_l \sin\delta_{l'} \, P_l(\cos\theta) P_{l'}(\cos\theta)$$

- Total scattering cross section

$$\sigma = \int d\Omega \, |f(\theta)|^2$$

Now, $d\Omega = \sin\theta \, d\theta \, d\phi$

$$\sigma = \int_0^{2\pi} d\phi \int_0^{\pi} \sin\theta \, d\theta \, (\ldots) = 2\pi \int_{+1}^{-1} \left(-d(\cos\theta)\right) (\ldots)$$

$$\sigma = 2\pi \int_{-1}^{1} d(\cos\theta)\,(\ldots\ldots)$$

$$= \frac{2\pi}{k^2} \sum_{l,l'} (2l+1)(2l'+1)(\sin\delta_l)(\sin\delta_{l'})\, e^{i(\delta_l-\delta_{l'})}$$

$$\underbrace{\int_{-1}^{1} P_l(\cos\theta)\, P_{l'}(\cos\theta)\, d(\cos\theta)}_{\frac{2}{2l+1}\delta_{ll'}}$$

$$\Rightarrow \sigma = \frac{4\pi}{k^2} \sum_l (2l+1)\sin^2\delta_l = \sum_l \sigma_l$$

Forward Scattering is almost like penetrating through the impurity

$f(\cos\theta = 1) \rightarrow$ scattering amplitude in this case.

$\theta = 0$

$$\mathrm{Im}\left[ f(\theta=0)\right] = \sum_l \frac{(2l+1)}{k} \sin^2\delta_l$$

$\square$ Optical Theorem :

$$\sigma = \frac{4\pi}{k} \sum_l \frac{(2l+1)}{k} \sin^2\delta_l = \sum_l \sigma_l$$

$$= \frac{4\pi}{k}\, \mathrm{Im}\left[ f(\theta=0)\right]$$

The forward amplitude only gives whole $\sigma$!

▫ Now we consider the Hard Sphere problem:

▷ Potential: $$V(r) = \begin{cases} \infty, & r < r_0 \\ 0, & r > r_0 \end{cases}$$

Boundary condition is, upto $r_0$ (radius of scattering), $\Psi = 0$ [node of the wavefunction].

Because we will never reach centre

$$R_l(r) = A_l j_l(kr) + B_l n_l(kr)$$

$$R_l(r = r_0) = 0$$

$$\frac{B_l}{A_l} = -\frac{j_l(kr)}{n_l(kr)}$$

Let, $A_l = \alpha \cos\delta_l$

$B_l = \alpha \sin\delta_l$

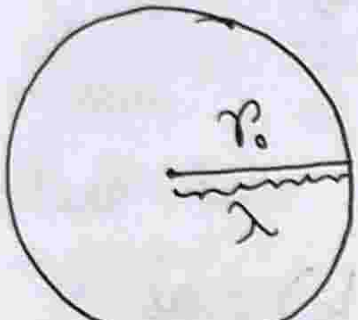
$$R_l(r)\Big|_{r\to\infty} = \left(\frac{\alpha}{kr}\right)\left(\sin\left(kr - \frac{l\pi}{2}\right) + \cos\left(kr - \frac{l\pi}{2}\right)\sin\delta_l\right)$$

$$\Rightarrow R_l(r)\Big|_{r\to\infty} = \left(\frac{\alpha}{kr}\right)\sin\left(kr - \left(\frac{\pi}{2} - \delta_l\right)\right)$$

Let us look at the $l=0$ contribution.

$$\delta_{l=0} = \tan^{-1}\left(\frac{\left(\frac{\sin(kr_0)}{kr_0}\right)}{\left(-\frac{\cos(kr_0)}{kr_0}\right)}\right) = -\tan^{-1}(\tan(kr_0))$$

$\hookrightarrow$ s-wave




$$\boxed{\Rightarrow \delta_{l=0} = -kr_0}$$

$$\Rightarrow \delta_{l=0} = -\frac{2\pi}{\lambda} r_0$$

$$\Rightarrow \delta(l=0) = -2\pi\left(\frac{r_0}{\lambda}\right)$$

So, ~~amount by which~~ no. of waves are "pushed out" from the centre (in all directions), is $k = \frac{r_0}{\lambda}$, where $\lambda \equiv$ de-broglie wavelength. This gives us the phase shift accumulated by the s-wave (outgoing from centre. $k = r_0/\lambda \rightarrow$ no. of wavelengths fitted in $r_0$.

The $\delta$ - phase shift

$$\delta_{l=0} = -kr_0, \text{ if } k \to 0 \text{ [low momenta]}$$

$$\Rightarrow \delta_{l=0} \to 0$$

$$\sigma = \frac{4\pi}{k^2}\sum_{l=0}^{\infty}(2l+1)\sin^2\delta_l$$

$$\boxed{\delta_l \sim k^n \;;\quad n > 2 \text{ for all } l \neq 0}$$

$\hookrightarrow$ If this is true, $\sigma$ sum truncates at $l=0$ in the limit $k \to 0$ [Hence $\delta_{l=0} \to 0$]

$$\sigma \simeq \frac{4\pi}{k^2} \sin^2 \underbrace{\delta_{l=0}}_{\to\ \delta_{l=0} \to 0}$$

$$\sigma = \frac{4\pi}{k^2} \left(\delta_{l=0}\right)^2 = \frac{4\pi}{k^2} (k r_0)^2$$

This is consistent $\Rightarrow$ $\boxed{\sigma = 4\pi r_0^2}$

for low energy limit (low k), as s-wave is spherical, where $r_0$ is the radius of the potential. No wonder scattering cross-section area $\sigma = 4\pi r_0^2$.

Now, $\boxed{\left.\dfrac{j_l(kr)}{n_l(kr)}\right|_{k \to 0,\, \text{finite } r} \Rightarrow (kr)^{2l+1}}$. Recall,

$$\delta_l \cong \tan^{-1}\left((k r_0)^{2l+1}\right) \leftarrow [\text{when } k \to 0 \text{ for finite } r]$$

$$\sigma = \frac{4\pi}{k^2} (2l+1) \delta_l^2$$

$$\sigma = \frac{4\pi}{k^2} (2l+1) (k r_0)^{4l+2}$$

| | |
|---|---|
| So, if $l = 0$, $\sigma = 4\pi r_0^2$, when $k \to 0$ | For $l \neq 0$, when $k \to 0$, powers of k do not cancel out, and $\sigma = 0$ |

$\to$ Only S-wave channels contribute.

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## Optical Theorem

Complete information of scattering is encoded in the forward scattering (imaginary part of it) only.

$$\sigma = \frac{4\pi}{k} \operatorname{Im} [f(\theta = 0)]$$

We saw for hard sphere potential, scattering cross section was exactly equal to surface area of the scattering sphere.

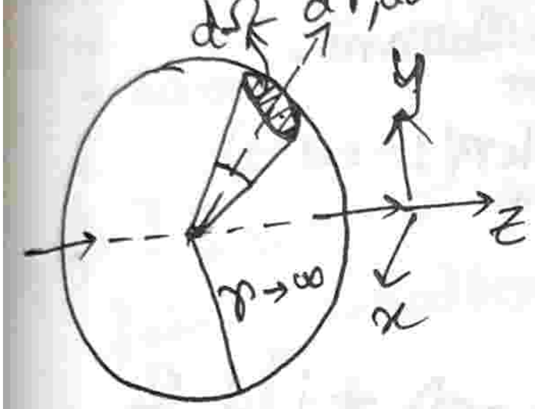
$$\left[-\frac{\hbar^2}{2m}\vec{\nabla}^2 + V(\vec{r})\right]\Psi(\vec{r}) = E\Psi(\vec{r}) \quad \Big| \; E = \frac{\hbar^2 k^2}{2m}$$

$$\psi(\vec{r}) \overset{r\to\infty}{\simeq} e^{ikz} + f(\theta, \phi)\frac{e^{ikr}}{r}$$

$$\vec{j}(\vec{r}) = \frac{\hbar}{2im}\left(\Psi^* \vec{\nabla}\Psi - \Psi \vec{\nabla}\Psi^*\right)$$

$$\vec{J}_{incoming} = \hat{e}_z \frac{\hbar k}{m}$$

$$j_{scatter} = \frac{\hbar k}{m} |f(\theta, \phi)|^2 \frac{\hat{e}_r}{r}$$




$j_{scatter}$ has to be cancelled by some term, otherwise it will look like the centre is emanating current out of nowhere, so current is not conserved. The term is nothing but interference current, which is the exact $-$ve of the scattered current. Depletion happens due to destructive interference between $j_{incoming}$ and $j_{scattered}$.

$$\boxed{\int j_{total} \cdot ds = 0}$$

inc, scattered, interference between incident and scattered

Putting in $j(\vec{r})$;

$$j(\vec{r}) = \frac{\hbar}{2im}\left((\Psi_I^* + \Psi_s^*)\nabla(\Psi_I + \Psi_s) + \cdots\right)$$

↳ We do all calculations in $r \to \infty$ limit.

$$-\vec{j}_{interference} \xrightarrow{r\to\infty} \frac{\hbar}{2im} f(\theta,\phi)\left[\frac{ke}{r}(\hat{e}_r+\hat{e}_z) + \frac{ie^{ik(r-z)}}{r^2}\hat{e}_r + \underbrace{h.c}_{\text{hermition conjugate}}\right]$$

↳ Interference between incident beam and radially scattered beam.

$$\vec{j}_{interference} \cdot \vec{dS} = \frac{\hbar}{2m} f(\theta,\phi) e^{ikr(1-\cos\theta)} \times [kr(1+\cos\theta)+i]\,d\Omega + h.c$$

$\hat{r}$

Exercise: $2^{nd}$ term $\int [\underbrace{kr}_{\gamma}(1+\underbrace{\cos\theta}_{x})+i]\,d\Omega$

is zero, Hayein???

$kr \to \gamma$ , $\cos\theta \to x$

$\gamma \to \infty$

$r \to \infty$

$$\lim_{\gamma\to\infty} \int_{-1}^{+1} (1+x) f(x) e^{i\gamma(1-x)} dx = 2i\underbrace{f(1)}_{f(\cos(\theta=0))}$$

Net current through the total surface:

$$I_{interference} = \lim_{r\to\infty} \int_0^{2\pi} d\varphi \int_{-1}^{+1} d(\cos\theta) \times \vec{j} \cdot d\vec{S}$$

$$= \frac{\hbar}{m} 2\pi i f(\cos 0) + h.c$$

$$I_{inter} = -\frac{\hbar}{m} 4\pi \, \mathrm{Im}\,[f(\theta=0)]$$

$\Rightarrow I_{inter}$ is exactly the $-$ve of the total scattering cross section.

The sphere is considered to have $r\to\infty$, which is much larger than range of influence of the potential, incoming and scattered waves are "free" states asymptotically, so absence of any bound states makes sure no electrons that come in are trapped — hence whatever wave comes in goes out.

Current interference and density interference are not the same:

$$\psi^*\psi = \rho(\vec{r}) = \psi_I^*\psi_I + \psi_s^*\psi_s + 2\,\mathrm{Re}[\psi_I^*\psi_s]$$

$[\psi = \psi_I + \psi_s]$

- Phase freedom (A continuous symmetry)

S.E. $\Psi$ invariant $\xrightarrow[\psi \to e^{i\theta}\psi]{}$ Associated with some conservation $\rightarrow$ particle number conservation

- Eg of number of particles not being conserved is pair production by high energy photons to $e^-$ and $e^+$. We can not do this with single particle spaces. We use Fock space, which is a new kind of vector space that can be thought of as a direct sum of single particle space, double particle space and so on. (here space means Hilbert space). Also zero particle Hilbert space. Annihilation and creation operators in Fock space reduce/increase number of particles by 1 in any state.

- Maxwell's Equations

$$\vec{\nabla}\cdot\vec{E} = 0 \; ; \; \vec{\nabla}\cdot\vec{B} = 0$$

$$\vec{\nabla}\times\vec{E} = -\frac{\partial\vec{B}}{\partial t} \; ; \; \vec{\nabla}\times\vec{B} = \frac{1}{c^2}\frac{\partial E}{\partial t}$$

$$\vec{\nabla}^2 \vec{E} - \frac{1}{c^2}\frac{\partial^2 \vec{E}}{\partial t^2} = 0 \quad \text{—} ①$$

$$\vec{\nabla}^2 B - \frac{1}{c^2}\frac{\partial^2 \vec{B}}{\partial t^2} = 0 \quad \text{—} ②$$

$$\vec{E}(\vec{x}, t) = \sum_m \underbrace{f_m(t)}_{\text{encodes time evolution}} \underbrace{\vec{u}_m(x)}_{\text{normal modes of the electric field – static}}$$ [This is in 1-D

- normal modes are stationary states within a box which obey the boundary conditions.

$$\nabla^2 u_m = -k_m^2 u_m, \quad \vec{\nabla} \cdot \vec{u}_m = 0$$

- Orthonormalization condition for any normal mode

---

$$\int \vec{u}_m(x) \vec{u}_n(x)\, d^3x = \delta_{m,n}$$

$$\sum_m \frac{d^2 f_m}{dt^2} + c^2 k_m^2 f_m(t) = 0$$

↳ Equation of motion of a harmonic oscillator. So this is a stack of independent harmonic oscillators (countably infinite) with different frequencies (excited states)
→ how many photons have been stacked

# Advanced Quantum Mechanics

Maxwell's equations in free space :

$\vec{\nabla}.\vec{E}=0$ , $\vec{\nabla}.\vec{B}=0$ , $\vec{\nabla}\times\vec{B} = \frac{1}{c^2}\frac{\partial E}{\partial t}$ , $\vec{\nabla}\times\vec{E} = -\frac{\partial \vec{B}}{\partial t}$

Rewriting in symmetric form.

$\nabla.\vec{E}=0$ , $\nabla.(c\vec{B})=0$, $\vec{\nabla}\times(c\vec{B}) = \frac{1}{c}\frac{\partial \vec{E}}{\partial t}$, $\vec{\nabla}\times\vec{E} = -\frac{1}{c}\frac{\partial(c\vec{B})}{\partial t}$

From these, we obtain wave equation:

$$\nabla^2\vec{B} = \frac{\partial^2 B}{\partial t^2} \quad ; \quad \nabla^2\vec{E} = \frac{\partial^2\vec{E}}{\partial t^2}$$

We expand electric and magnetic fields into normal modes:

$$\vec{E}(\vec{x},t) = \sum_m f_m(t)\,\vec{u}_m(\vec{x}) \; ; \quad \vec{B} = \sum_n h_n(t)\,\vec{u}_n(x)$$

The mode functions satisfy:

$$\nabla^2\vec{u}_m = -k_m^2\vec{u}_m \quad , \quad \nabla.\vec{u}_m = 0$$

For a metallic box : Boundary condition is:

$$\hat{n}\times\vec{u}_m = 0$$

These modes form a complete orthogonal set:

$$\int \vec{u}_m(x).\,\vec{u}_{m'}(x)\,d^3x = \delta_{mm'}$$

Substituting this eq expansion into wave eq$^n$ gives.

$$\sum_m\left(\frac{d^2 f_m(t)}{dt^2} + c^2k_m^2 f_m(t)\right) = 0$$

$\because$ Modes are linearly independent, each term must vanish: $\frac{\partial^2 f_m(t)}{\partial t^2} + c^2k_m^2 f_m(t) = 0$, with $\omega_m = ck_m$

Using Maxwell's equations, we relate the electric and magnetic field:

$$\vec{\nabla}\times\vec{E} = \sum_m f_m(t)\,\vec{\nabla}\times\vec{u}_m = -\frac{\partial\vec{B}}{\partial t}$$

$$\vec{\nabla}\times\vec{B} = \frac{\partial\vec{E}}{\partial t}$$ [P.T.O]

which leads to:

$$\sum_m h_m(t)\vec{\nabla}\times(\vec{\nabla}\times\vec{u}_m) = \sum_m \frac{df_m(t)}{dt}\vec{u}_m$$

Using $\nabla\cdot\vec{u}_m = 0$, this becomes:

$$-\sum_m h_m(t)\nabla^2\vec{u}_m = \sum_m \frac{df_m(t)}{dt}\vec{u}_m$$

The total energy of the field is given by the Hamiltonian:

$$H = \frac{1}{8\pi}\int(\vec{E}^2 + \vec{B}^2)\,d^3x$$

~~⊕~~ Substituting mode expansions:

$$H = \sum_{n,m}\left(f_n f_m\int\vec{u}_n\cdot\vec{u}_m\,d^3x + h_n h_m\int(\vec{\nabla}\times\vec{u}_n)\cdot(\vec{\nabla}\times\vec{u}_m)\,d^3x\right)$$

Using orthogonality:

$$\int\vec{u}_n\cdot\vec{u}_m\,d^3x = \delta_{nm},\quad \int(\vec{\nabla}\times\vec{u}_n)\cdot(\vec{\nabla}\times\vec{u}_m)\,d^3x = k_n^2\,\delta_{nm}$$

We obtain: $H = \frac{1}{2}\sum_n \frac{1}{4\pi}\left(f_n^2 + k_n^2 h_n^2\right)$

Now as $\vec{\nabla}\times\vec{E} = -\frac{\partial\vec{B}}{\partial t} = -\sum_n \frac{dh_n}{dt}\vec{u}_n$

$$\Rightarrow \sum_n f_n(t)\,\nabla\times\vec{u}_n = -\sum_n \frac{dh_n}{dt}\vec{u}_n$$

$$\Rightarrow \frac{dh_n}{dt} = -f_n$$

∴ We quantise the system by introducing ~~commutation~~ commutation relations between $f_n$ and $h_n$, we define,

$$f_n = \sqrt{2\pi\omega_n\hbar}\,(a_n^\dagger + a_n)$$

$$h_n = i\sqrt{\frac{m\omega_n\hbar}{2}}\,(a_n - a_n^\dagger)$$

with $m=1$, $f_n = x$

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Can we write down quantum state of an electric or magnetic field with optimum fluctuations.

→ Optimal uncertainty, wavepacket.

→ $\Delta x \Delta p = \hbar$

For harmonic oscillator, ground state is one such example. Other such states are "coherent" states. Coherent states are infinite number of states that form a complete basis. Each eigenstate of Hamiltonian are linearly independent (perpendicular to each other), but the elements of set of coherent states are not linearly independent, ie., we have to pay attention to their coefficients.

$$\hat{E}(\vec{r},t) = \sum_m \sqrt{2\hbar\pi}\ \omega_n \left[\hat{a}^{+}_m(t) + \hat{a}_m(t)\right] u_m(\vec{x})$$

$$\langle \hat{E}(x,t)\rangle = 0 \ ; \quad \langle m| a^{+}_m |m\rangle = 0 \ ; \langle m| a_m |m\rangle = 0$$

$$\langle \vec{E}\cdot\vec{E}\rangle \neq 0$$

S.H.O. $\quad [\hat{p},\hat{x}] = -i\hbar \ ; \ [a, \hat{a}^{+}] = 1$

- $a = \dfrac{1}{\sqrt{2m\omega\hbar}}(m\omega\hat{x} + i\hat{p})$

$\downarrow$

- $a^{+} = \dfrac{1}{\sqrt{2m\omega\hbar}}(m\omega\hat{x} - i\hat{p})$

- $H = \hbar\omega\left(a^{+}a + \underbrace{\tfrac{1}{2}}_{\text{Zero point energy}}\right)$

$\hat{N} = a^{+}a$

$[N, a^{+}] = +1\,a^{+}$

$[N, a] = -1\,a$

↳ $\hat{N}$ works on occupancy-basis.

$$\langle n|N|n\rangle = \langle n|n|n\rangle = \underbrace{n}_{\text{not -ve},\ n\geq 0}\cancel{\langle n|n\rangle}^{1}$$

$$\Rightarrow \langle \underbrace{n|}_{\langle\psi|}a^{+}a\underbrace{|n\rangle}_{|\psi\rangle} \geq 0$$

- Now, let us consider the ground state wavefunction:

$$a|\psi_0\rangle = 0 \Rightarrow \frac{1}{\sqrt{2m\omega\hbar}}\left(m\omega x + i\frac{\hbar}{i}\frac{d}{dx}\right)\psi_0(x) = 0$$

[P.T.O]

$$\left(\frac{m\omega}{\hbar}x + \frac{d}{dx}\right)\psi_0(x) = 0$$

Mera Gaussian Alag Hai 👍

$$\Rightarrow \int \frac{d\psi_0}{\psi_0} = -\int dx \frac{m\omega}{\hbar}x$$

$$\Rightarrow \psi_0(x) = N \exp\left(-\frac{m\omega}{2\hbar}x^2\right) \rightarrow \text{Optimum Gaussian} \rightarrow \text{not any gaussian}$$

$$\Rightarrow \int_{-\infty}^{\infty} |\psi_0(x)|^2 dx = 1 \Rightarrow N = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4}$$

$$\therefore \psi_0(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} \exp\left(-\frac{m\omega}{2\hbar}x^2\right)$$

$\Rightarrow$ To find $\Delta x \Delta p$, we need to find $(\langle x^2\rangle - \langle x\rangle^2)$ and $(\langle p^2\rangle - \langle p\rangle^2)$

- For $\hat{x}$: $\langle\psi_n|\hat{x}|\psi_n\rangle \simeq \langle\psi_n|(a^\dagger + a)|\psi_n\rangle = 0$
  - $\langle\psi_n|\hat{x}^2|\psi_n\rangle \simeq \langle\psi_n|(a^\dagger + a)(a^\dagger + a)|\psi_n\rangle$

$$\simeq \langle\psi_n|(2\hat{N}+1)|\psi_n\rangle$$

$$\simeq \frac{\hbar}{m\omega}\left(n + \frac{1}{2}\right) = x_0^2\left(n + \frac{1}{2}\right)$$

$x_0 \equiv$ Extent of Harmonic Oscillator / Size of H.O.

$$(\Delta x) = \sqrt{\langle x^2\rangle - \langle x\rangle^2} = x_0\sqrt{\left(n + \frac{1}{2}\right)}$$

$\Delta x$ going to zero in the classical limit

Sudarshan Got hit by the Sudarshan Chakra twice 👍

- For $\hat{p}$: $\langle\hat{p}^2\rangle = \frac{\hbar^2}{x_0^2}\left(n + \frac{1}{2}\right)$

$a|\alpha\rangle = \alpha|\alpha\rangle$; $\alpha \in \mathbb{Z} = re^{i\theta}$; $r = 0$

$\hookrightarrow$ eigenstate of annihilation operator $\hat{a}$ with eigenvalue $\alpha$, a complex number.

Unitary operators map Hermitian matrix to Hermitian matrix: $\hat{U}^{-1}\hat{H}\hat{U} = \tilde{H}$. Similarity transforms do not change eigenvalues. That means we can take the whole set of Hermitian matrices (which forms a vector space) over the field of real numbers. Space of Hermitian matrices is 4-D ⇒ decomposed into $\sigma_x, \sigma_y, \sigma_z, \mathbb{I}$. So, if we use a non-unitary operator on a Hermitian operator, we get a Hermitian matrix, but with same eigenvalues as the original Hermitian matrix, due to it being a similarity transform.

- For each eigenvalue of such operators, there ~~is~~ are two eigenstates → left and right eigenstates.

$$\left.\begin{array}{l} H|\Psi\rangle = \alpha|\Psi\rangle \\ \langle\Psi|H = \alpha\langle\Psi| \end{array}\right\}$$ Daggering one does not give the other. That property was only relevant for Hermitian matrices.

- <u>Assumption:</u> $\langle\alpha|\alpha\rangle = 1$

$$\langle\alpha|H|\alpha\rangle = \hbar\omega\langle\alpha|a^\dagger a + \tfrac{1}{2}|\alpha\rangle$$
$$= \hbar\omega\left(\langle\alpha|a^\dagger a|\alpha\rangle + \langle\alpha|\tfrac{1}{2}|\alpha\rangle\right)$$
$$= \hbar\omega\left(|\alpha|^2 + \tfrac{1}{2}\right)$$

$$\boxed{\begin{array}{l} a|\alpha\rangle = \alpha|\alpha\rangle \\ \langle\alpha|a^\dagger = \alpha^*\langle\alpha| \\ \langle\alpha|a^\dagger a|\alpha\rangle \\ = \alpha^*\alpha = |\alpha|^2 \end{array}}$$

Now we need the freedom, where one operator translates and another rotates.
→ Full Freedom on state.

- $U(\theta) = e^{-i\theta\hat{N}}$ → Phase shift operator.

$$U(\theta)^\dagger a U(\theta) = a e^{i\theta}$$
$$\frac{d}{d\theta}\left[U^\dagger(\theta)\, a\, U(\theta)\right] = iU^\dagger(\theta)\,[\hat{N}, a]\,U(\theta)$$
$$= -iU^\dagger(\theta)\, a\, U(\theta) = \text{LHS}$$

RHS ⇒ $aie^{i\theta}$.

$$\therefore\ U a|\alpha\rangle = U_\theta a U_\theta^\dagger U_\theta|\alpha\rangle = aU|\alpha\rangle$$
$$\therefore\ aU|\alpha\rangle = \alpha e^{-i\theta}|\alpha\rangle$$